

EASTSOUND SEWER AND WATER DISTRICT
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

NO.	REVISIONS	BY	DATE
1	NEW STAGING AREA	JGC	5/10/22
2	ADD INF. SAMPLE SUMP	JGC	5/10/22
3	ADD UV DISINFECTION	JGC	5/10/22
4	ADD TRENCH DRAINS	JGC	5/10/22
5	ADD INTERIM PLAN OF OP.	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

WILSON
ENGINEERING





DESIGNED BY: JGC
DRAWN BY: SNR
CHECKED BY: AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

COVER SHEET

DATE: 11-02-2022

SHEET: G0.1

SCALE: AS SHOWN

JOB NUMBER: 1 OF 96

AREA MAP- NOT TO SCALE

VICINITY MAP- NOT TO SCALE

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CONTACT INFORMATION

WILSON ENGINEERING, LLC
JEFF CHRISTNER, PE & SAM RODRIGUEZ, EIT
PHONE 360.733.6100

OWNER
EASTSOUND SEWER AND WATER DISTRICT
JASON BRADSHAW, GENERAL MANAGER

PROJECT DATUM

HORIZONTAL: NAD 83/91

VERTICAL: NGVD 1929

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PROTECTION OF THE ENVIRONMENT

NO CONSTRUCTION-RELATED ACTIVITY SHALL CONTRIBUTE TO THE DEGRADATION OF THE ENVIRONMENT, ALLOW MATERIAL TO ENTER SURFACE OR GROUND WATERS, OR ALLOW PARTICULATE EMISSIONS TO THE ATMOSPHERE, WHICH EXCEED STATE OR FEDERAL STANDARDS. ANY ACTIONS THAT POTENTIALLY ALLOW A DISCHARGE TO STATE WATERS MUST HAVE PRIOR APPROVAL OF THE WASHINGTON STATE DEPARTMENT OF ECOLOGY.

PROJECT FUNDING INFORMATION

ECOLOGY SRF FUNDING NO. WQC-2021-EaSoWD-00208

CDBG FUNDING CONTRACT NO. 21-62210-036

THIS PROJECT IS FUNDED IN PART
BY THE WASHINGTON STATE
DEPARTMENT OF ECOLOGY

BID SET

NOV 2, 2022

11-02-2022

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 G0.3 NOTES & SPECIFICATIONS.DWG - 11/2/2022 3:20 PM - Jeff Smith

NOTES & SPECIFICATIONS

GENERAL NOTES

- THE CONTRACTOR SHALL COORDINATE WITH EASTSOUND SEWER AND WATER DISTRICT ON ALL CHANGES TO EXISTING WASTEWATER FACILITY OPERATIONS. IT IS ESSENTIAL THAT EASTSOUND SEWER AND WATER DISTRICT OPERATIONS ARE NOT INTERRUPTED BY CONTRACTOR ACTIVITIES.
- CONTRACTOR AGREES THAT HE SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR JOB SITE CONDITIONS DURING THE COURSE OF CONSTRUCTION OF THIS PROJECT, INCLUDING SAFETY OF ALL PERSONS AND PROPERTY, THAT THIS REQUIREMENT SHALL APPLY CONTINUOUSLY AND NOT BE LIMITED TO NORMAL WORKING HOURS, AND THAT THE CONTRACTOR SHALL DEFEND, INDEMNIFY, AND HOLD THE EASTSOUND SEWER AND WATER DISTRICT AND THE ENGINEER HARMLESS FROM ANY AND ALL LIABILITY, REAL OR ALLEGED, IN CONNECTION WITH THE PERFORMANCE OF WORK ON THIS PROJECT, EXCEPTING FOR LIABILITY ARISING FROM THE SOLE NEGLIGENCE OF THE EASTSOUND WATER AND SEWER DISTRICT OR THE ENGINEER.
- IF THERE ARE ANY DISCREPANCIES BETWEEN DIMENSIONS IN DRAWING AND EXISTING CONDITIONS WHICH WILL AFFECT THE WORK, THE CONTRACTOR SHALL BRING SUCH DISCREPANCIES TO THE ATTENTION OF THE ENGINEER FOR ADJUSTMENT BEFORE PROCEEDING WITH THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER FITTING OF ALL WORK AND FOR THE COORDINATION OF ALL TRADES, SUBCONTRACTORS, AND PERSONS ENGAGED UPON THIS CONTRACT.
- CONTRACTOR SHALL BE RESPONSIBLE TO REPAIR OR REPLACE ANY EXISTING IMPROVEMENTS OR UNDERGROUND FACILITIES DAMAGED BY HIM, HIS SUBCONTRACTORS, OR HIS MATERIAL SUPPLIERS WITHIN 48 HOURS OF THE DAMAGE OCCURRENCE AND/OR AS REQUIRED BY THE CONSTRUCTION INSPECTOR.
- EXISTING UTILITIES AND UNDERGROUND FACILITIES INDICATED ARE FOR INFORMATION ONLY. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY THE LOCATION AND DEPTH. NEITHER THE EASTSOUND SEWER AND WATER DISTRICT NOR THE ENGINEER ASSUMES RESPONSIBILITY THAT THE UTILITIES AND UNDERGROUND FACILITIES INDICATED WILL BE THE UTILITIES AND UNDERGROUND FACILITIES ENCOUNTERED.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING THE ELEVATIONS OF THE EXISTING STORM DRAINS, SEWERS, AND WATER TO BE EXTENDED, CROSSED, OR CONNECTED TO PRIOR TO COMMENCING THE WORK. NOTIFY ENGINEER IF ACTUAL IS DIFFERENT FROM PLANS.
- ALL TRENCHES AND EXCAVATIONS SHALL BE CONSTRUCTED IN STRICT COMPLIANCE WITH THE APPLICABLE SECTIONS OF FEDERAL OSHA REQUIREMENTS AND OTHER APPLICABLE SAFETY ORDINANCES. CONTRACTOR SHALL BEAR FULL RESPONSIBILITY FOR TRENCH SHORING DESIGN AND INSTALLATION.
- THE CONTRACTOR IS RESPONSIBLE FOR MATCHING EXISTING STREETS, SURROUNDING LANDSCAPE, AND OTHER IMPROVEMENTS WITH A SMOOTH TRANSITION IN PAVING, CURBS, GUTTERS, SIDEWALKS, GRADING, ETC, AND TO AVOID ANY ABRUPT OR APPARENT CHANGES IN GRADES OR CROSS SLOPES, LOW SPOTS, OR HAZARDOUS CONDITIONS.
- THE ENGINEER WILL PROVIDE CONSTRUCTION STACKING FOR THE PROJECT. SEE SUPPLEMENTAL CONDITIONS, NO. 7 SURVEYS, PERMITS, REGULATIONS.
- A 1-WEEK NOTIFICATION TO THE EASTSOUND SEWER AND WATER DISTRICT IS REQUIRED PRIOR TO ANY UTILITY SERVICE DISRUPTION. CONTRACTOR MUST PROVIDE TEMPORARY EQUIPMENT (POWER GENERATOR, PUMPS, PIPING, ETC) AND AN APPROVED PLAN FOR ALL WASTEWATER TREATMENT PLANT DISRUPTIONS. SEE SUMMARY OF WORK (SPECIFICATION SECTION 01 11 00.
- ALL APPROVALS AND PERMITS REQUIRED BY SAN JUAN COUNTY SHALL BE OBTAINED BY THE CONTRACTOR PRIOR TO THE START OF CONSTRUCTION.

SCHEDULE NOTES:

- ALL SITE EARTH WORK, CONCRETE WORK AND ASPHALT PAVEMENT REPAIR TO BE PRIORITIZED DURING DRY WEATHER MONTHS.

SURVEY MONUMENTATION:

CONTRACTOR SHALL BE FULLY RESPONSIBLE FOR OBTAINING PERMITS FROM THE WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES FOR REMOVING AND REPLACING ALL SURVEY MONUMENTATION THAT MAY BE AFFECTED BY CONSTRUCTION ACTIVITY, PURSUANT TO WAC 332-120. APPLICATIONS MUST BE COMPLETED BY A REGISTERED LAND SURVEYOR. APPLICATIONS FOR PERMITS TO REMOVE MONUMENTS MAY BE OBTAINED FROM THE WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES (DNR), OR BY CONTACTING THEIR OFFICE BY TELEPHONE AT (206) 902-1190.

WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES
PUBLIC LAND SURVEY OFFICE
1111 WASHINGTON STREET S.E.
OLYMPIA, WASHINGTON 98504-7060

UPON COMPLETION OF CONSTRUCTION, ALL MONUMENTS DISPLACED, REMOVED, OR DESTROYED SHALL BE REPLACED BY A REGISTERED LAND SURVEYOR, AT THE COST OF THE CONTRACTOR, PURSUANT TO THESE REGULATIONS. THE APPROPRIATE FORMS FOR REPLACEMENT OF SAID MONUMENTS SHALL BE COMPLETED AND FILED WITH DNR AT THE CONTRACTOR'S EXPENSE.

CONSTRUCTION INSPECTIONS:

CONTRACTOR MUST COORDINATE REQUIRED INSPECTIONS WITH THE COUNTY BUILDING INSPECTION OFFICIALS. IN ADDITION, CONTRACTOR IS TO PROVIDE OWNER AND ENGINEER 24-HOUR (MINIMUM) NOTICE FOR SCHEDULED INSPECTIONS. REQUIRED INSPECTIONS INCLUDE THE FOLLOWING ITEMS:

- BUILDING FOOTINGS
- CONCRETE STRUCTURES
- ELECTRICAL INSPECTIONS
- FINAL OVERALL INSPECTION

CONTRACTOR MUST PROVIDE OWNER AND ENGINEER 24-HOUR (MINIMUM) NOTICE FOR ALL SPECIAL INSPECTIONS IDENTIFIED ON S1.1 - SPECIAL INSPECTIONS, TESTING, AND STRUCTURAL OBSERVATIONS.

GENERAL NOTES, CONT.

PROTECTION OF THE ENVIRONMENT

NO CONSTRUCTION RELATED ACTIVITY SHALL CONTRIBUTE TO THE DEGRADATION OF THE ENVIRONMENT, ALLOW MATERIAL TO ENTER SURFACE OR GROUND WATERS, OR ALLOW PARTICULATE EMISSIONS TO THE ATMOSPHERE, WHICH EXCEED STATE OR FEDERAL STANDARDS. ANY ACTIONS THAT POTENTIALLY ALLOW A DISCHARGE TO STATE WATERS MUST HAVE PRIOR APPROVAL OF THE STATE OF WASHINGTON, DEPARTMENT OF ECOLOGY

OPERATION PLAN

WWTP UGRADE CONSTRUCTION - GENERAL NOTES

- THE EXISTING WASTEWATER TREATMENT PLANT (WWTP) SHALL NOT BE INTERRUPTED DURING CONSTRUCTION EXCEPT AS DESCRIBED BELOW. THE CONTRACTOR SHALL BE RESPONSIBLE FOR SEQUENCING CONSTRUCTION ACTIVITIES SO THAT NO INTERRUPTION OCCURS.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ALL TEMPORARY BY-PASS PUMPING, PIPING, HOLDING TANKS AND ADDITIONAL STORAGE NECESSARY TO KEEP SEWER AND WWTP SYSTEM UNINTERRUPTED DURING CONSTRUCTION ACTIVITIES.
- THE PLAN ASSUMES THAT ALL BASIN, BUILDINGS, EQUIPMENT, AND PIPING WILL BE INSTALLED/TESTED AND READY TO BE CONNECTED PRIOR TO SWITCH-OVER. THIS IS A SUGGESTED PLAN AND DOES NOT RELIEVE THE CONTRACTOR FROM OBLIGATIONS TO AVOID INTERRUPTION:
- EMERGENCY CONTACTS:

ESWD: JASON BRADSHAW, GENERAL MANAGER, CELL: 620-441-4006
WILSON ENG: JEFF CHRISTNER, PROJECT ENGINEER, CELL: 360-303-5129

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



WILSON
ENGINEERING



DESIGNED BY
JGC

DRAWN BY
SNR

CHECKED BY
AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

NOTES & SPECIFICATIONS

DATE

SCALE

AS SHOWN

JOB NUMBER

SHEET

3 OF 96



BID SET

11-02-2022

NOV 2, 2022

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020_070 G0.4 WAC COMPLIANCE.DWG - 11/2/2022 3:20 PM - Jeff Smith

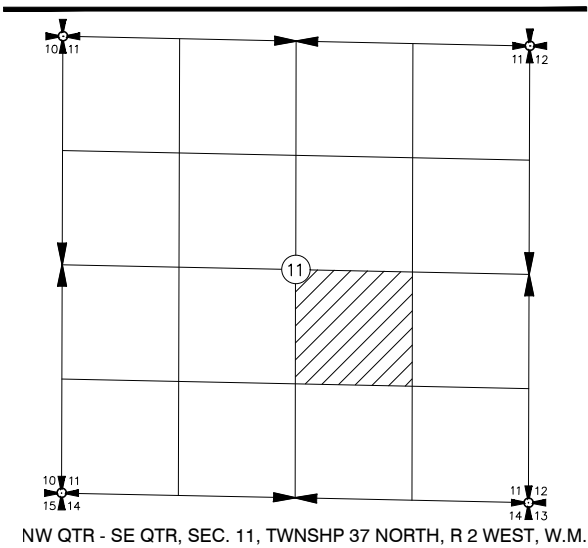
NOV 2, 2022
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EASTSOUND SEWER AND WATER DISTRICT

WASTEWATER TREATMENT PLANT DESIGN

W.A.C. 332-130 COMPLIANCE SHEET

SECTIONAL INDEX DATA



NOTICE TO USER

EFFECTIVE JANUARY 13, 2019, ALL TOPOGRAPHIC MAPS PREPARED BY A LICENSED SURVEYOR IN THE STATE OF WASHINGTON, AND SUBJECT TO THE LICENSURE AND PRACTICE REQUIREMENTS ESTABLISHED BY THE WASHINGTON STATE BOARD OF REGISTRATION FOR ENGINEERS AND LAND SURVEYORS, MUST INCLUDE THE DESCRIPTIVE NOTES AND METADATA ENUMERATED UNDER W.A.C 332-130-145 AND ITS APPURTENANT SECTIONS OF 332-130. THIS EXHIBIT IS INTENDED TO ADDRESS THE STATUTORY REQUIREMENTS STIPULATED BY THIS W.A.C DIRECTIVE.

W.A.C. 332-130-145 REQUIRED DATA

1.E: THIS SURVEY WAS PREPARED UNDER THE DIRECT SUPERVISION OF:

PAUL J. DARROW, WA PLS #50697
SR. PROJECT SURVEYOR
WILSON ENGINEERING LLC
805 DUPONT STREET, SUITE 7
BELLINGHAM, WA 98225
360-733-6100 (EXT. 243)
pdarrow@wilsonengineering.com

2.A: BASIS OF ELEVATIONS: ELEVATION VALUES AND CONTOURS DEPICTED ON THIS SURVEY ARE BASED UPON HOLDING AS FIXED THE NGVD29 DATUM, PER BENCHMARK NO. 20 RLM, ACCORDING TO THE PERPETUATION OF SAN JUAN COUNTY BENCHMARK #00165, HAVING AN ELEVATION OF 34.28, PER LAND CORNER RECORD # 2000 0303018.

2.B: PURPOSE OF SURVEY: WILSON ENGINEERING PERFORMED THIS SURVEY DURING JANUARY OF 2013 WITH ADDITIONAL DATA AND CHANGES BEING SURVEYED IN MARCH OF 2021, AT THE REQUEST OF EASTSOUND SEWER AND WATER DISTRICT PURSUANT TO WASTEWATER TREATMENT PLANT UPGRADE DESIGN. THIS SURVEY WAS PREPARED WITH THE BENEFIT OF A TITLE REPORT, AND THE DEPICTED PARCEL BOUNDARY SHOULD BE CONSIDERED AUTHORITATIVE.

2.C: SOURCE OF CONTOURS: THE CONTOURS DEPICTED ON THIS SURVEY WERE DERIVED BASED ON DIRECT FIELD OBSERVATIONS.

2.D: CONTOUR INTERVAL LABELING: MAJOR CONTOURS AT 1-FOOT INTERVALS HAVE BEEN EXPLICITLY LABELED.

2.E: DESCRIPTION OF BENCHMARKS SET PURSUANT TO THIS SURVEY: REFER TO THE ACCOMPANYING "CONTROL TABLE" FOR COORDINATES, ELEVATION, AND DESCRIPTION OF ON-SITE CONTROL SET PURSUANT TO THIS SURVEY.

2.F: ELEVATION AND/OR CONTOUR ACCURACY: IF CONTOURS HAVE BEEN DEPICTED ON THE FACE OF THIS SURVEY, IT IS ANTICIPATED THAT 90% OF ANY MEASURED ELEVATION VALUE, IF OBSERVED RELATIVE TO THE CONTROL POINTS SPECIFICALLY ENUMERATED IN THE ACCOMPANYING CONTROL TABLE, WILL BE, IN FACT, WITHIN ONE-HALF OF THE MINOR-CONTOUR INTERVAL DEPICTED HEREON. SPECIFIC ELEVATIONS DEPICTED HEREON, IF ANY, ARE EXPECTED TO BE WITHIN ONE INTEGRAL VALUE OF THE FINAL DEPICTED SIGNIFICANT FIGURE. THAT IS, 90% OF ELEVATIONS EXPRESSED TO THE TENTH-FOOT, SHOULD BE WITHIN 0.1 FEET OF THAT VALUE, IF OBSERVED RELATIVE TO THE SURVEY CONTROL SPECIFICALLY ENUMERATED IN THE ACCOMPANYING CONTROL TABLE. IF OFF-SITE CONTROL IS EMPLOYED, EVEN CONTROL PURPORTING TO BE ON THE SAME DATUM OR BASED ON THE SAME OFF-SITE BENCHMARK, THEN NO ABSOLUTE STATEMENT REGARDING THE ACCURACY OF THE DEPICTED POINTS CAN BE MADE, AND VALUES SO OBSERVED ARE OUTSIDE OF THIS SURVEY'S AUTHORITY OR INTEREST.

2.G: STATEMENT OF USE: AS NOTED IN SECTION 2.B, THIS SURVEY WAS PREPARED FOR THE SPECIFIC PURPOSE OF WASTEWATER TREATMENT PLAN UPGRADE DESIGN. IN THE COURSE OF PREPARING THIS SURVEY, PURSUANT TO THIS PURPOSE, ANCILLARY DATA NECESSARY TO ACCOMPLISH THIS SURVEYS INTENDED PURPOSE MAY HAVE BEEN CAPTURED. IN THE CASE OF THIS SURVEY A NUMBER OF BUILDING ENVELOPES ON ADJACENT PARCELS WERE CAPTURED, BUT THE DEPICTION OF SAME SHOULD NOT BE CONSIDERED AUTHORITATIVE.

2.H: SOURCE OF CONTROLLING BOUNDARY INFORMATION: THE OWNERSHIP BOUNDARIES DEPICTED ON THIS SURVEY ARE BASED UPON SOME, OR ALL, OF THE DOCUMENTS ENUMERATED IN THE ACCOMPANYING "REFERENCE DOCUMENTS" AS THEREIN CHARACTERIZED. BEARINGS HAVE BEEN TRANSLATED AND/OR ROTATED FROM THE RECORD VALUES TO CONFORM WITH THE CURRENT DATUM OF THIS SURVEY.

3.A: SOURCE OF DEPICTED UTILITY INFORMATION: UTILITY LINES DEPICTED ON THIS SURVEY ARE BASED UPON DIRECT OBSERVATION OF SURFACE VISIBLE STRUCTURES AND LOCATE MARKS MADE BY UTILITY LOCATING PROFESSIONALS.

3.B: ACCURACY OF DEPICTED UTILITY INFORMATION: WILSON ENGINEERING DOES NOT PROVIDE FOR-HIRE UTILITY LOCATION AND/OR MARKING SERVICES, AND CAN NOT INDEPENDENTLY ASCERTAIN THE ACCURACY OF ANY DEPICTED UTILITY THAT WAS NOT DIRECTLY OBSERVED IN THE COURSE OF THIS SURVEY.

3.C: STATEMENT OF LIMITATIONS REGARDING UTILITY-DEPICTION ACCURACY: EASTSOUND SEWER AND WATER DISTRICT HAS BEEN NOTIFIED THAT WILSON CAN NOT, AND DOES NOT, GUARANTEE THE ACCURACY, AT ANY LEVEL, OF DEPICTED UTILITIES BASED ON THIRD-PARTY PAINT MARKS OR RECORD INFORMATION.

CONTROL NOTES

HORIZONTAL DATUM: BASIS OF COORDINATES, BEARINGS AND CALCULATIONS ARE NAD 83/91 WASHINGTON STATE PLANE NORTH ZONE BASED ON TIES TO MONUMENTS AS SHOWN PER RECORD OF SURVEY ON FILE IN BOOK 16 OF SURVEYS AT PAGE 99, RECORDS OF SAN JUAN COUNTY, WASHINGTON.

BASIS OF COORDINATES: COORDINATION AND MENSURATION ARE LOCAL GROUND VALUES, BASED UPON HOLDING THE PUBLISHED NAD83/91 POSITION FOR THE 3" BRASS DISC IN CASE AT PROJECTED CENTERLINE OF MT. BAKER ROAD, SOUTH OF THE AIRPORT CENTER BUILDING. DISC BEARS THE STAMP "SAN JUAN COUNTY SURVEY CONTROL MONUMENT 00165". SAID MONUMENT HAS THE FOLLOWING PUBLISHED POSITION:

NORTHING = 628,007.20 USFT
EASTING = 1,138,590.22 USFT

BASIS OF BEARINGS: BEARINGS ARE BASED UPON HOLDING THE CALCULATED INVERSE BETWEEN SAN JUAN COUNTY MONUMENTS "00165" AND THE BRASS DISK PRESUMED TO BE THAT MONUMENT MARKING THE E 16TH CORNER BETWEEN SECTIONS 11 AND 14, PER THE AFOREMENTIONED RECORD OF SURVEY. SAID INVERSE BEING S 88° 16' 54" E, AT A DISTANCE OF 1301.67' (GRID). THE PUBLISHED WASHINGTON STATE PLANE (NORTH ZONE) VALUES FOR THE 16TH-CORNER MONUMENT ARE AS FOLLOWS:

NORTHING = 627,968.17 USFT
EASTING = 1,139,891.30 USFT

VERTICAL DATUM:
NGVD 29

BASIS OF ELEVATIONS: ELEVATIONS SHOWN ON THIS SURVEY ARE BASED ON THE USGS BENCH MARK DESIGNATED 20 RLM (BRASS DISC IN CONCRETE) AT THE SOUTH END OF THE RUNWAY AT ORCAS ISLAND AIRPORT, HAVING AN ELEVATION OF 31.0 FEET (NGVD 29), PER THE PERPETUATION OF SAN JUAN COUNTY BENCHMARK #00165, HAVING AN ELEVATION OF 34.28, PER LAND CORNER RECORD #2000 0303018.

ON-SITE SURVEY CONTROL TABLE

POINT	NORTHING	EASTING	ELEVATION	DESCRIPTION
205	629,477.19	1,139,257.92	23.74	REBAR AND PLASTIC CAP
206	629,622.13	1,139,287.90	22.40	REBAR AND PLASTIC CAP
208	629,620.32	1,139,364.78	23.16	PK NAIL AND WASHER
211	629,528.31	1,139,360.16	23.28	PK NAIL AND WASHER

SURVEYOR'S CERTIFICATE

I HEREBY CERTIFY THAT I AM A LICENSED LAND SURVEYOR IN THE STATE OF WASHINGTON, THAT THIS MAP IS BASED ON AN ACTUAL FIELD SURVEY DONE BY ME OR UNDER MY DIRECT SUPERVISION AND THAT ALL DATA SHOWN HEREON ACTUALLY EXISTS IN THE LOCATIONS SHOWN AT THE TIME OF THIS SURVEY. THIS TOPOGRAPHIC MAP WAS DONE AT THE REQUEST OF EASTSOUND SEWER AND WATER DISTRICT IN 2021.

PAUL JONATHAN DARROW, P.L.S. NO. 50697 DATE

LEGAL DESCRIPTION PER AFN. 2021-1022007

THE NORTH 144 FEET OF THE WEST 148 FEET OF THE SOUTH HALF OF THE SOUTHEAST QUARTER OF THE NORTHWEST QUARTER OF THE SOUTHEAST QUARTER OF SECTION 11, TOWNSHIP 37 NORTH, RANGE 2 WEST, W.M., IN SAN JUAN COUNTY, WASHINGTON.

TOGETHER WITH THE SOUTH 10 FEET OF THE NORTH 154 FEET OF THE EAST 112 FEET OF THE WEST 148 FEET OF THE SOUTH HALF OF THE SOUTHEAST QUARTER OF THE NORTHWEST QUARTER OF THE SOUTHEAST QUARTER OF SECTION 11, TOWNSHIP 37 NORTH, RANGE 2 WEST, W.M., IN SAN JUAN COUNTY, WASHINGTON.

RECORD DOCUMENTS

- BOOK 3, PAGE 61 - PLAT OF SURVEY FOR EASTSOUND WATER DISTRICT
- BOOK 18, PAGE 131 - RECORD OF SURVEY FOR PORT OF ORCAS
- BOOK 16, PAGE 99 - RECORD OF SURVEY FOR SAN JUAN COUNTY

SURVEYOR'S NOTES

1. THIS TOPOGRAPHIC SURVEY WAS ORIGINALLY PERFORMED IN JANUARY 2013. ADDITIONAL DATA WAS SURVEYED IN MARCH 2021. ALL MONUMENTS SHOWN HEREON WERE VISITED DURING THE COURSE OF THIS SURVEY UNLESS OTHERWISE NOTED. THIS SURVEY CORRECTLY SHOWS THE AREA OF THE SUBJECT PROPERTY AND THE LOCATION OF ALL STRUCTURES AND IMPROVEMENTS SITUATED ON THE SUBJECT PROPERTY.

2. ANGULAR AND LINEAR MEASUREMENTS WERE COLLECTED USING A COMBINATION OF GPS AND CONVENTIONAL METHODOLOGIES. PRIMARY CONTROL WAS COLLECTED USING TRIMBLE 5700 SURVEY-GRADE GPS RECEIVERS OPERATING IN NETWORKED RTK MODE. FROM GPS CONTROL, A TRIMBLE S7 ROBOTIC TOTAL STATION WAS USED TO TIE SECONDARY CONTROL POINTS, COLLECT TOPOGRAPHIC DATA AND MEASURE LOT CORNERS.

3. THE LOCATION OF UTILITIES SHOWN HEREON ARE FROM OBSERVED EVIDENCE OF ABOVE GROUND APPURTENANCES TOGETHER WITH MARKINGS BY UTILITY COMPANIES.



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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

W.A.C. 332-130 COMPLIANCE SHEET

DATE

SCALE

AS SHOWN

JOB NUMBER

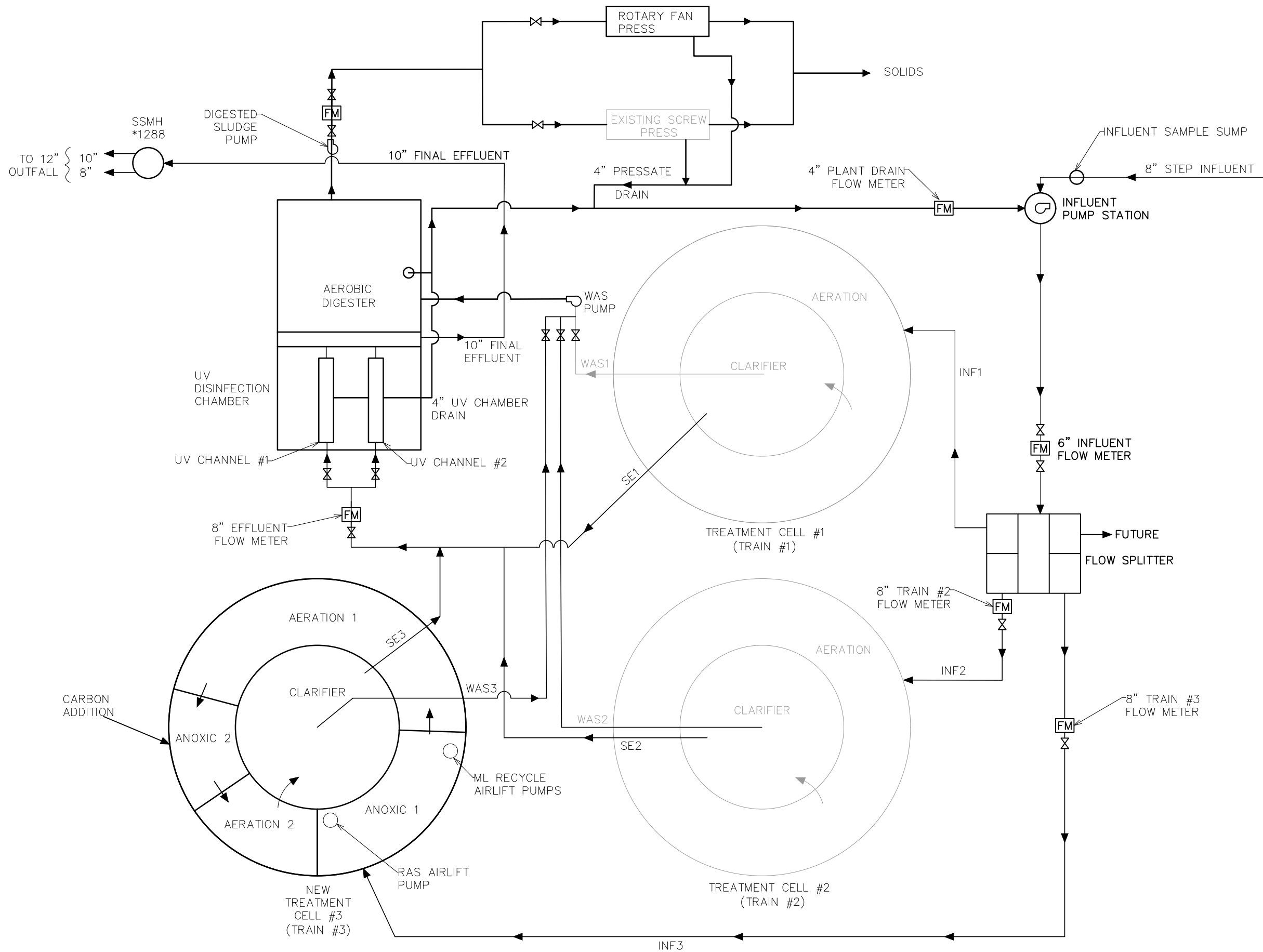
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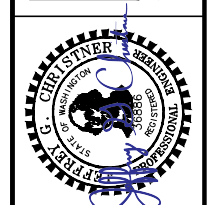
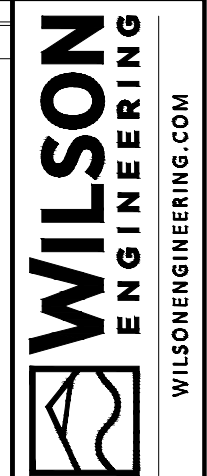
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NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



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	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I	
DATE	SCALE	JOB NUMBER
	AS SHOWN	96

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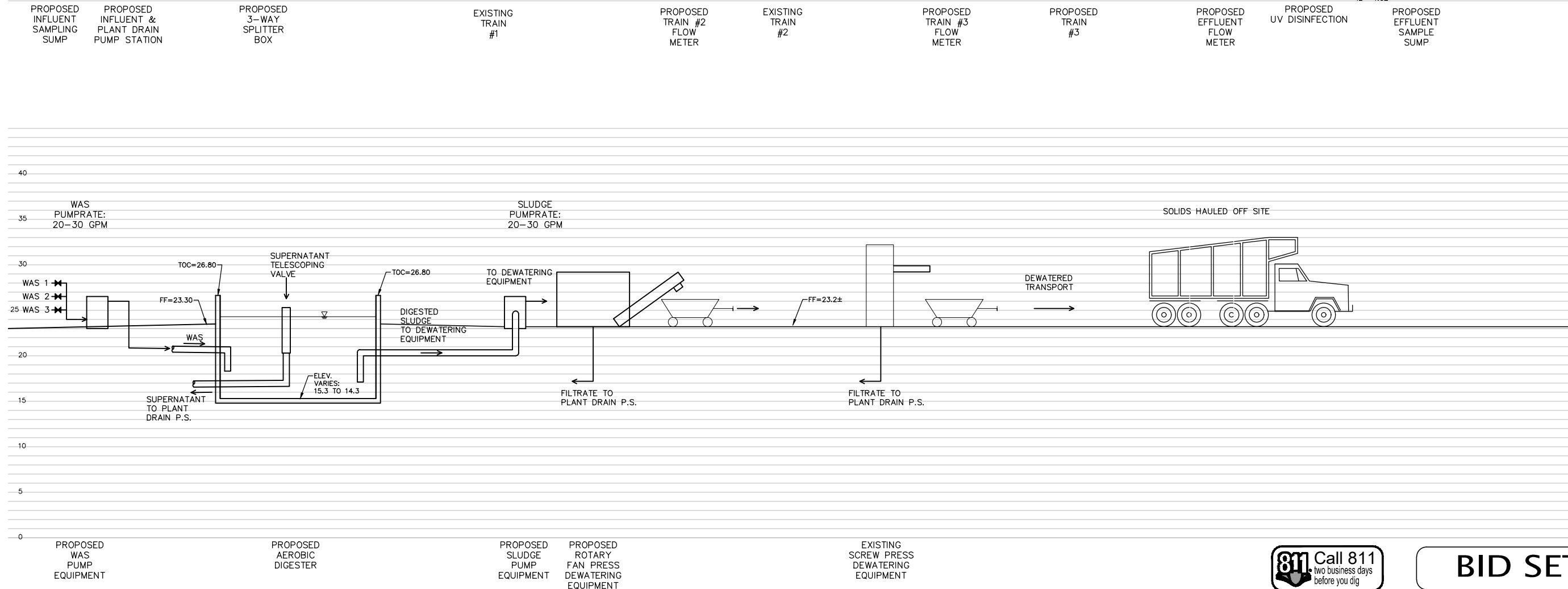
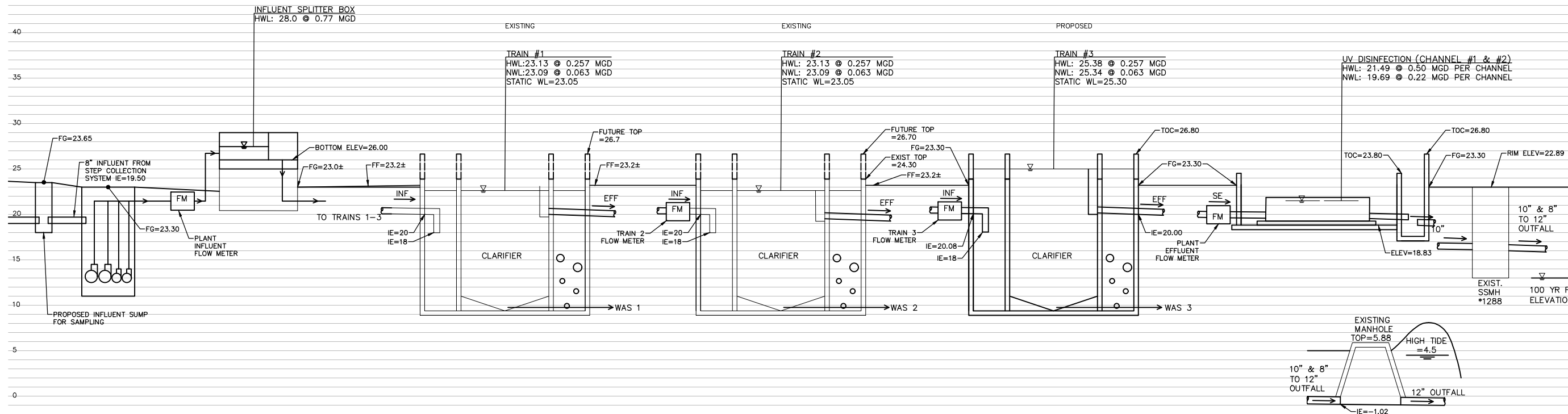
LEGEND	
	ISOLATION VALVE
	FLOW DIRECTION
	FLOW METER
	AIRLIFT PUMP



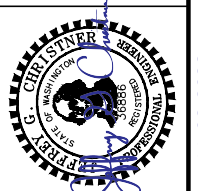
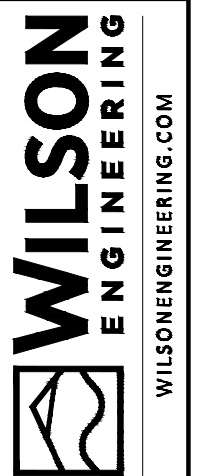
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NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
HYDRAULIC PROFILE

DATE	SCALE	JOB NUMBER
	AS SHOWN	96
SHEET		6 OF 96



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3	ADD UV DISINFECTION	JGC	5-10-22

WASTEWATER TREATMENT PLANT DESIGN INFORMATION

START-UP DESIGN INFORMATION
 (AFTER PHASE I IS COMPLETED)

FLOW:

ADF	= 0.10 MGD
MMF	= 0.12 MGD
MAX. DAY	= 0.30 MGD
PEAK HR	= 0.49 MGD

AVERAGE DAILY LOADS:

BOD	= 189 mg/L
TSS	= 46 mg/L
Ammonia-N	= 47 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 47 mg/L
TKN	= 62 mg/L
Alkalinity	= 328 mg/L

MAX MONTH LOADS:

BOD	= 250 mg/L
TSS	= 59 mg/L
Ammonia-N	= 62 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 62 mg/L
TKN	= 82 mg/L
Alkalinity	= 433 mg/L

CLARIFIER INFORMATION:

NO. OF UNITS:	3
DIAMETER:	18 FT
SWD:	CELL 1: 13.25 FT, CELL 2: 13.25' FT, CELL 3: 15.50'
RAS:	0.5Q TO 1.0Q

CLARIFIERS

ADF SOR:	197 GAL / DAY / SF
MMF SOR:	236 GAL / DAY / SF
MAX DAY:	588 GAL / DAY / SF

AERATION BASIN INFORMATION:

NO. OF UNITS: 3

TRAIN #1 (EXISTING)

VOLUME:	0.054 MG
ADF HRT:	38.9 HOURS
MMF HRT:	32.4 HOURS
MAX DAY:	13.0 HOURS

TRAIN #2 (EXISTING)

VOLUME:	0.054 MG
ADF HRT:	38.9 HOURS
MMF HRT:	32.4 HOURS
MAX DAY:	13.0 HOURS

TRAIN #3 (PROPOSED)

VOLUME:	0.0397 MG
ADF HRT:	28.1 HOURS
MMF HRT:	23.3 HOURS
MAX DAY:	9.4 HOURS

ANOXIC BASIN INFORMATION

NO. OF UNITS: 1

TRAIN #4 (PROPOSED)

VOLUME:	0.0245 MG
ADF HRT:	18 HOURS
MMF HRT:	15 HOURS
MAX DAY:	6 HOURS

UV DISINFECTION

MONTHLY GEOMETRIC MEAN: 200 FC/100 ML
 WEEKLY GEOMETRIC MEAN: 400 FC/100 ML

CAPACITY OF 1 CHANNEL: 0.5 MGD
 CAPACITY OF 2 CHANNELS : 1.0 MGD

AEROBIC DIGESTER

VOL: 15,000 GAL
 ADF STORAGE AT 2.0% SOLIDS = 4.5 WEEKS
 MMF STORAGE AT 2.0% SOLIDS = 2.9 WEEKS
 MMF STORAGE AT 1.0% SOLIDS = 1.5 WEEKS

BUILD-OUT DESIGN INFORMATION
 (YR - 2040 AFTER PHASE II IS COMPLETED)

FLOW:

ADF	= 0.19 MGD
MMF	= 0.22 MGD
MAX. DAY	= 0.46 MGD
PEAK HR	= 0.77 MGD

AVERAGE DAILY LOADS:

BOD	= 181 mg/L
TSS	= 44 mg/L
Ammonia-N	= 45 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 45 mg/L
TKN	= 60 mg/L
Alkalinity	= 314 mg/L

MAX MONTH LOADS:

BOD	= 250 mg/L
TSS	= 59 mg/L
Ammonia-N	= 62 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 62 mg/L
TKN	= 82 mg/L
Alkalinity	= 433 mg/L

CLARIFIER INFORMATION:

NO. OF UNITS:	3
DIAMETER:	18 FT
SWD:	15.50 FT
RAS:	0.5Q TO 1.0Q

CLARIFIERS

ADF SOR:	249 GAL / DAY / SF
MMF SOR:	286 GAL / DAY / SF
MAX DAY:	603 GAL / DAY / SF
PEAK HR:	1008 GAL / DAY / SF

AERATION & ANOXIC BASIN INFORMATION:

NO. OF TRAINS: 3
 NO. OF AERATION BASINS PER TRAIN: 2
 NO. OF ANOXIC BASINS PER TRAIN: 2
 TOTAL VOLUME OF AERATION & ANOXIC BASINS PER TRAIN: 0.0642 MG
 MMF HRT: 21 HOURS

UV DISINFECTION

MONTHLY GEOMETRIC MEAN: 200 FC/100 ML
 WEEKLY GEOMETRIC MEAN: 400 FC/100 ML

CAPACITY OF 1 CHANNEL: 0.5 MGD
 CAPACITY OF 2 CHANNELS : 1.0 MGD

AEROBIC DIGESTER

VOL: 15,000 GAL
 ADF STORAGE AT 2.0% SOLIDS (WITH CARBON ADDITION) = 1.9 WEEKS
 MMF STORAGE AT 2.0% SOLIDS (WITH CARBON ADDITION) = 1.2 WEEKS

PROCESS DESIGN CRITERIA

TRAIN 3 AERATION BASIN: DESIGN INFORMATION

- AERATION BASIN MIXING = MEDIUM BUBBLE AERATION MIXING
- ANOXIC BASIN MIXER = SUBMERSIBLE MIXERS
 - ONE 5 HP SUBMERSIBLE MIXER IN ANOXIC ZONE #1: 15,686 GALLONS.
 - ONE 5 HP SUBMERSIBLE MIXER IN ANOXIC ZONE #2: 8,789 GALLONS.
- AERATION BASINS
 - AERATION ZONE #1: 29,793 GALLONS, 8 DIFFUSERS. DESIGN: 24 SCFM/DIFFUSER, 192 SCFM TOTAL
 - AERATION ZONE #2: 9,889 GALLONS, 2 DIFFUSERS. DESIGN: 24 SCFM/DIFFUSER, 48 SCFM TOTAL
 - TOTAL AERATION AIR = 240 SCFM
- AIR LIFT PUMPS
 - AIR NEEDED FOR RAS AIRLIFT PUMP = 15 SCFM. PIPE SIZE = 3-INCH. FLOW = 51 GPM. TARGET = 1Q
 - AIR NEEDED FOR NITRATE RECYCLE AIRLIFT PUMP = 61 SCFM. PIPE SIZE = 6-INCH. FLOW = 203 GPM. TARGET = 4Q
 - TOTAL AIRLIFT AIR = 76 SCFM
- TOTAL AIR REQUIRED = 316 SCFM AT 9.2 PSIG
- SRT = 15 DAYS (YR-2040 MMF)
- HRT = 21 HOURS (YR-2040 MMF)
- MLSS = 3000 MG/L (RANGE = 2000 TO 4000 MG/L)
- F/M = 0.10 TO 0.20 (LBS INFLUENT BOD/LBS MLVSS)
- VOLUMETRIC LOAD = 10 TO 30 (LBS. INFLUENT BOD/KCF DAY)

UV DISINFECTION: DESIGN INFORMATION

- UV DOSAGE = 30,000 MJ/CM2
- DESIGN UV TRANSMITTANCE = 50% OR GREATER
- LAMPS: 2 PER MODULE, 8 MODULES PER CHANNEL, 2 CHANNELS TOTAL
- HYDRAULIC CAPACITY = 0.5 MGD PER CHANNEL, TOTAL CAPACITY = 1.0 MGD

AEROBIC DIGESTER: DESIGN INFORMATION

- AERATION: COARSE BUBBLE. CAPACITY = 20 CFM/DIFFUSER X 6 DIFFUSERS = 120 CFM TOTAL.
- DIGESTER VOLUME = 15,000 GALLONS = 2,000 CF
- AIR NEEDED FOR MIXING = 30 CFM PER KCF OF STORAGE VOLUME = 30 X 2 = 60 CFM
- MLSS RANGE = 2000 TO 4000 MG/L
- WWTP MCRT RANGE = 5 TO 15 DAYS
- DIGESTER SOLIDS RETENTION TIME = 7 DAYS TO 28 DAYS (1 TO 4 WEEKS)
- DECANT CONTROL: MANUAL T-VALVE ADJUSTMENTS
- WEEKLY DECANT VOLUME = 50% TO 75% OF WEEKLY WAS VOLUME (WAS = 10,000 TO 18,000 GAL/WK; DECANT = 5000 TO 13,500 GAL/WK)

INFLUENT PUMP STATION: SIZE, TYPE, DESIGN CAPACITY

- 2 SMALL SUBMERSIBLE PUMPS. CAPACITY OF EACH PUMP = 120 TO 220 GPM
- 2 LARGE SUBMERSIBLE PUMPS. CAPACITY OF EACH PUMP = 320 TO 500 GPM
- TOTAL FIRM PS CAPACITY (WITH 1 SMALL AND 1 LARGE PUMP OUT OF SERVICE) = 535 GPM (0.77 MGD)

FLOW METER INFO: SIZE, TYPE, DESIGN CAPACITY

- INFLUENT: 6", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 10 FPS = 0 TO 880 GPM (1.27 MGD)
- TRAIN #2: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 3 FPS = 0 TO 470 GPM (0.68 MGD)
- TRAIN #3: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 3 FPS = 0 TO 470 GPM (0.68 MGD)
- EFFLUENT: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 6 FPS = 0 TO 940 GPM (1.35 MGD)
- PLANT DRAIN: 4", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 5 FPS = 0 TO 200 GPM (0.29 MGD)
- 2W WATER: 1", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 20 FPS = 0 TO 25 GPM

RAS RECYCLE: DESIGN INFORMATION

AIR NEEDED FOR RAS AIRLIFT PUMP = 15 SCFM. PIPE SIZE = 3-INCH. FLOW = 51 GPM. TARGET = 1Q

NITRATE RECYCLE: DESIGN INFORMATION

AIR NEEDED FOR NITRATE RECYCLE AIRLIFT PUMP = 61 SCFM. PIPE SIZE = 6-INCH. FLOW = 203 GPM. TARGET = 4Q

WAS PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP (SWING PUMP)
- PUMP STYLE: HOSE PUMP (PERISTALTIC)
- PUMP CAPACITY: 0 TO 40 GPM (ANTICIPATED SETPOINT RANGE 10 GPM TO 30 GPM)

DIGESTER PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP (SWING PUMP)
- PUMP STYLE: HOSE PUMP (PERISTALTIC)
- PUMP CAPACITY: 0 TO 40 GPM (ANTICIPATED SETPOINT RANGE 12 GPM TO 20 GPM)

ROTARY FAN PRESS: DESIGN INFORMATION

- NUMBER OF DEWATERING UNITS: 1 UNIT
- DEWATERING UNIT STYLE: ROTARY FAN SYSTEM
- DEWATERING UNIT INPUT CAPACITY: 12 GPM AT 2% SOLIDS, 20 GPM AT 0.5% SOLIDS
- DEWATERING UNIT OUTPUT CAPACITY:
 - 12% TO 16% CAKE SOLIDS
 - WET CAKE THROUGHPUT: UP TO 1000 WET LBS. PER HOUR
 - DRY CAKE THROUGHPUT: UP TO 120 DRY LBS. PER HOUR

2W WATER SYSTEM PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP
- PUMP STYLE: SUBMERSIBLE PUMP
- PUMP CAPACITY: 20 TO 80 GPM (VFD CONTROL)



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WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DESIGN CRITERIA

DATE

SCALE

AS SHOWN

JOB NUMBER

SHEET

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OF

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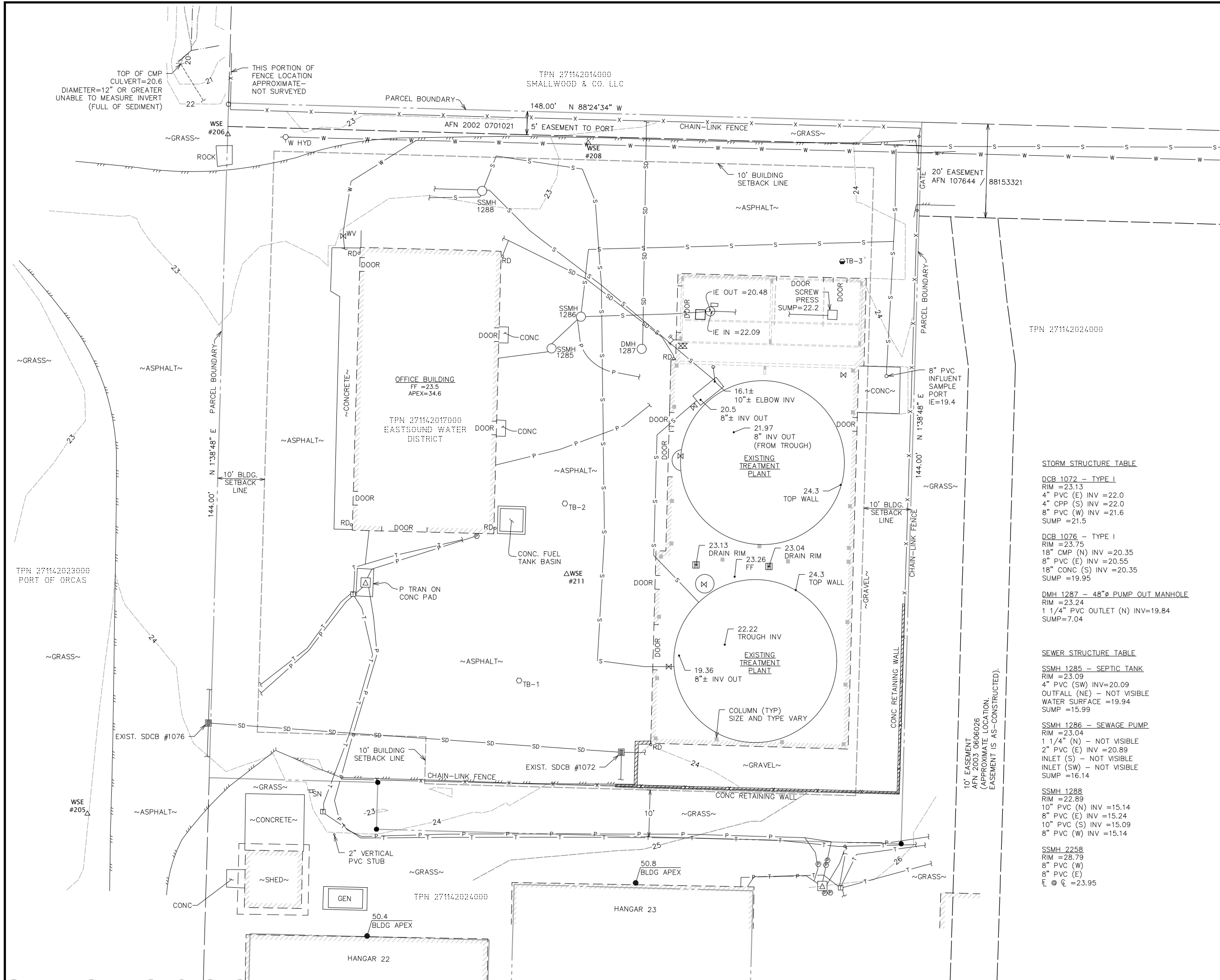
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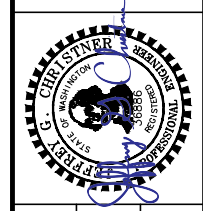
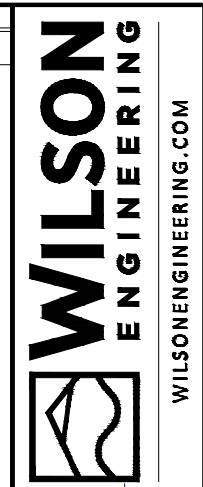


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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I	EXISTING CONDITIONS

SHEET	DATE	
	SCALE	
	JOB NUMBER	

STORM STRUCTURE TABLE

DCB 1072 - TYPE I
RIM = 23.13
4" PVC (E) INV = 22.0
4" CPP (S) INV = 22.0
8" PVC (W) INV = 21.6
SUMP = 21.5

DCB 1076 - TYPE I
RIM =23.75
18" CMP (N) INV =20.35
8" PVC (E) INV =20.55
18" CONC (S) INV =20.35
SUMP =19.95

DMH 1287 - 48"Ø PUMP OUT MANHOLE
RIM =23.24
1 1/4" PVC OUTLET (N) INV=19.84
SUMP=7.04

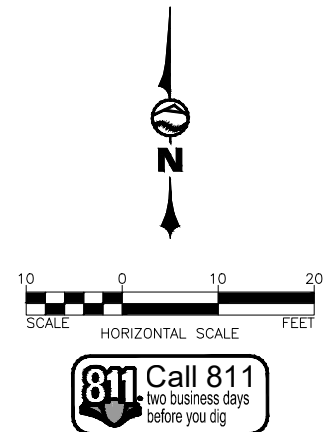
SEWER STRUCTURE TABLE

SSMH 1285 - SEPTIC TANK
RIM =23.09
4" PVC (SW) INV=20.09
OUTFALL (NE) - NOT VISIBLE
WATER SURFACE =19.94
SUMP =15.99

SSMH 1286 - SEWAGE PUMP
RIM = 23.04
1 1/4" (N) - NOT VISIBLE
2" PVC (E) INV = 20.89
INLET (S) - NOT VISIBLE
INLET (SW) - NOT VISIBLE
SUMP = 16.14

SSMH 1288
RIM = 22.89
10" PVC (N) INV = 15.14
8" PVC (E) INV = 15.24
10" PVC (S) INV = 15.09
8" PVC (W) INV = 15.14

SSMH 2258
RIM = 28.79
8" PVC (W)
8" PVC (E)
E @ C = 23.95



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REQUIRED TESC ELEMENTS

ELEMENT 1: PRESERVE VEGETATION/MARK CLEARING LIMITS

A) BEFORE BEGINNING LAND DISTURBING ACTIVITIES, INCLUDING CLEARING AND GRADING, CLEARLY MARK ALL CLEARING LIMITS, SENSITIVE AREAS AND THEIR BUFFERS, AND TREES THAT ARE TO BE PRESERVED WITHIN THE CONSTRUCTION AREA.
B) RETAIN THE DUFF LAYER, NATIVE TOP SOIL, AND NATURAL VEGETATION IN AN UNDISTURBED STATE TO THE MAXIMUM DEGREE PRACTICABLE.

ELEMENT 2: ESTABLISH CONSTRUCTION ACCESS

A) LIMIT CONSTRUCTION VEHICLE ACCESS AND EXIT TO ONE ROUTE, IF POSSIBLE.
B) STABILIZE ACCESS POINTS WITH A PAD OF QUARRY SPALLS, CRUSHED ROCK, OR OTHER EQUIVALENT BMPS, TO MINIMIZE TRACKING OF SEDIMENT ONTO PUBLIC ROADS.
C) LOCATE WHEEL WASH OR TIRE BATHS ON SITE, IF THE STABILIZED CONSTRUCTION ENTRANCE IS NOT EFFECTIVE IN PREVENTING TRACKING SEDIMENT ONTO ROADS.
D) IF SEDIMENT IS TRACKED OFF SITE, CLEAN THE AFFECTED ROADWAY THOROUGHLY AT THE END OF EACH DAY, OR MORE FREQUENTLY AS NECESSARY (FOR EXAMPLE, DURING WET WEATHER). REMOVE SEDIMENT FROM ROADS BY SHOVELING, SWEEPING, OR PICK UP AND TRANSPORT THE SEDIMENT TO A CONTROLLED SEDIMENT DISPOSAL AREA.
E) CONDUCT STREET WASHING ONLY AFTER SEDIMENT IS REMOVED IN ACCORDANCE WITH THE ABOVE ITEM.
F) CONTROL STREET WASH WASTEWATER BY PUMPING BACK ON-SITE, OR OTHERWISE PREVENT IT FROM DISCHARGING INTO SYSTEMS TRIBUTARY TO WATERS OF THE STATE.

ELEMENT 3: CONTROL FLOW RATES

A) THIS ELEMENT IS NOT REQUIRED ON SITE THAT ARE THIS SMALL.

ELEMENT 4: INSTALL SEDIMENT CONTROLS

A) DESIGN, INSTALL, AND MAINTAIN EFFECTIVE EROSION CONTROLS AND SEDIMENT CONTROLS TO MINIMIZE THE DISCHARGE OF POLLUTANTS.
B) CONSTRUCT SEDIMENT CONTROL BMPS (SEDIMENT, TRAPS, SILT FENCE, INLET PROTECTION, ETC.) AS ONE OF THE FIRST STEPS IN GRADING. THESE BMPS SHALL BE FUNCTIONAL BEFORE OTHER LAND DISTURBING ACTIVITIES TAKE PLACE.
C) MINIMIZE SEDIMENT DISCHARGES FROM THE SITE. THE DESIGN, INSTALLATION AND MAINTENANCE OF EROSION AND SEDIMENT CONTROLS MUST ADDRESS FACTORS SUCH AS THE AMOUNT, FREQUENCY, INTENSITY AND DURATION OF PRECIPITATION, THE NATURE OF RESULTING STORMWATER RUNOFF, AND SOIL CHARACTERISTICS, INCLUDING THE RANGE OF SOIL PARTICLE SIZES EXPECTED TO BE PRESENT ON THE SITE.
D) DIRECT STORMWATER RUNOFF FROM DISTURBED AREAS THROUGH A SEDIMENT REMOVAL BMP, BEFORE THE RUNOFF LEAVES A CONSTRUCTION SITE. RUNOFF FROM FULLY STABILIZED AREAS MAY BE DISCHARGED WITHOUT A SEDIMENT REMOVAL BMP.
E) BMP C140 DUST CONTROL. CONTRACTOR SHALL KEEP DUST FROM CONSTRUCTION ACTIVITIES AND EXPOSED SOILS TO A MINIMUM.

ELEMENT 5: STABILIZE SOILS

A) STABILIZE EXPOSED AND UNWORKED SOILS BY APPLICATION OF EFFECTIVE BMPS THAT PREVENT EROSION. APPLICABLE BMPS INCLUDE, BUT ARE NOT LIMITED TO: TEMPORARY AND PERMANENT SEEDING, SODDING, MULCHING, PLASTIC COVERING, EROSION CONTROL FABRICS AND MATTING, SOIL APPLICATION OF POLYACRYLAMIDE (PAM), THE EARLY APPLICATION OF GRAVEL BASE EARLY ON AREAS TO BE PAVED, AND DUST CONTROL.
B) CONTROL STORMWATER VOLUME AND VELOCITY WITHIN THE SITE TO MINIMIZE SOIL EROSION.
C) SOILS MUST NOT REMAIN EXPOSED AND UNWORKED FOR MORE THAN THE TIME PERIODS SET FORTH BELOW TO PREVENT EROSION:
– DURING THE DRY SEASON (MAY 1 – SEPT. 30): 7 DAYS
– DURING THE WET SEASON (OCTOBER 1 – APRIL 30): 2 DAYS
D) STABILIZE SOILS AT THE END OF THE SHIFT BEFORE A HOLIDAY OR WEEKEND IF NEEDED BASED ON THE WEATHER FORECAST.
E) STABILIZE SOIL STOCKPILES FROM EROSION, PROTECTED WITH SEDIMENT TRAPPING MEASURES, AND WHERE POSSIBLE, BE LOCATED AWAY FROM STORM DRAIN INLETS, WATERWAYS AND DRAINAGE CHANNELS.
F) MINIMIZE THE AMOUNT OF SOIL EXPOSED DURING CONSTRUCTION ACTIVITY.
G) MINIMIZE THE DISTURBANCE OF STEEP SLOPES.
H) PRESERVE TOPSOIL AND MINIMIZE COMPACTING OF THESE AREAS.

ELEMENT 6: PROTECT SLOPES

A) DESIGN AND CONSTRUCT CUT-AND-FILL SLOPES IN A MANNER TO MINIMIZE EROSION. APPLICABLE PRACTICES INCLUDE, BUT ARE NOT LIMITED TO, REDUCING CONTINUOUS LENGTH OF SLOPE WITH TERRACING AND DIVERSIONS, REDUCING SLOPE STEEPNESS, AND ROUGHENING SLOPE SURFACES (FOR EXAMPLE, TRACK WALKING).
B) DIVERT OFF-SITE STORMWATER (RUN-ON) OR GROUND WATER AWAY FROM SLOPES AND DISTURBED AREAS WITH INTERCEPTOR DIKES, PIPES AND/OR SWALES. OFF-SITE STORMWATER SHOULD BE MANAGED SEPARATELY FROM STORMWATER GENERATED ON THE SITE.
C) AT THE TOP OF SLOPES, COLLECT DRAINAGE IN PIPE SLOPE DRAINS OR PROTECTED CHANNELS TO PREVENT EROSION.
D) PLACE EXCAVATED MATERIAL ON THE UPHILL SIDE OF TRENCHES, CONSISTENT WITH SAFETY AND SPACE CONSIDERATIONS.
E) PLACE CHECK DAMS AT REGULAR INTERVALS WITHIN CONSTRUCTED CHANNELS THAT ARE CUT DOWN A SLOPE.

ELEMENT 7: PROTECT DRAIN INLETS

A) PROTECT ALL STORM DRAIN INLETS MADE OPERABLE DURING CONSTRUCTION SO THAT STORMWATER RUNOFF SHALL NOT ENTER THE CONVEYANCE SYSTEM WITHOUT FIRST BEING FILTERED OR TREATED TO REMOVE SEDIMENT.
B) CLEAN OR REMOVE AND REPLACE INLET PROTECTION DEVICES WHEN SEDIMENT HAS FILLED ONE-THIRD OF THE AVAILABLE STORAGE (UNLESS A DIFFERENT STANDARD IS SPECIFIED BY THE PRODUCT MANUFACTURER).

ELEMENT 8: STABILIZE CHANNELS AND OUTLETS

A) DESIGN, CONSTRUCT, AND STABILIZE ALL ON-SITE CONVEYANCE CHANNELS TO PREVENT EROSION.
B) PROVIDE STABILIZATION, INCLUDING ARMORING MATERIAL, ADEQUATE TO PREVENT EROSION OF OUTLETS, ADJACENT STREAM BANKS, SLOPES AND DOWNSTREAM REACHES AT THE OUTLETS OF ALL CONVEYANCE SYSTEMS.

ELEMENT 9: CONTROL POLLUTANTS

A) DESIGN, INSTALL, IMPLEMENT AND MAINTAIN EFFECTIVE POLLUTION PREVENTION MEASURES TO MINIMIZE THE DISCHARGE OF POLLUTANTS.

B) HANDLE AND DISPOSE OF ALL POLLUTANTS, INCLUDING WASTE MATERIALS AND DEMOLITION DEBRIS THAT OCCUR ON-SITE IN A MANNER THAT DOES NOT CAUSE CONTAMINATION OF STORMWATER.
C) PROVIDE COVER, CONTAINMENT, AND PROTECTION FROM VANDALISM FOR ALL CHEMICALS, LIQUID PRODUCTS, PETROLEUM PRODUCTS, AND OTHER MATERIALS THAT HAVE THE POTENTIAL TO POSE A THREAT TO HUMAN HEALTH OR THE ENVIRONMENT. ON-SITE FUELING TANKS MUST INCLUDE SECONDARY CONTAINMENT. SECONDARY CONTAINMENT MEANS PLACING TANKS OR CONTAINERS WITHIN AN IMPERVIOUS STRUCTURE CAPABLE OF CONTAINING 110% OF THE VOLUME CONTAINED IN THE LARGEST TAKE WITHIN THE CONTAINMENT STRUCTURE. DOUBLE-WALLED TANKS DO NOT REQUIRE ADDITIONAL SECONDARY CONTAINMENT.
D) CONDUCT MAINTENANCE, FUELING, AND REPAIR OF HEAVY EQUIPMENT AND VEHICLES USING SPILL PREVENTION AND CONTROL MEASURES. CLEAN CONTAMINATED SURFACES IMMEDIATELY FOLLOWING ANY SPILL INCIDENT.
E) DISCHARGE WHEEL WASH OR TIRE BATH WASTEWATER TO A SEPARATE ON-SITE TREATMENT SYSTEM THAT PREVENTS DISCHARGE TO SURFACE WATER, SUCH AS CLOSED-LOOP RECIRCULATION OR UPLAND APPLICATION, OR TO THE SANITARY SEWER, WITH LOCAL SEWER DISTRICT APPROVAL.
F) APPLY FERTILIZERS AND PESTICIDES IN A MANNER AND AT APPLICATION RATES THAT WILL NOT RESULT IN LOSS OF CHEMICAL TO STORMWATER RUNOFF. FOLLOW MANUFACTURERS’ LABEL REQUIREMENTS FOR APPLICATION RATES AND PROCEDURES.
G) USE BMPS TO PREVENT CONTAMINATION OF STORMWATER RUNOFF BY PH MODIFYING SOURCES. THE SOURCES FOR THIS CONTAMINATION INCLUDE, BUT ARE NOT LIMITED TO: BULK CEMENT, CEMENT KILN DUST, FLY ASH, NEW CONCRETE WASHING AND CURING WATERS, WASTE STREAMS GENERATED FROM CONCRETE GRINDING AND SAWING, EXPOSED AGGREGATE PROCESSES, DEWATERING CONCRETE VAULTS, CONCRETE PUMPING AND MIXER WASHOUT WATERS.
H) ADJUST THE PH OF STORMWATER IF NECESSARY TO PREVENT VIOLATIONS OF WATER QUALITY STANDARDS.
I) ASSURE THAT WASHOUT OF CONCRETE TRUCKS IS PERFORMED OFF-SITE OR IN DESIGNATED CONCRETE WASHOUT AREAS ONLY. DO NOT WASH OUT CONCRETE TRUCKS ONTO THE GROUND, OR INTO STORM DRAINS, OPEN DITCHES, STREETS, OR STREAMS. DO NOT DUMP EXCESS CONCRETE ON-SITE, EXCEPT IN DESIGNATED CONCRETE WASHOUT AREAS. CONCRETE SPILLAGE OR CONCRETE DISCHARGE TO SURFACE WATERS OF THE STATE IS PROHIBITED.
J) OBTAIN WRITTEN APPROVAL FROM ECOLOGY BEFORE USING CHEMICAL TREATMENT OTHER THAN CO2 OR DRY ICE TO ADJUST PH.
ELEMENT 10: CONTROL DE-WATERING
A) DISCHARGE FOUNDATION, VAULT, AND TRENCH DE-WATERING WATER, WHICH HAS SIMILAR CHARACTERISTICS TO STORMWATER RUNOFF AT THE SITE, INTO A SEDIMENT REMOVAL BMP BEFORE DISCHARGING OFFSITE.
B) CLEAN, NON-TURBID DE-WATERING WATER, SUCH AS WELL-POINT GROUND WATER, CAN BE DISCHARGED TO SYSTEMS TRIBUTARY TO, OR DIRECTLY INTO SURFACE WATERS OF THE STATE PROVIDED THE DE-WATERING FLOW DOES NOT CAUSE EROSION OR FLOODING OF RECEIVING WATERS. DO NOT ROUTE CLEAN DEWATERING WATER THROUGH STORMWATER SEDIMENT PONDS.
C) HANDLE HIGHLY TURBID OR OTHERWISE CONTAMINATED DEWATERING WATER SEPARATELY FROM STORMWATER.
D) OTHER TREATMENT OR DISPOSAL OPTIONS MAY INCLUDE:
– TRANSPORT OFF-SITE IN A VEHICLE, SUCH AS A VACUUM FLUSH TRUCK, FOR LEGAL DISPOSAL IN A MANNER THAT DOES NOT POLLUTE STATE WATERS.
– ECOLOGY-APPROVED ON-SITE CHEMICAL TREATMENT OR OTHER SUITABLE TREATMENT TECHNOLOGIES.
– SANITARY OR COMBINED SEWER DISCHARGE WITH LOCAL SEWER DISTRICT APPROVAL, IF THERE IS NO OTHER OPTION.
– USE OF A SETTLING TANKS WITH OUTFALL TO A DITCH OR SWALE FOR SMALL VOLUMES OF LOCALIZED DEWATERING.
ELEMENT 11: MAINTAIN BMPS
A) MAINTAIN AND REPAIR ALL TEMPORARY AND PERMANENT EROSION AND SEDIMENT CONTROL BMPS AS NEEDED TO ASSURE CONTINUED PERFORMANCE OF THEIR INTENDED FUNCTION IN ACCORDANCE WITH BMP SPECIFICATIONS.
B) REMOVE ALL TEMPORARY EROSION AND SEDIMENT CONTROL BMPS WITHIN 30 DAYS AFTER ACHIEVING FINAL SITE STABILIZATION OR AFTER THE TEMPORARY BMPS ARE NO LONGER NEEDED.
ELEMENT 12: MANAGE THE PROJECT
A) PHASE LAND DISTURBANCE TO THE MAXIMUM DEGREE PRACTICABLE AND TAKE INTO ACCOUNT SEASONAL WORK LIMITATIONS.
B) THIS PROJECT IS NOT REQUIRED TO OBTAIN A CONSTRUCTION STORMWATER GENERAL PERMIT.
C) PROJECTS THAT DISTURB ONE OR MORE ACRES MUST HAVE SITE INSPECTIONS CONDUCTED BY A CERTIFIED EROSION AND SEDIMENT CONTROL LEAD (CESCL). PROJECT SITES DISTURBING LESS THAN ONE ACRE MAY HAVE A CESCL OR A PERSON WITHOUT CESCL CERTIFICATION CONDUCT INSPECTIONS. BY THE INITIATION OF CONSTRUCTION, THE SWPPP MUST IDENTIFY THE CESCL OR INSPECTOR, WHO MUST BE PRESENT ON-SITE OR ON-CALL AT ALL TIMES.
D) THE CESCL OR INSPECTOR (PROJECT SITES LESS THAN ONE ACRE) MUST HAVE THE SKILLS TO ASSESS THE:
– SITE CONDITIONS AND CONSTRUCTION ACTIVITIES THAT COULD IMPACT THE QUALITY OF STORMWATER.
– EFFECTIVENESS OF EROSION AND SEDIMENT CONTROL MEASURES USED TO CONTROL THE QUALITY OF STORMWATER DISCHARGES.
E) THE CESCL OR INSPECTOR MUST EXAMINE STORMWATER VISUALLY FOR THE PRESENCE OF SUSPENDED SEDIMENT, TURBIDITY, DISCOLORATION, AND OIL SHEEN. THEY MUST EVALUATE THE EFFECTIVENESS OF BMPS AND DETERMINE IF IT IS NECESSARY TO INSTALL, MAINTAIN, OR REPAIR BMPS TO IMPROVE THE QUALITY OF STORMWATER DISCHARGES. BASED ON THE RESULTS OF THE INSPECTION, CONSTRUCTION SITE OPERATORS MUST CORRECT THE PROBLEMS IDENTIFIED BY:
– REVIEWING THE SWPPP FOR COMPLIANCE WITH THE 13 CONSTRUCTION SWPPP ELEMENTS AND MAKING APPROPRIATE REVISIONS WITHIN 7 DAYS OF THE INSPECTION.
– IMMEDIATELY BEGINNING THE PROCESS OF FULLY IMPLEMENTING AND MAINTAINING APPROPRIATE SOURCE CONTROL AND/OR TREATMENT BMPS AS SOON AS POSSIBLE, ADDRESSING THE PROBLEMS NOT LATER THAN WITHIN 10 DAYS OF THE INSPECTION. IF INSTALLATION OF NECESSARY TREATMENT BMPS IS NOT FEASIBLE WITHIN 10 DAYS, THE CONSTRUCTION SITE OPERATOR MAY REQUEST AN EXTENSION WITHIN THE INITIAL 10-DAY RESPONSE PERIOD.
– DOCUMENTING BMP IMPLEMENTATION AND MAINTENANCE IN THE SITE LOG BOOK (SITES LARGER THAN 1 ACRE).
F) THE CESCL OR INSPECTOR MUST INSPECT ALL AREAS DISTURBED BY CONSTRUCTION

ACTIVITIES, ALL BMPS, AND ALL STORMWATER DISCHARGE POINTS AT LEAST ONCE EVERY CALENDAR WEEK AND WITHIN 24 HOURS OF ANY DISCHARGE FROM THE SITE. (FOR PURPOSES OF THIS CONDITION, INDIVIDUAL DISCHARGE EVENTS THAT LAST MORE THAN ONE DAY DO NOT REQUIRE DAILY INSPECTIONS. FOR EXAMPLE, IF A STORMWATER POND DISCHARGES CONTINUOUSLY OVER THE COURSE OF A WEEK, ONLY ONE INSPECTION IS REQUIRED THAT WEEK.) THE CESCL OR INSPECTOR MAY REDUCE THE INSPECTION FREQUENCY FOR TEMPORARY STABILIZED, INACTIVE SITES TO ONCE EVERY CALENDAR MONTH.

ELEMENT 13: PROTECT LOW IMPACT DEVELOPMENT BMPS

A) THERE ARE NOT LID BMPS ON THIS SITE.

GENERAL NOTES

1. BMPS: BEST MANAGEMENT PRACTICES (BMPS) REFERRED TO ON THIS PLAN AND IN THESE NOTES SHALL BE CONSTRUCTED AND MAINTAINED AS DESCRIBED IN MOST RECENT VERSION OF THE DEPARTMENT OF ECOLOGY’S STORMWATER MANAGEMENT MANUAL FOR WESTERN WASHINGTON (SWMMWW) SECTION ON CONSTRUCTION STORMWATER POLLUTION PREVENTION.
2. EXTENT: THE EXTENT OF EROSION AND SEDIMENTATION CONTROL MEASURES IS DEPENDENT ON WEATHER CONDITIONS, SITE SLOPES, LENGTH OF TIME GROUND IS LEFT EXPOSED, AND THE AREA OF EXPOSED GROUND. THE CONTRACTOR SHALL AT ALL TIMES MINIMIZE THE RISK OF SITE EROSION BY CAREFUL SCHEDULING AND BY IMPLEMENTING AND MAINTAINING BMPS UNTIL THE SITE IS PERMANENTLY STABILIZED.
3. VEGETATION: EXISTING VEGETATION SHALL BE PRESERVED WHERE ATTAINABLE.
4. ADJACENT PROPERTIES: PROPERTIES ADJACENT TO THE PROJECT SHALL BE PROTECTED FROM SEDIMENT DEPOSITION.
5. REMOVAL OF BMPS: ALL TEMPORARY EROSION AND SEDIMENT CONTROL BMPS SHALL BE REMOVED WITHIN 30 DAYS AFTER FINAL SITE STABILIZATION IS ACHIEVED OR AFTER THE TEMPORARY BMPS ARE NO LONGER NEEDED. TRAPPED SEDIMENT SHALL BE REMOVED OR STABILIZED ON-SITE. DISTURBED SOIL AREAS RESULTING FROM REMOVAL SHALL BE PERMANENTLY STABILIZED.
6. OTHER REQUIREMENTS: THE ENGINEER, OWNER, DEPARTMENT OF ECOLOGY, OR OTHER AGENCIES MAY REQUIRE BMPS IN ADDITION TO WHAT IS SHOWN ON THIS PLAN IF NECESSARY TO PREVENT VIOLATIONS OF SURFACE WATER QUALITY. THE CONTRACTOR SHALL IMPLEMENT THE BMPS AS REQUIRED.
7. THE IMPLEMENTATION OF THESE TESC PLANS AND THE CONSTRUCTION, MAINTENANCE, REPLACEMENT, AND UPGRADING OF THESE TESC FACILITIES IS THE RESPONSIBILITY OF THE CONTRACTOR UNTIL ALL CONSTRUCTION IS COMPLETED AND APPROVED AND VEGETATION/FINAL SURFACING IS ESTABLISHED.
8. THE TESC FACILITIES SHOWN ON THIS PLAN MUST BE CONSTRUCTED IN CONJUNCTION WITH ALL CLEARING AND GRADING ACTIVITIES, AND IN SUCH A MANNER AS TO INSURE THAT SEDIMENT AND SEDIMENT LADEN WATER DO NOT ENTER THE DRAINAGE SYSTEM, ROADWAYS, OR VIOLATE APPLICABLE WATER STANDARDS.
9. THE TESC FACILITIES SHOWN ON THIS PLAN ARE THE MINIMUM REQUIREMENTS FOR ANTICIPATED SITE CONDITIONS. DURING THE CONSTRUCTION PERIOD, THESE ESC FACILITIES SHALL BE UPGRADED AS NEEDED FOR UNEXPECTED STORM EVENTS AND TO ENSURE THAT SEDIMENT AND SEDIMENT LADEN WATER DO NOT LEAVE THE SITE.
10. OBTAIN PERMISSION AND/OR PERMIT APPROVAL AS REQUIRED FROM EASTSOUND WATER AND SEWER DISTRICT PRIOR TO ANY DISCHARGES TO THE SANITARY SEWER SYSTEM.
11. CONVEYANCE BYPASS: IF REQUIRED, PROVISION FOR BYPASS OF STORMWATER CONVEYANCE SHALL BE PROVIDED. BYPASS SHALL BE INSTALLED WHILE WORK IS IN PROGRESS AT A PARTICULAR LOCATION, BUT MATERIALS AND EQUIPMENT FOR IMMEDIATE INSTALLATION SHALL BE ON HAND. BYPASS SHALL BE IN PLACE WHILE SITE IS UNATTENDED FOR GREATER THAN 12 HOURS.
12. PUBLIC RIGHTS-OF WAY SHALL BE KEPT IN A CLEAN AND SERVICEABLE CONDITION AT ALL TIMES. IN THE EVENT MATERIALS ARE INADVERTENTLY DEPOSITED ON ROADWAYS, THE MATERIAL SHALL BE PROMPTLY REMOVED. MATERIALS ARE TO BE SWEEP AND REMOVED PRIOR TO ANY STREET FLUSHING. PUBLIC AND PRIVATE DRAINAGE WAYS SHALL BE PROTECTED FROM POLLUTION. NO MATERIAL IS TO BE DISCHARGED TO OR DEPOSITED IN STORMWATER SYSTEMS THAT MAY RESULT IN VIOLATION OF STATE OR FEDERAL WATER QUALITY STANDARDS.
13. ALL CATCH BASINS WITHIN THE PROJECT LIMITS SHALL BE CLEANED OUT AT THE COMPLETION OF THE PROJECT AND ANY MATERIAL REMOVED SHALL BE PROPERLY DISPOSED OF.
14. CONCRETE PROCESS WATER AND SLURRY FROM PAVEMENT CUTTINGS SHALL NOT BE ALLOWED TO ENTER ANY DRAINAGE CONVEYANCE. SLURRY/CUTTINGS SHALL NOT REMAIN ON PAVEMENT OVERNIGHT. ALL SAWCUTTING AND SURFACING OPERATIONS SHALL USE BMP C152 FOR POLLUTION PREVENTION.
15. PROJECT DESCRIPTION
a.TOTAL PROJECT AREA. 0.45 ACRES
b.TOTAL PROPOSED IMPERVIOUS AREA. 0.04 ACRES
c.TOTAL PROPOSED AREA TO BE DISTURBED. 0.30 ACRES
d.TOTAL VOLUME OF PROPOSED CUT/FILL. CUT = 3382 CY, FILL = 1598 CY. (ESTIMATED VOLUME BETWEEN EXISTING GROUND SURFACE AND PROPOSED FINISH GRADES FOR BASE BID AND ALL ALTERNATES. NO SHRINK OR SWELL FACTORS APPLIED.) VOLUMES ARE FOR TESC NARRATIVE ONLY AND ARE NOT INTENDED FOR BIDDING OR TAKE-OFF PURPOSES.
16. EXISTING SITE CONDITIONS
a.DESCRPTION OF EXISTING TOPOGRAPHY. EXISTING SLOPES VARY SLIGHTLY THROUGHOUT THE PROJECT SITE. MOST AREAS ARE FLAT TO SLOPED UP TO 2 PERCENT.
b.DESCRPTION OF EXISTING VEGETATION. AREAS ALONG THE NORTH AND SOUTH PROPERTY LINES ARE GRASS TURF. THERE ARE NO TREES OR SHRUBS ON THE SITE.
17. DESCRIPTION OF EXISTING DRAINAGE. RUNOFF FROM THE SITE FLOWS GENERALLY FROM SOUTHEAST TO NORTHWEST TO WHERE IT ENTERS A DITCH ON THE ADJOINING SITE ALONG THE NORTH BOUNDARY.

18. ADJACENT AREAS
a.DESCRPTION OF ADJACENT AREAS WHICH MAY BE AFFECTED BY SITE DISTURBANCE:
I.STREAMS. NONE.
II. LAKES. NONE.
III. WETLANDS. NONE.
IV. RESIDENTIAL AREAS. NONE.
V. ROADS. CESSNA ROAD, UNNAMED AIRPORT ACCESS ROADS.

b.DESCRPTION OF THE DOWNSTREAM DRAINAGE PATH LEADING FROM THE SITE TO THE RECEIVING BODY OF WATER (MINIMUM DISTANCE OF 400 YARDS). RUNOFF FROM THE SITE WILL CONTINUE WITH THE SAME PATTERN THAT EXISTS. RUNOFF WILL BE ROUTED TO A DITCH THAT IS JUST NORTHWEST OF THE SITE. RUNOFF IN THE DITCH IS CONVEYED NORTH THROUGH A SERIES OF DITCHES AND CULVERTS THROUGH PORT PROPERTY. IT APPEARS THAT THE OUTFALL MAY BE TO PUGET SOUND FURTHER TO THE NORTH.

19. CRITICAL AREAS
a.DESCRPTION OF CRITICAL AREAS THAT ARE ON OR ADJACENT TO THE SITE. THERE ARE NO CRITICAL AREAS WITHIN THE PROJECT SITE. THE SITE IS SURROUNDED BY DEVELOPMENT ON ALL SIDES. IT IS UNCLEAR IF THERE ARE CRITICAL AREAS NEAR THE SITE.
b.DESCRPTION OF SPECIAL REQUIREMENTS FOR WORKING IN OR NEAR CRITICAL AREAS. ALL WORK ON THIS PROJECT WILL BE DONE IN ACCORDANCE WITH ALL APPLICABLE LOCAL, STATE, AND FEDERAL PERMITS AS APPLICABLE. NO KNOWN CRITICAL AREAS ARE WITHIN OR ADJACENT TO THE PROJECT LIMITS.

20. DESCRIPTION OF ONSITE SOILS
a.THE FOLLOWING SOIL DESCRIPTIONS WERE TAKEN FROM THE USDA NRCS WEB SOIL SURVEY (HTTP://WEBSOILSURVEY.NRCS.USDA.GOV/APP/). SOILS IN THE WORK AREAS ARE CLASSIFIED AS:
–ROCHE-KILLEBREW COMPLEX. 2 TO 10 PERCENT SLOPES.

21. EROSION PROBLEM AREAS.
NO EROSION PROBLEM AREAS ARE EXPECTED.

22. CONSTRUCTION PHASING.
a.CONSTRUCTION SEQUENCE. THE CONSTRUCTION SEQUENCE WILL GENERALLY BE AS FOLLOWS;
• INSTALL EROSION PREVENTION FACILITIES: SILT FENCE, CONSTRUCTION ENTRANCE, INLET PROTECTION;
• REMOVE PORTIONS OF THE PAVEMENT TO INSTALL UNDERGROUND AND ABOVE GROUND SEWER STRUCTURES. CLEAR AND GRADE AREAS FOR BUILDING ADDITION AND RETAINING WALL.
• INSTALL ABOVE AND BELOW GROUND SEWER STRUCTURES, BUILDING ADDITION, AND RETAINING WALL
• REMOVE REMAINDER OF PAVEMENT AND INSTALL STORMWATER CONVEYANCE FACILITIES
• CLEAR NEW PAVING AREAS
• PAVE SITE WITH ASPHALT AND PROVIDE FINAL STABILIZATION TO LANDSCAPE AREAS.
b.CONSTRUCTION PHASING. CONTRACTOR IS ENCOURAGED TO PHASE WORK SUCH THAT DISTURBED AREAS ARE OPENED UP, IMPROVEMENTS INSTALLED, AND PERMANENTLY STABILIZED IN SECTIONS SUCH TO MINIMIZE POTENTIAL FOR EROSION.

23. CONSTRUCTION SCHEDULE
a.PROVIDE A PROPOSED CONSTRUCTION SCHEDULE. CONSTRUCTION IS EXPECTED TO OCCUR STARTING IN SPRING OF 2023 AND ENDING IN FALL OF 2024.
b.WET SEASON CONSTRUCTION ACTIVITIES. WET SEASON CONSTRUCTION ACTIVITIES ARE EXPECTED INCREASED DILIGENCE WILL EMPLOYED TO PREVENT EROSION AND PROTECT DOWNSTREAM WATERS AND PROPERTIES FROM SEDIMENT. ALL WET SEASON HEIGHTENED TESC ACTIVITIES WILL BE IMPLEMENTED IN ACCORDANCE WITH THE SWMMWW.
I.PROPOSED WET SEASON CONSTRUCTION RESTRAINTS FOR ENVIRONMENTALLY SENSITIVE/CRITICAL AREAS. THERE ARE NO KNOWN ENVIRONMENTALLY SENSITIVE /CRITICAL AREAS WITHIN THE SITE, SO THERE ARE NO ADDITIONAL MEASURES TO BE UTILIZED.

24. ENGINEERING CALCULATIONS
a.SEDIMENT PONDS/TRAPS. NONE USED.
b.DIVERSIONS. NONE ARE ANTICIPATED
c.WATERWAYS. NONE USED.
d.RUNOFF/STORMWATER DETENTION CALCULATIONS. NONE NEEDED.

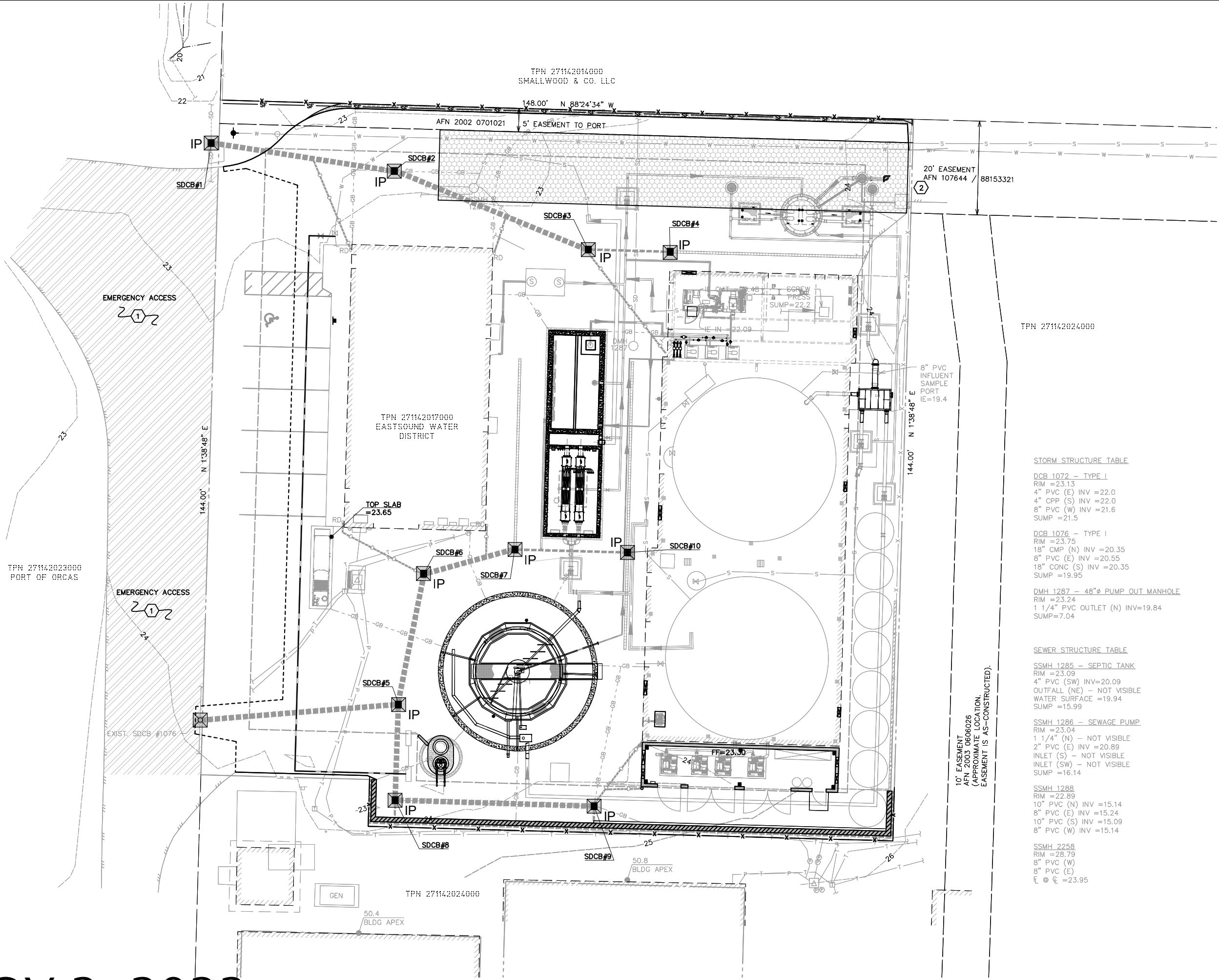


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EASTSOUND SEWER AND WATER DISTRICT		WASHINGTON	
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I			
DATE	SCALE	JOB NUMBER	
C0.2	AS SHOWN	96	
SHEET		9 OF 96	
T.E.S.C. NARRATIVE			

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NO.	REVISIONS	BY	DATE

LEGEND

= STABILIZED CONSTRUCTION ENTRANCE PER $\frac{4}{C0.4}$

= EMERGENCY ACCESS

= SILT FENCE PER $\frac{3}{C0.4}$

IP = INLET PROTECTION PER $\frac{2}{C0.4}$

CD = CHECK DAM PER $\frac{5}{C0.4}$

KEYED NOTES

① = EMERGENCY ACCESS. NO PARKING, BLOCKING, STOCKPILING, STAGING ALLOWED. THIS AREA CAN BE USED FOR VEHICLE TURN AROUND

② = SOLE INGRESS & EGRESS LOCATION

STORM STRUCTURE TABLE

DCB 1072 - TYPE I
RIM = 23.13
4" PVC (E) INV = 22.0
4" CPP (S) INV = 22.0
8" PVC (W) INV = 21.6
SUMP = 21.5

DCB 1076 - TYPE I
RIM = 23.75
18" CMP (N) INV = 20.35
8" PVC (E) INV = 20.55
18" CONC (S) INV = 20.35
SUMP = 19.95

DMH 1287 - 48" Ø PUMP OUT MANHOLE
RIM = 23.24
1 1/4" PVC OUTLET (N) INV = 19.84
SUMP = 7.04

SEWER STRUCTURE TABLE

SSMH 1285 - SEPTIC TANK
RIM = 23.09
4" PVC (SW) INV = 20.09
OUTFALL (NE) - NOT VISIBLE
WATER SURFACE = 19.94
SUMP = 15.99

SSMH 1286 - SEWAGE PUMP
RIM = 23.04
1 1/4" (N) - NOT VISIBLE
2" PVC (E) INV = 20.89
INLET (S) - NOT VISIBLE
INLET (SW) - NOT VISIBLE
SUMP = 16.14

SSMH 1288
RIM = 22.89
10" PVC (N) INV = 15.14
8" PVC (E) INV = 15.24
10" PVC (S) INV = 15.09
8" PVC (W) INV = 15.14

SSMH 2258
RIM = 28.79
8" PVC (W)
8" PVC (E)
E @ E = 23.95

SHEET NOTE:

1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.

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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

T.E.S.C. PLAN

DATE
SCALE
JOB NUMBER

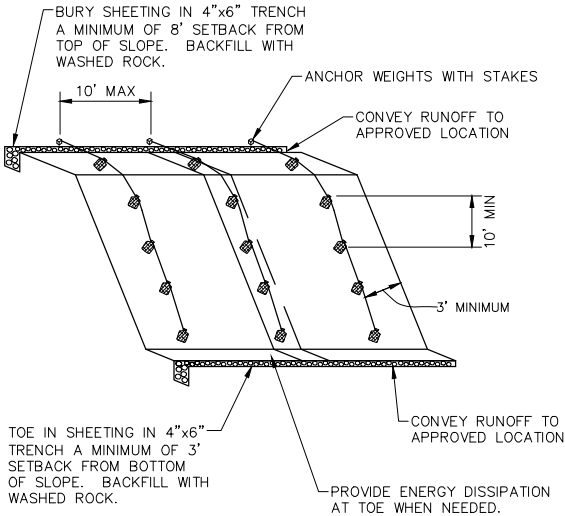
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NOV 2, 2022

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NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

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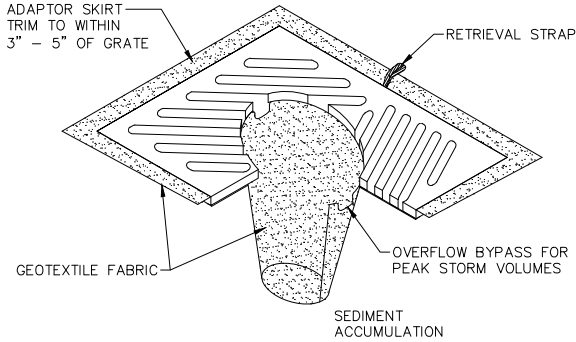


NOTES:

1. TIRES, SANDBAGS, OR EQUIVALENT MAY BE USED TO WEIGH DOWN PLASTIC SHEETING.
2. SEAMS BETWEEN SHEETS MUST OVERLAP A MINIMUM OF 12" AND BE WEIGHTED OR TAPED.
3. PLASTIC SHEETING SHALL HAVE A MINIMUM THICKNESS OF 6 MIL.
4. DUE TO RAPID RUNOFF CAUSED BY PLASTIC SHEETING, THIS METHOD SHALL NOT BE USED UPSLOPE OF AREAS THAT MIGHT BE ADVERSELY IMPACTED BY CONCENTRATED RUNOFF.
5. CONSTRUCT BERM OR SWALE AT TOP OF SLOPE AS DIRECTED BY THE CLEARING AND GRADING INSPECTOR.
6. CONSTRUCT DITCH AT BASE OF SLOPE IF NEEDED, AND DISCHARGE TO APPROVED LOCATION.

PLASTIC COVERING FOR SLOPES & STOCKPILES

NOT TO SCALE



NOTES:

1. INSERT SHALL BE INSTALLED PRIOR TO CLEARING AND GRADING ACTIVITY, OR UPON PLACEMENT OF A NEW CATCH BASIN.
2. SEDIMENT SHALL BE REMOVED FROM THE UNIT WHEN IT BECOMES HALF FULL.
3. SEDIMENT REMOVAL SHALL BE ACCOMPLISHED BY REMOVING THE INSERT, EMPTYING, AND RE-INSERTING IT INTO THE CATCH BASIN.

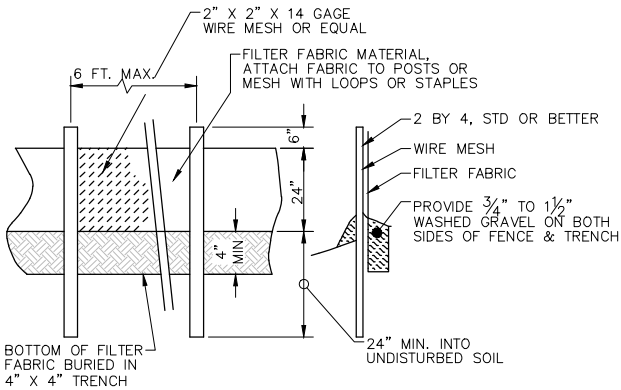
CATCH BASIN INSERT

NOT TO SCALE

NOTE: BASED ON FIGURE II-3.1: "STABILIZED CONSTRUCTION ACCESS" IN THE 2019 STORMWATER MANAGEMENT MANUAL FOR WESTERN WASHINGTON, REVISED JULY 2019, VOL II - CHAPTER 3 - PAGE 279

NOTES:

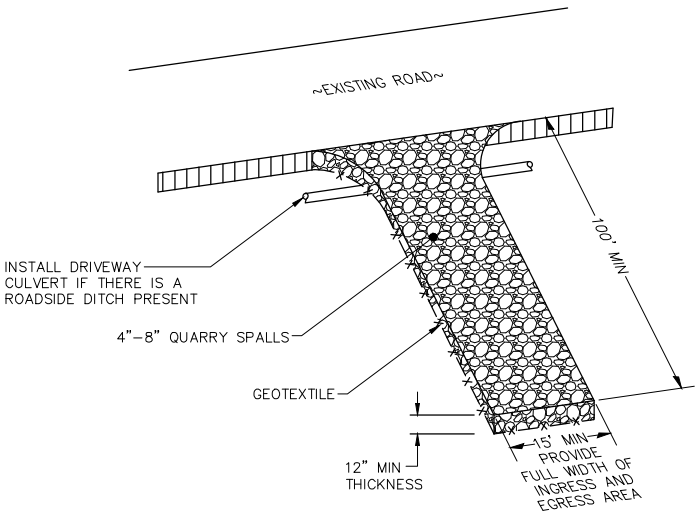
1. DRIVEWAY SHALL MEET THE REQUIREMENTS OF THE PERMITTING AGENCY.
2. IT IS RECOMMENDED THAT THE ACCESS BE CROWNED SO THAT RUNOFF DRAINS OFF THE PAD.



NOTE: SEE WSDOT STANDARD PLAN I-30.10-02 OR I-30.15-02, FOR HIGH VISIBILITY FENCE, SEE WSDOT STANDARD PLAN I-30.16-01 OR I-30.17-01.

FILTER FABRIC FENCE

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STABILIZED CONSTRUCTION ENTRANCE

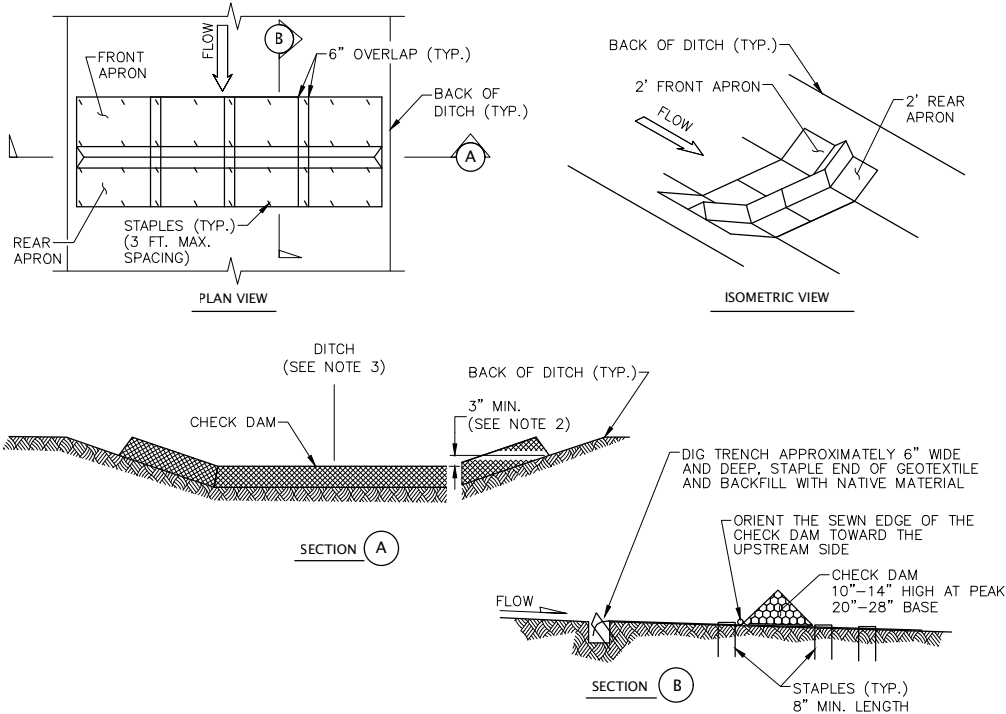
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BMP C208 - GEOTEXTILE ENCASED CHECK DAM

PURPOSE: GEOTEXTILE ENCASED CHECK DAMS MAY BE USED AS CHECK DAMS, FOR PERIMETER PROTECTION, FOR TEMPORARY SOIL STOCKPILE PROTECTION, FOR DROP INLET PROTECTION, OR AS A TEMPORARY INTERCEPTOR DIKE.

INSTALLATION: INSTALL WITH ENDS CURVED UP TO PREVENT WATER FROM FLOWING AROUND ENDS. THE FABRIC FLAPS AND CHECK DAM UNITS ARE ATTACHED TO THE GROUND WITH WIRE STAPLES. WIRE STAPLES SHOULD BE NO. 11 GAUGE AND SHOULD BE 200MM TO 300MM IN LENGTH. WHEN MULTIPLE UNITS ARE INSTALLED, THE SLEEVE OF FABRIC AT THE END OF THE UNIT SHALL OVERLAP THE ABUTTING UNIT AND BE STAPLED. CHECK DAMS SHOULD BE LOCATED AND INSTALLED AS SOON AS CONSTRUCTION WILL ALLOW AND SHOULD BE PLACED PERPENDICULAR TO THE FLOW OF WATER. WHEN USED AS CHECK DAMS, THE LEADING EDGE MUST BE SECURED WITH ROCKS, SANDBAGS, OR A SMALL KEY SLOT AND STAPLES. IN THE CASE OF GRASS-LINED DITCHES AND SWALES, CHECK DAMS AND ACCUMULATED SEDIMENT SHALL BE REMOVED WHEN THE GRASS HAS MATURED SUFFICIENTLY TO PROTECT THE DITCH OR SWALE UNLESS THE SLOPE OF THE SWALE IS GREATER THAN 4 PERCENT. THE AREA BENEATH THE CHECK DAMS SHALL BE SEEDED AND MULCHED IMMEDIATELY AFTER DAM REMOVAL.

MAINTENANCE: CHECK DAMS SHALL BE MONITORED FOR PERFORMANCE AND SEDIMENT ACCUMULATION DURING AND AFTER EACH RUNOFF PRODUCING RAINFALL. SEDIMENT SHALL BE REMOVED WHEN IT REACHES ONE HALF THE HEIGHT OF THE DAM. ANTICIPATE SUBMERGENCE AND DEPOSITION ABOVE THE TRIANGULAR SILT DAM AND EROSION FROM HIGH FLOWS AROUND THE EDGES OF THE DAM. IMMEDIATELY REPAIR ANY DAMAGE OR ANY UNDERCUTTING OF THE DAM.



NOTES:

1. GEOTEXTILE ENCASED CHECK DAMS SHALL MEET THE REQUIREMENTS OF STANDARD SPECIFICATIONS 8-01.3(6)A AND 9-14.5(4).
2. INSTALL THE SLOPED ENDS OF THE CHECK DAM A MINIMUM OF 3" HIGHER THAN THE TOP OF THE CHECK DAM IN THE CHANNEL TO ENSURE THAT WATER FLOWS OVER THE DAM AND NOT AROUND IT.
3. FLAT BOTTOM DITCH DESIGN SHOWN, CHECK DAM INSTALLATION DETAILS ARE SIMILAR FOR "V" BOTTOM DITCHES.
4. PERFORM MAINTENANCE IN ACCORDANCE WITH STANDARD SPECIFICATION 8-01.3(15).

NOTE:

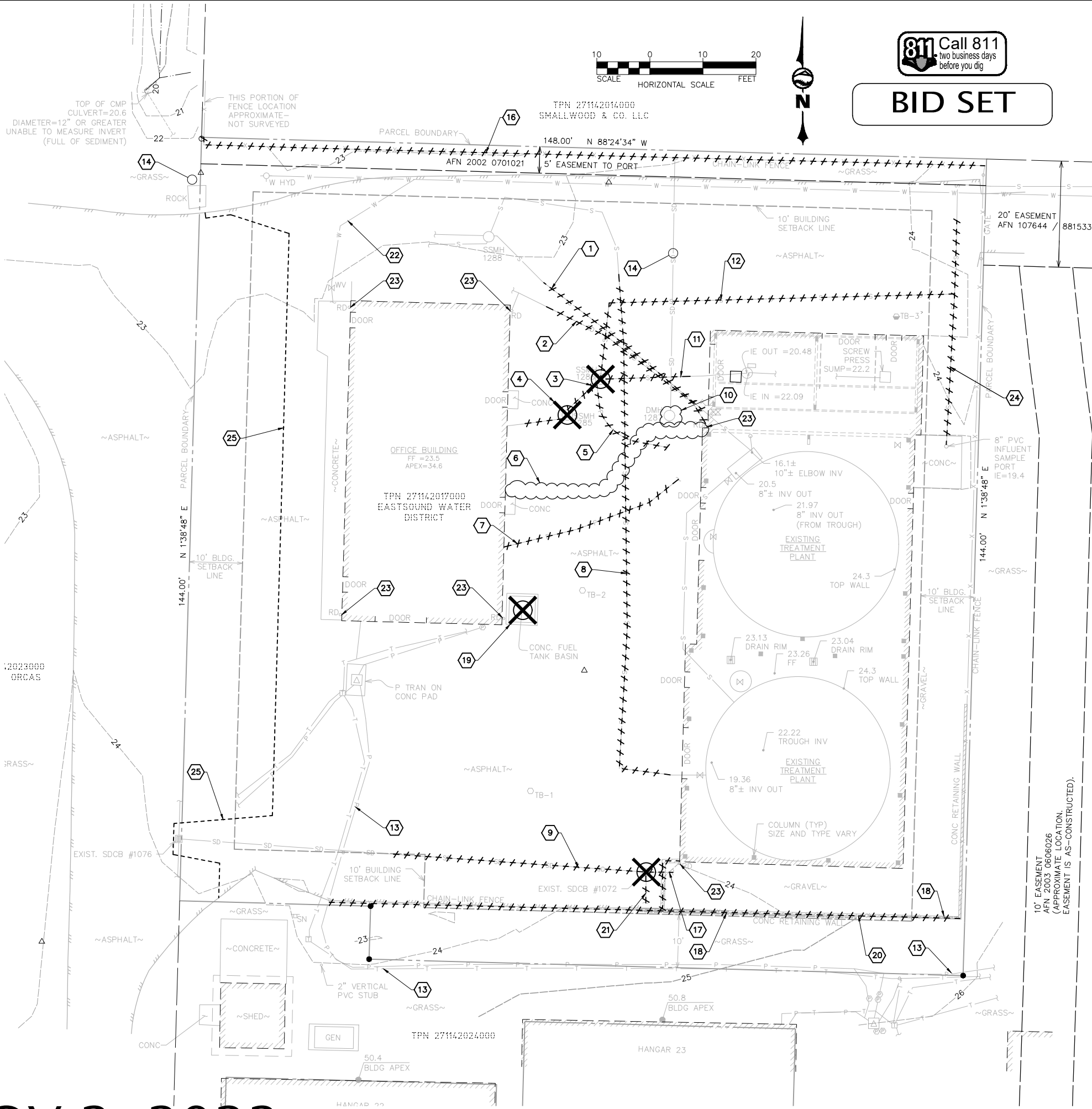
THE CONTRACTOR SHALL BE RESPONSIBLE FOR UPDATING COVERAGE UNDER THE DOE'S "CONSTRUCTION STORMWATER GENERAL PERMIT". CONTACT MELINDA WILSON, PERMIT ADMINISTRATOR AT DOE (360-870-8290), EMAIL: MELINDA.WILSON@ECY.WA.GOV) FOR ADDITIONAL INFORMATION.

GEOTEXTILE ENCASED CHECK DAM

NOT TO SCALE

SHEET		DATE		EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY					
11 OF 96		SCALE		WASHINGTON		DRAWN BY				WILSONENGINEERING.COM	
		AS SHOWN		SAN JUAN COUNTY		JGS					
		JOB NUMBER		WASTEWATER TREATMENT PLANT UPGRADE – PHASE I		CHECKED BY					
				T.E.S.C. DETAILS		AWL					

PLOT SETTINGS: Adobe PDF1,pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 CO.5 DEMOLITION PLANDWG - 11/2/2022 3:21 PM - Jeff Smith



811 Call 811
 two business days
 before you dig

BID SET

KEYED NOTES:

1. ABANDON AND REMOVE 40 LF OF 8" EFFLUENT (OUTFALL) PIPE AFTER INTERIM 8" HDPE PIPE IS INSTALLED AND TESTED. SEE SHEET C0.7 INTERIM PLAN OF OPERATION
2. ABANDON AND REMOVE 40 LF OF 4" ROOF DRAIN PIPE AFTER NEW 4" ROOF DRAIN GRAVITY PIPE IS INSTALLED AND TESTED. SEE C2 STORM DRAINAGE SHEETS
3. REMOVE EXISTING SUMP PUMP STATION AFTER NEW PLANT DRAIN PUMP STATION IS INSTALLED AND TESTED (OR PROVIDE TEMPORARY PUMPS, PIPING, AND BASINS AS NEEDED)
4. REMOVE EXISTING SEPTIC TANK AFTER NEW SEPTIC TANK AND INFLUENT P.S. ARE INSTALLED AND TESTED. PROVIDE TEMPORARY PUMPS AND BASINS AS NEEDED
5. MAINTAIN POWER TO EXISTING SUMP P.S. UNTIL NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED
6. MAINTAIN EXISTING 2" WATER SERVICE UNTIL NEW WATER SERVICE IS INSTALLED AND TESTED. SEE C1 SITE PIPING SHEETS
7. RE-ROUTE ELECTRICAL AS NEEDED. SEE ELECTRICAL SHEETS
8. ABANDON AND REMOVE 60 LF OF 10" EFFLUENT (OUTFALL) PIPE AFTER INTERIM 8" HDPE PIPE IS INSTALLED AND TESTED. PROVIDE TEMPORARY RESTRAINED PLUG AT PIPE OPENING UNTIL NEW 10" EFFLUENT IS CONNECTED.
9. REMOVE EXISTING 87 LF STORM DRAIN & CATCH BASIN
10. PROTECT EXISTING STORM DRAIN PUMP STATION MANHOLE AND DISCHARGE PIPING
11. ABANDON AND REMOVE 20 LF OF 6" DRAIN PIPE TO SUMP PUMP STATION AFTER NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED
12. ABANDON AND REMOVE 60 LF OF 13" SUMP PUMP FORCE MAIN PIPE, AFTER NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED. PLUG CONNECTION AT 8" INFLUENT GRAVITY SEWER PIPE (EAST SIDE). SEE C1 SITE PIPING SHEETS
13. POTHOLE POWER & TELECOMMUNICATIONS PRIOR TO INSTALLING STORM SEWER OR RETAINING WALL OR FENCE. IF IN CONFLICT, RELOCATE AS NEEDED TO BE CLEAR OF PROPOSED FACILITIES. BED & BACKFILL SIMILAR TO STORM SEWER PIPE MIN. RELOCATED FACILITIES TO BE INSTALLED W/ 2' OF COVER MIN. ADDITIONAL COVER MAY BE REQUIRED TO AVOID CONFLICT WITH OTHER FACILITIES
14. POTHOLE EXISTING STORM SEWER AND REPORT PIPE LOCATION & ELEVATION TO THE ENGINEER PRIOR TO GRADING THE SITE OR INSTALLING ANY STORM SEWER FACILITIES.
15. EXISTING HOLDING TANKS TO BE REMOVED DURING CONSTRUCTION & RELOCATED ON SITE PER OWNER
16. REMOVE 149 LF OF CHAIN LINK FENCE
17. REMOVE 7' OF STORM DRAIN CONNECTED TO ROOF DRAIN
18. REMOVE 61 LF OF EXISTING RETAINING WALL. MAKE CLEAN CUT AT EAST END SUITABLE FOR MATCHING TO PROPOSED BLOCK WALL
19. EXISTING GENERATOR FUEL TANK AND CONCRETE FUEL SPILL BASIN TO BE REMOVED AND RESTORED AS ASPHALT PAVEMENT.
20. REMOVE 119 LF OF CHAIN LINK FENCE
21. REMOVE 8 LF OF 4" STORM DRAIN
22. RELOCATE WATER TO BE 1' CLEAR BELOW BOTTOM OF STORM DRAIN
23. DISCONNECT ROOF DRAINS AND PREPARE FOR CONNECTION TO PROPOSED DRAINS
24. ABANDON & REMOVE EXISTING 8" INFLUENT PIPE PER CONSTRUCTION SEQUENCE NOTES (SHEET C3.1)
25. SAWCUT ASPHALT PAVEMENT

CONSTRUCTION PHASING:

NOTE: ALL CONSTRUCTION PHASING STEPS SHALL BE COORDINATED WITH THE OWNER AND ENGINEER.

1. PERFORM DEMOLITION ITEMS 1 & 2.
2. LEAVING EXISTING INTERNAL PIPING IN PLACE AND OPERATIONAL, INSTALL NEW CONCRETE WALLS, WEIR GATES, AND IMPROVEMENTS TO EXISTING INTERIOR WALLS.
3. TIME CRITICAL STEP: PLUG OUTLET PIPE AT UPSTREAM MANHOLE AND BYPASS ALL FLOW. CLOSE INFLUENT VALVES TO PLANTS 1 & 2. THIS WILL ALLOW THE PORTION OF PIPE WITHIN THE BASINS TO BE PUMPED DRY BEFORE CUTTING. PUMP WASTEWATER FROM THE INFLUENT SAMPLE PORT TO TREATMENT PLANT #1 AND PUMP ALL FLOW FROM UPSTREAM MANHOLE TO TREATMENT PLANTS 1 AND 2. CONTRACTOR TO PROVIDE A TEMPORARY BYPASS SYSTEM PER SPECS. DURING THIS TIME PIPING MAY BE CUT, INSTALL FLOW METER PIPING, AND DISCHARGE CONNECTIONS CAN BE MADE, EXISTING PIPE PENETRATIONS SHALL BE FILLED AND ALLOWED ADEQUATE TIME TO CURE.
4. BEFORE THE INFLUENT BASIN IS READY TO ACCEPT WASTEWATER THE FOLLOWING CHECKLIST MUST BE COMPLETED:
 - a. INTERIOR CONCRETE WALLS AND COATING ARE CURED AND READY TO ACCEPT WASTEWATER
 - b. PIPE PENETRATIONS HAVE BEEN SEALED AND GROUTING HAS CURED.
 - c. ALL THREE WEIR GATES ARE INSTALLED AND SET TO THE APPROPRIATE LEVELS. WEIR GATES #1 AND #3 SHOULD BE COMPLETELY CLOSED (IN THE HIGH POSITION).
 - d. DISCHARGE PIPING FROM DISCHARGE BASIN #2 IS COMPLETE.
 - e. INFLUENT VALVE TO PLANTS ARE OPEN
5. REMOVE UPSTREAM MANHOLE PLUG TO BEGIN ACCEPTING WASTEWATER.

DEMOLITION SCHEDULE:

1. REMOVE & DISPOSE OF EXISTING 8" SLAB OVER ORIGINAL PRE-AERATION BASIN. BASIN & SLAB EXTEND UNDER PLANT BUILDING. CONTRACTOR SHALL SAW CUT CONCRETE SLAB AS CLOSE TO EAST FACE OF BUILDING AS POSSIBLE, LEAVING PORTION OF 8" SLAB THAT EXTENDS UNDER BUILDING.
2. LEAVING EXISTING INTERNAL PIPING INTACT, CONTRACTOR SHALL REMOVE AND DISPOSE OF ALL EXISTING FILL FROM EXISTING BASIN. SUPPORT INTERNAL PIPING AS NECESSARY.
3. PER CONSTRUCTION PHASING SCHEDULE, CONTRACTOR SHALL REMOVE INTERNAL PIPING AFTER NEW FLOW SPLITTER IS OPERATIONAL.

GENERAL NOTES:

CONTRACTOR TO PREPARE A DEMOLITION PLAN WITH SCHEDULE, AND PRESENT TO ENGINEER FOR COORDINATION AND APPROVAL PRIOR TO BEGINNING ANY DEMOLITION ACTIVITIES.

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

		WILSONENGINEERING.COM	
		11-02-2022	
DESIGNED BY	JGC	DRAWN BY	SNR
CHECKED BY	AWL		
EASTSOUND SEWER AND WATER DISTRICT		WASHINGTON	
SAN JUAN COUNTY		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I	
DATE		SCALE AS SHOWN	JOB NUMBER 12 OF 96
SHEET		DEMOLITION PLAN	

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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NO.	REVISIONS	BY	DATE
1	NEW STAGING AREA	JGC	5-10-22

LEGEND

— SF — = SILT FENCE PER 3 C0.4

↗ = EXISTING GROUND SLOPE

— > — = EXISTING DITCH FLOW

— > — = VEHICLE PATH

KEYED NOTES

1 = STAGING AREA

2 = EXISTING VEGETATED DITCH

SHEET NOTE:

1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.

30 0 30 60
SCALE HORIZONTAL SCALE FEET

811 Call 811
two business days
before you dig

BID SET

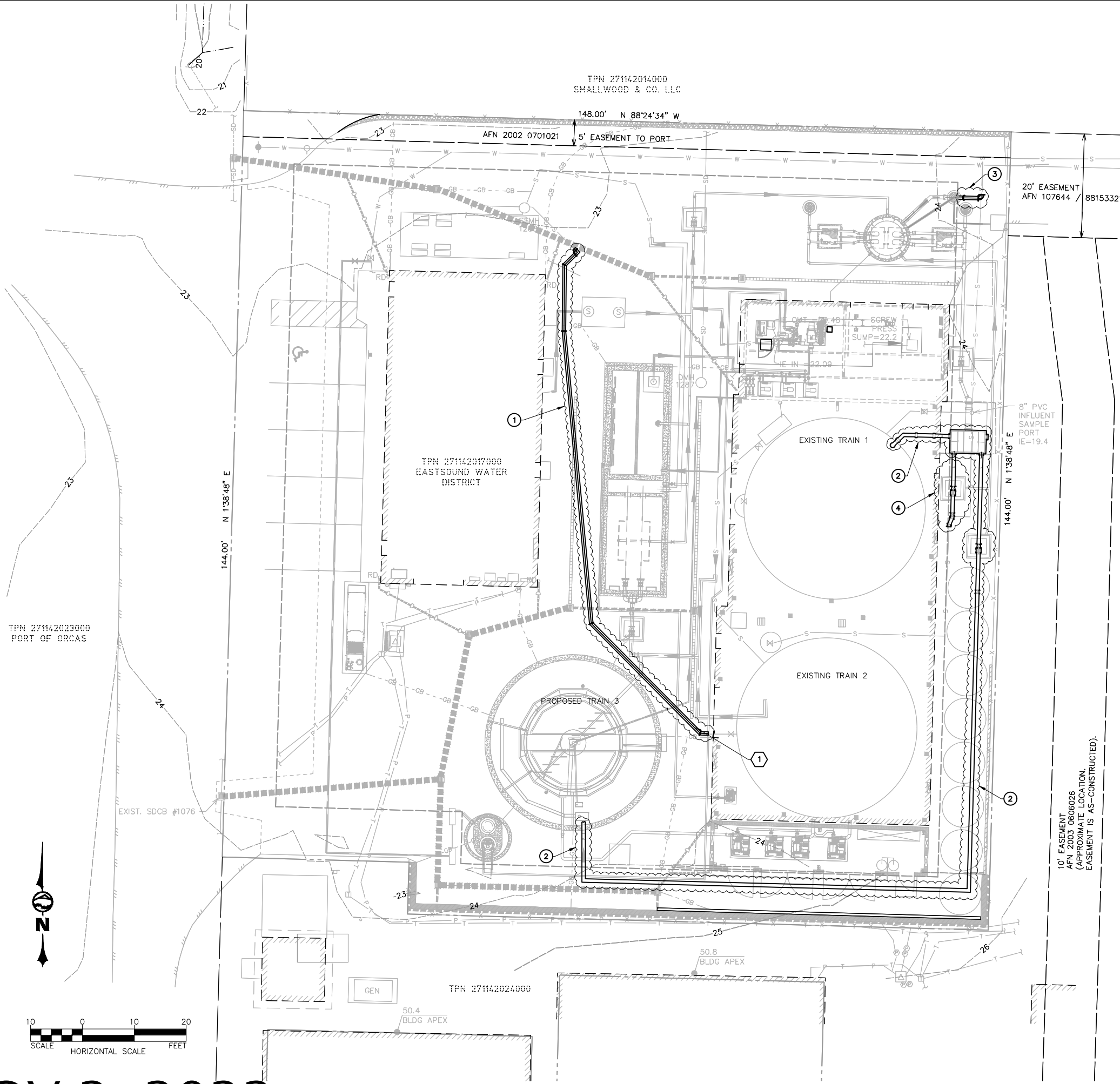
DESIGNED BY JGC		DRAWN BY JGS		CHECKED BY AWL	
DATE		SCALE		JOB NUMBER	
EASTSOUND SEWER AND WATER DISTRICT		SAN JUAN COUNTY		13 OF 96	
WASHINGTON		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		STAGING PLAN	

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- KEYED NOTES**
- ① = PROVIDE TEMPORARY 8" PIPING FROM THE EXISTING CHLORINE DISINFECTION OUTLET (PIPE DIAMETER = 10") TO THE EXISTING 8" GRAVITY SEWER (JUST UPSTREAM OF SSMH 1288). TEMPORARY PIPE IS TO BE FLEXIBLE HDPE PIPE WITH RESTRAINED BAUER END CONNECTIONS OR APPROVED EQUAL.
 - ENSURE POSITIVE DRAINAGE FOR TEMPORARY 8" PIPE. MIN SLOPE = 0.04%.
 - POT HOLE EXISTING PIPE AND PROVIDE PLANNED ALIGNMENT INFORMATION FOR REVIEW AND APPROVAL, PRIOR TO CONSTRUCTION.
 - ENSURE WATER TIGHT RESTRAINED CONNECTIONS ARE PROVIDED AT UPSTREAM AND DOWNSTREAM CONNECTIONS.

- CONSTRUCTION SEQUENCE:**
- NOTE: ALL CONSTRUCTION PHASING STEPS SHALL BE COORDINATED WITH THE OWNER AND ENGINEER.
- ① = PRIOR TO CONSTRUCTION, DIVERT TREATED EFFLUENT AROUND THE PROPOSED WORK AREA THROUGH A TEMPORARY 8" PIPE. SEE KEYED NOTE #1
 - ② = INSTALL NEW SPLITTER BOX AND PIPING TO TRAIN #1 AND TRAIN #3 WITHOUT DISTURBING EXISTING PIPING AND PLANT OPERATIONS.
 - ③ = WHEN STAGE 1 FACILITIES ARE COMPLETE, TESTED, ACCEPTED AND READY TO BE PLACED ON-LINE, CONNECT 8" INFLUENT TO NEW INFLUENT PUMP STATION AND DIRECT FLOW TO TRAIN #1.
 - ④ = DEWATER, ABANDON, AND REMOVE EXISTING INFLUENT PIPING CONFLICTS SO THAT TRAIN #2 PIPING AND FLOW METER VAULT CAN BE INSTALLED AND TRAIN #2 PLACED IN SERVICE. TRAIN #2 DOWNTIME IS TO BE LIMITED TO 2-DAYS MAX, AND TO BE SCHEDULED DURING DRY WEATHER CONDITIONS. COORDINATE WITH OWNER.
 - ⑤ = AFTER TREATED EFFLUENT FLOW HAS BEEN DIVERTED TO NEW UV DISINFECTION SYSTEM, PLUG & ABANDON THE TEMPORARY 8" PIPE (IDENTIFIED AS KEYED NOTE #1).
 - ⑥ = PLACE STAGE 2 FACILITIES ON-LINE.
 - ⑦ = PLACE STAGE 3 FACILITIES ON-LINE.

- STAGES:**
- STAGE 1: INFLUENT PUMP STATION, INFLUENT SAMPLE STATION, INFLUENT FLOW SPLITTER, AEROBIC DIGESTER AND BLOWER, SLUDGE PUMP SYSTEM, UV DISINFECTION SYSTEM, BLOWER BUILDING.
- STAGE 2: SLUDGE DEWATERING SYSTEM, 2W WATER SYSTEM.
- STAGE 3: TRAIN #3.

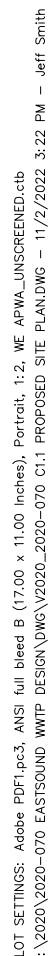
SHEET NOTES

1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.

NO.	REVISIONS	BY	DATE
5	ADD INTERIM PLAN OF OP.	JGC	5-10-22

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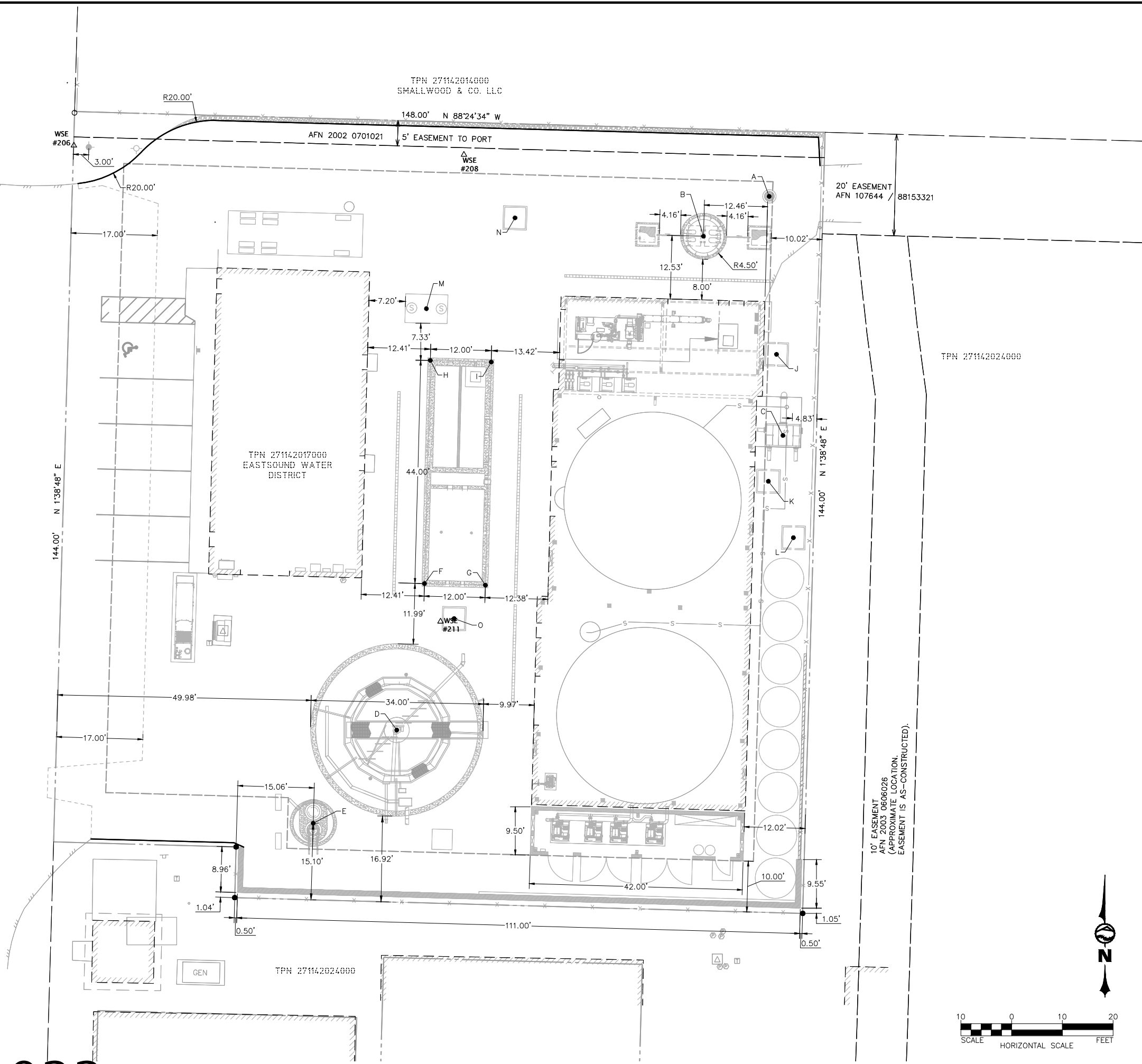
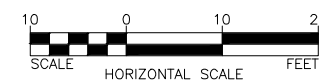
DESIGNED BY JGC	DRAWN BY SNR	CHECKED BY AWL	WILSON ENGINEERING WILSONENGINEERING.COM	
EASTSOUND SEWER AND WATER DISTRICT			11-02-2022	
SAN JUAN COUNTY			WASHINGTON	
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I			WWTP INTERIM PLAN OF OPERATION	
DATE	SCALE	AS SHOWN	JOB NUMBER	
			14 OF 96	



NOV 2, 2022

- 11-02-2022

O: EFFLUENT FM CENTER
N 629,528.89 E 1,139,362.85



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PORT OF ORCAS

WSE
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NOV 2. 2022



PIPE IDENTIFICATION LEGEND	
1	8" EXISTING EFFLUENT (DI GRAV.)
2	4" UV DISINFECTION DRAIN AND DIGESTER SUPERNATANT (DI GRAV.) SLOPE = 2% MIN.
3	8" INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
4	3" INFLUENT PUMP STATION AIR VENT (SS) OUTLET 8' ABOVE FINISH GRADE WITH GOOSENECK POINTING DOWN. ATTACH PIPE SECURELY TO EXTERIOR BUILDING WALL SLOPE = POSITIVE DRAINAGE THROUGHOUT
5	10" UV DISINFECTION EFFLUENT (DI GRAV.) SLOPE = 0.28% MIN.
6	4" BUILDING SEPTIC (SCH. 80 PVC GRAV.) CLOPE = 2% MIN.
7	2" DIGESTED SLUDGE (SCH. 80 PVC FM) SLOPE = POSITIVE THROUGHOUT
8	4" VAULT DRAIN (SCH. 80 PVC GRAV.) SLOPE = 2.0%
9	1" 2W WATER SYSTEM AIR VENT (SS) OUTLET 3' ABOVE GRADE WITH GOOSENECK POINTING DOWN SLOPE = POSITIVE DRAINAGE THROUGHOUT
10	2" 2W WATER LINE (SCH. 80 PVC FM)
11	8" TRAIN 3 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
12	2" WATER PIPE (SCH. 80 PVC FM)
13	8" TRAIN 2 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
14	8" TRAIN 1 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
15	2" TRAIN 3 WAS (SCH. 80 PVC GRAV.) SLOPE = POSITIVE DRAINAGE THROUGHOUT
16	8" TRAIN 2 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
17	8" TRAIN 3 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
18	6" INFLUENT TO FLOW SPLITTER (DI FM) SLOPE = POSITIVE THROUGHOUT
19	2" DIGESTER WAS INFLUENT (SCH. 80 PVC GRAV.)
20	6" CELL 3 AIR (SS)
21	4" DIGESTER AIR (SS)

- KEYED NOTES
- ① = CUT EXISTING 8" PIPE AND CONNECT NEW 8" INFLUENT PIPE TO TRAIN 2
- ② = EXTEND NEW 8" PIPE TO TRAIN 1. ABANDON AND PLUG EXISTING TRAIN 1 INFLUENT PIPE

SHEET NOTES

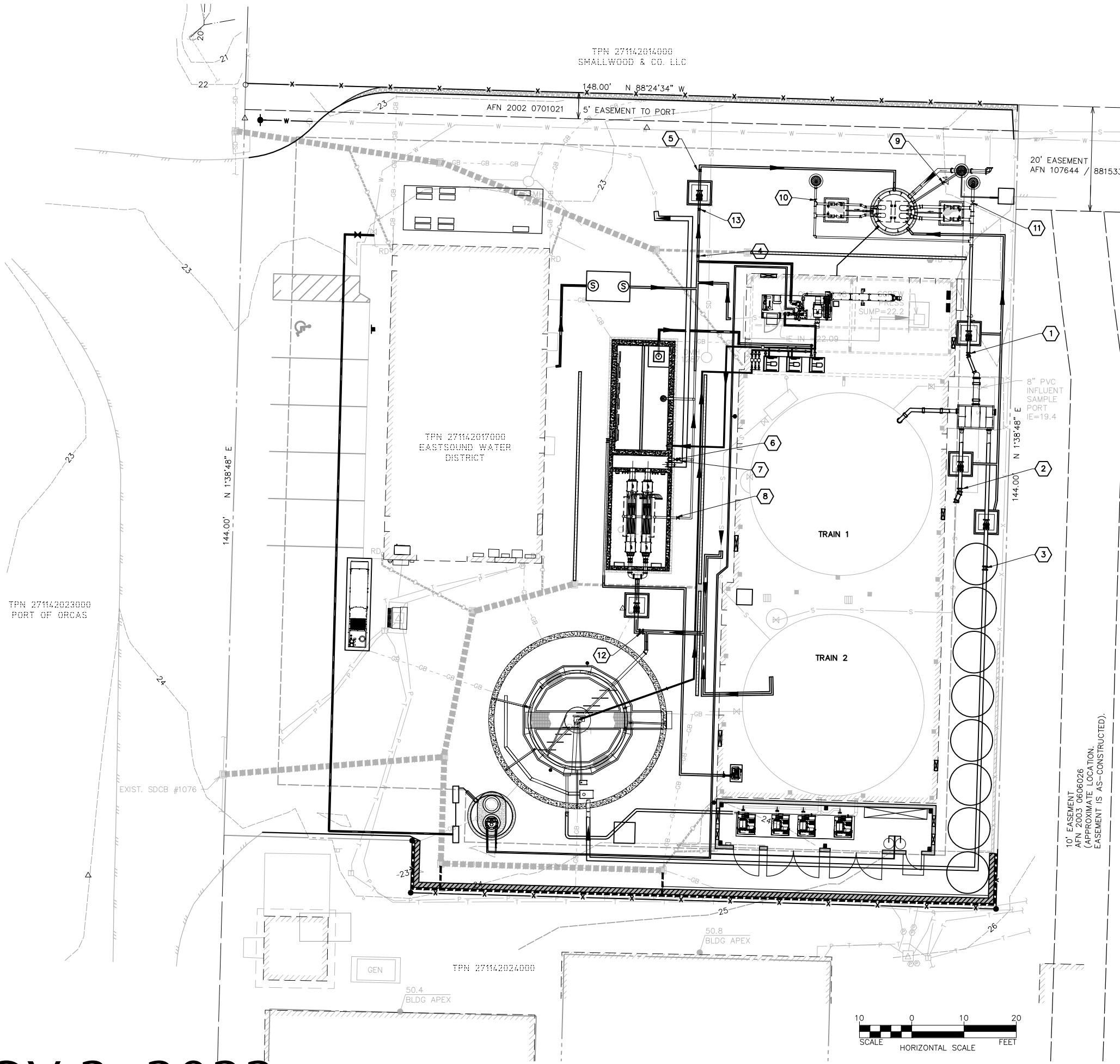
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2) SEE SHEETS C2.1–C2.4 FOR STORM GRADING, DRAINAGE, AND PIPING.

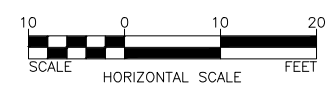


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BURIED VALVE IDENTIFICATION LEGEND	
1	6" GATE VALVE FOR INFLUENT FLOW METER ISOLATION
2	8" GATE VALVE FOR TRAIN 2 FLOW METER ISOLATION
3	8" GATE VALVE FOR TRAIN 3 FLOW METER ISOLATION
4	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
5	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
6	4" GATE VALVE FOR TREATED EFFLUENT OUTLET BOX DRAIN
7	4" GATE VALVE FOR TREATED EFFLUENT OUTLET BOX DRAIN
8	4" GATE VALVE FOR UV DISINFECTION DRAIN
9	4" GATE VALVE FOR SAMPLE SUMP DRAIN
10	4" GATE VALVE FOR PUMP STATION CLEAN OUT DRAIN
11	4" GATE VALVE FOR PUMP STATION CLEAN OUT DRAIN
12	8" PLUG VALVE FOR EFFLUENT FLOW METER ISOLATION
13	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION



- SHEET NOTES**
- 1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.
 - 2) SEE SHEETS C2.1-C2.4 FOR STORM GRADING, DRAINAGE, AND PIPING.



BID SET

NO.	REVISIONS	BY	DATE

DESIGNED BY

JCC

DRAWN BY

SNR

CHECKED BY

AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SITE PIPING PLAN WITH BURIED VALVE ID

DATE

SCALE

AS SHOWN

JOB NUMBER

18

OF

96

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11-02-2022

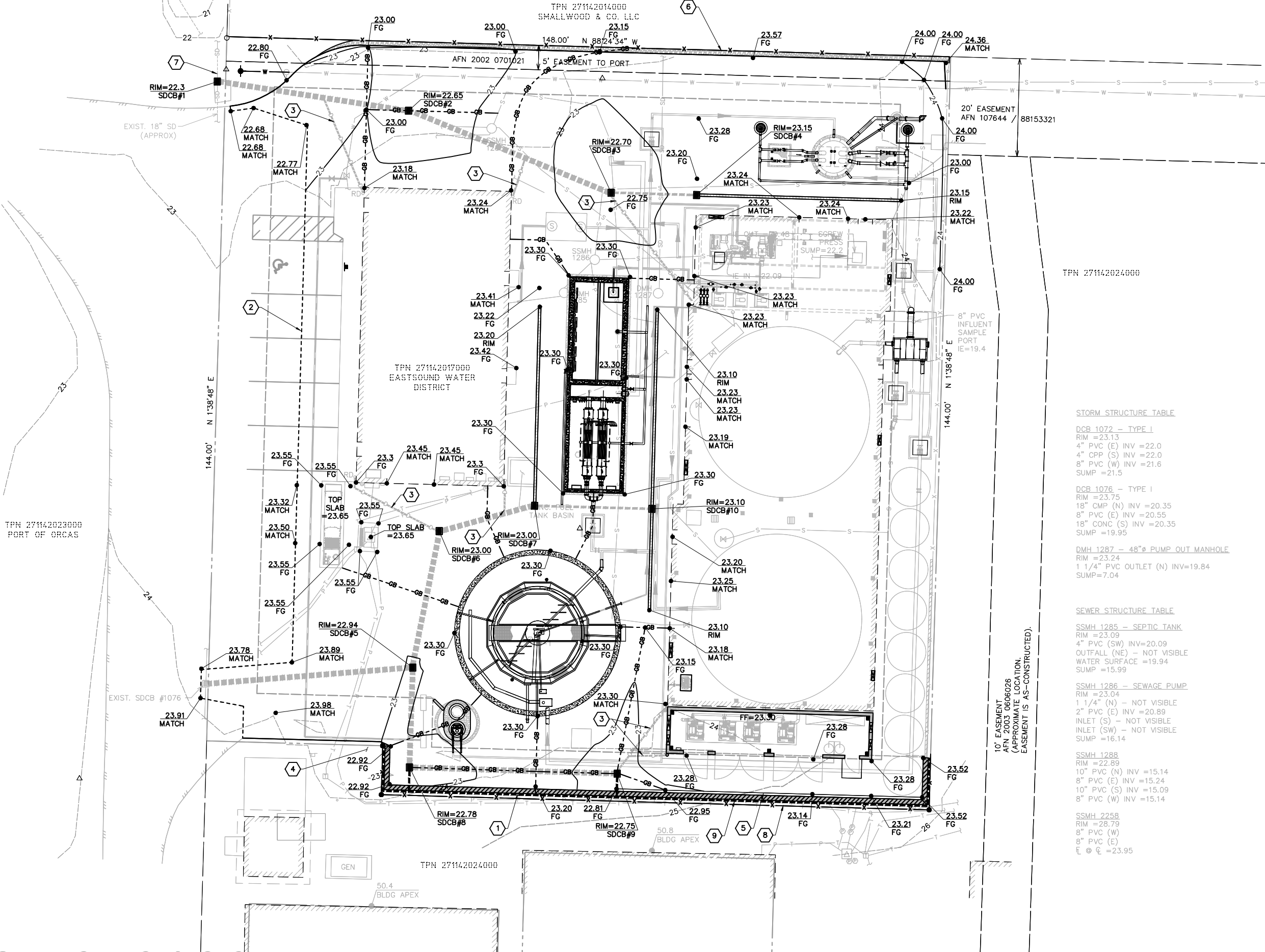
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TPN 271142023000
 PORT OF ORCAS



STORM STRUCTURE TABLE

DCB 1072 - TYPE I
 RIM = 23.13
 4" PVC (E) INV = 22.0
 4" CPP (S) INV = 22.0
 8" PVC (W) INV = 21.6
 SUMP = 21.5

DCB 1076 - TYPE I
 RIM = 23.75
 18" CMP (N) INV = 20.35
 8" PVC (E) INV = 20.55
 18" CONC (S) INV = 20.35
 SUMP = 19.95

DMH 1287 - 48" Ø PUMP OUT MANHOLE
 RIM = 23.24
 1 1/4" PVC OUTLET (N) INV = 19.84
 SUMP = 7.04

SEWER STRUCTURE TABLE

SSMH 1285 - SEPTIC TANK
 RIM = 23.09
 4" PVC (SW) INV = 20.09
 OUTFALL (NE) - NOT VISIBLE
 WATER SURFACE = 19.94
 SUMP = 15.99

SSMH 1286 - SEWAGE PUMP
 RIM = 23.04
 1 1/4" (N) - NOT VISIBLE
 2" PVC (E) INV = 20.89
 INLET (S) - NOT VISIBLE
 INLET (SW) - NOT VISIBLE
 SUMP = 16.14

SSMH 1288
 RIM = 22.89
 10" PVC (N) INV = 15.14
 8" PVC (E) INV = 15.24
 10" PVC (S) INV = 15.09
 8" PVC (W) INV = 15.14

SSMH 2258
 RIM = 28.79
 8" PVC (W)
 8" PVC (E)
 E @ E = 23.95

10' EASEMENT
 AFN 2003 0606026
 (APPROXIMATE LOCATION,
 EASEMENT IS AS-CONSTRUCTED).

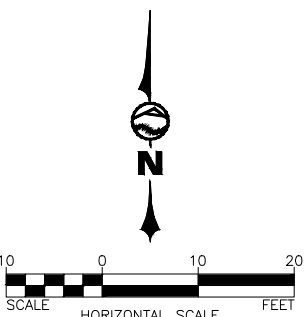
NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22

KEYED NOTES

- 1 = KEYSTONE BLOCK RETAINING WALL. SEE PLAN, PROFILE, AND DETAILS PER C2.5 & C7.8
- 2 = SAWCUT LINE PER DEMOLITION PLAN C0.5
- 3 = 4" DIA ROOF DRAIN
- 4 = GRADE AREA FROM FINISHED GRADES TO DAYLIGHT AT 2H:1V
- 5 = ASPHALT WEDGE CURB PER C4 & C7.7
- 6 = 149 LF OF 6' CHAIN LINK FENCE. CONNECT TO EXISTING.
- 7 = PRESUMED LOCATION OF EXISTING STORM SEWER. CATCH BASIN LOCATION TO BE ADJUSTED BASED ON POTHOLES RESULTS
- 8 = 134 LF OF 6' CHAIN LINK FENCE. CONNECT TO EXISTING
- 9 = FOOTING DRAIN

NOTES:

- 1) MATCH ELEVATION AT BUILDING DOORS IS 0.02' BELOW FINISHED FLOOR.
- 2) ALL CATCH BASINS TO HAVE HERRINGBONE GRATES.
- 3) ALL HATCHES AND COVERS FOR NON-STORMWATER ITEM 5 TO BE 0.02' MIN ABOVE SURROUNDING ASPHALT GRADES. IN CASES WHERE HATCHES & COVERS ARE GREATER THAN 0.02' ABOVE SURROUNDING ASPHALT, SLOPE ASPHALT AT 1H:1V, AS NEEDED TO ACHIEVE 0.02' DISTANCE.
- 4) SEE SHEET C0.1 FOR EXISTING CONDITIONS.



Call 811
 two business days
 before you dig

BID SET

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DESIGNED BY	JGC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT
 WASHINGTON
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
 SAN JUAN COUNTY
 GRADING & STORM DRAINAGE PLAN

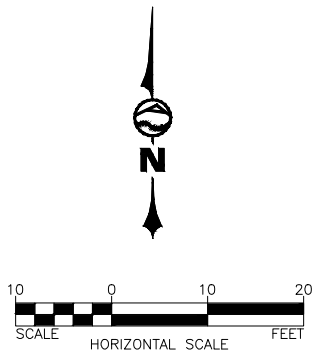
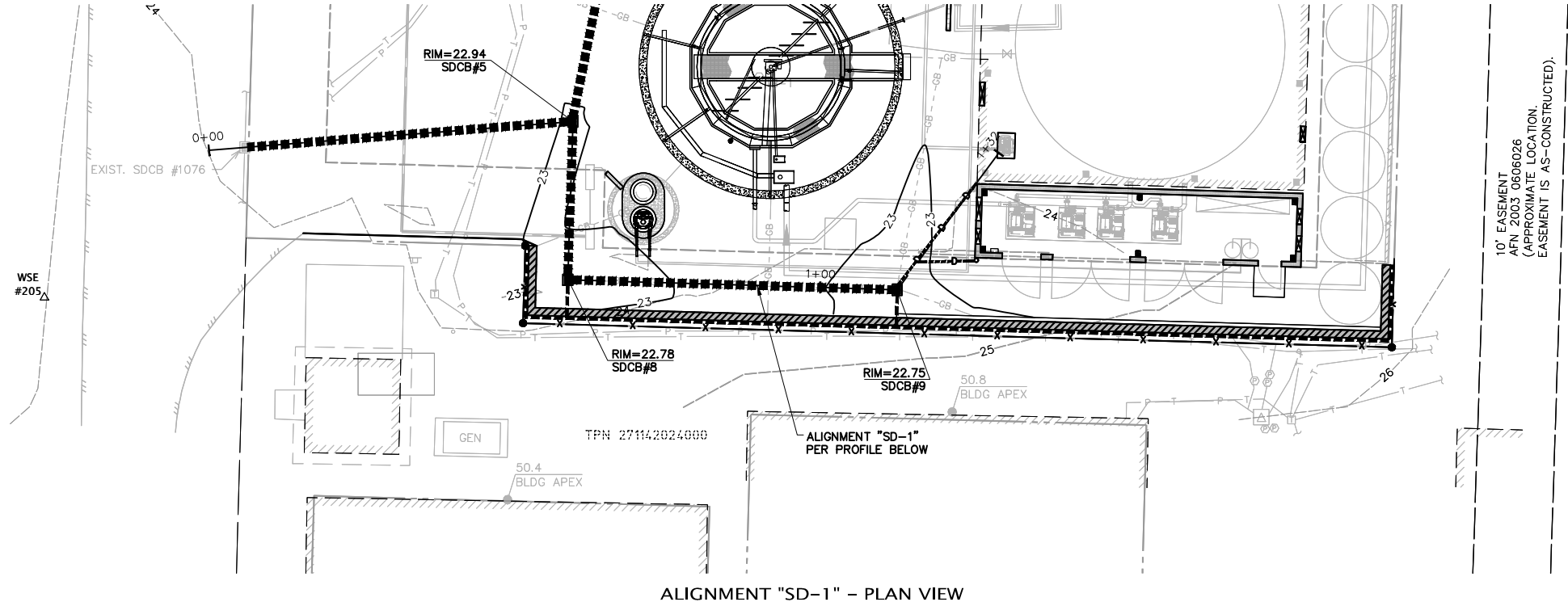
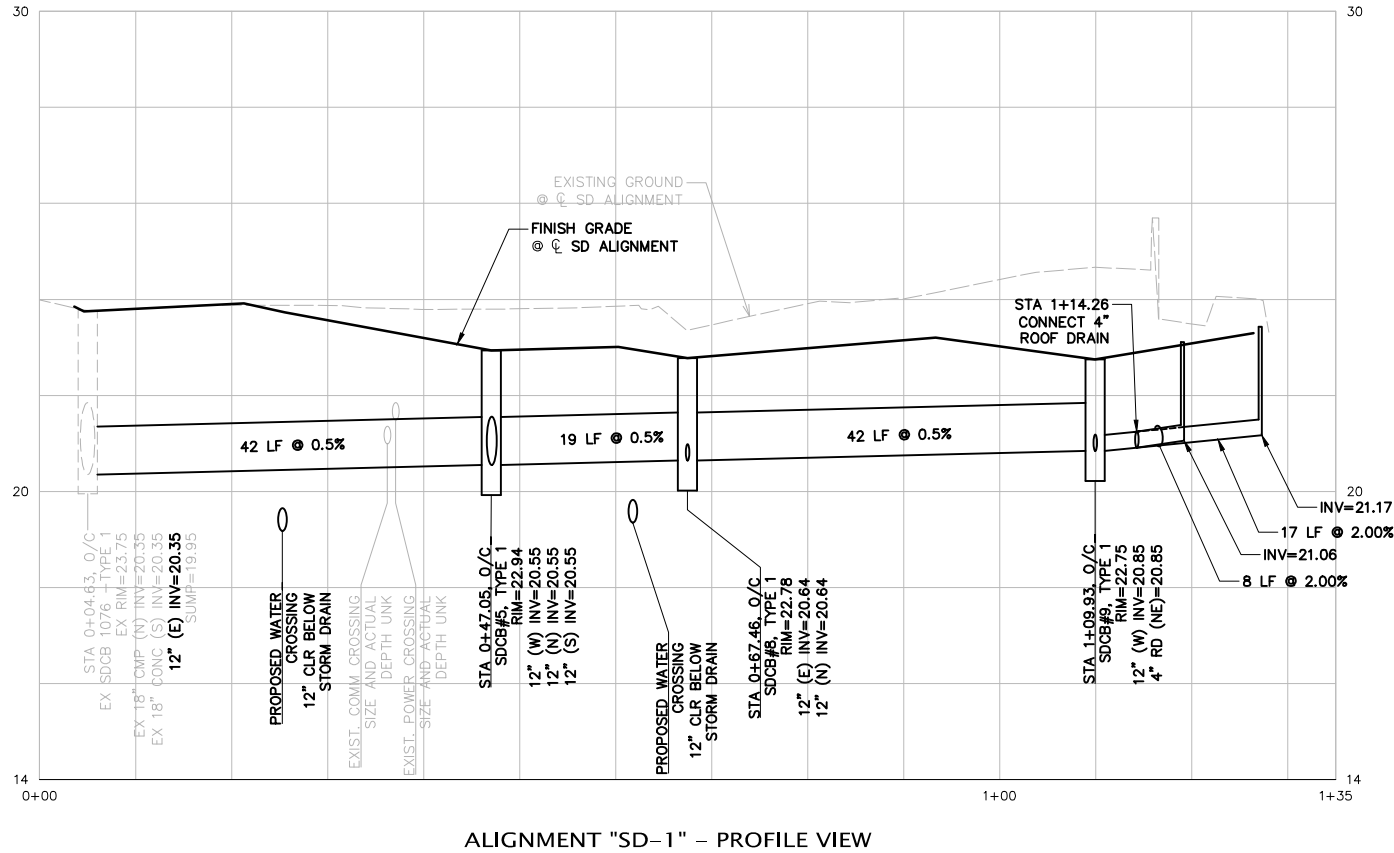
DATE	SCALE	AS SHOWN	JOB NUMBER
			20 OF 96

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11-02-2022

NOV 2, 2022

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 Inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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NO.	REVISIONS	BY	DATE



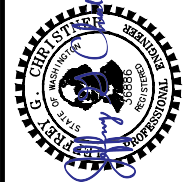
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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

STORM DRAINAGE PLAN & PROFILE

DESIGNED BY	JCC
DRAWN BY	JGS
CHECKED BY	AWL



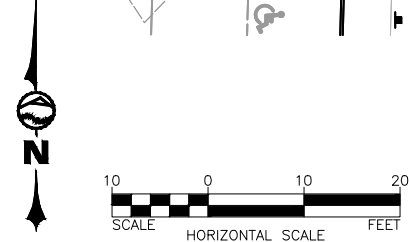
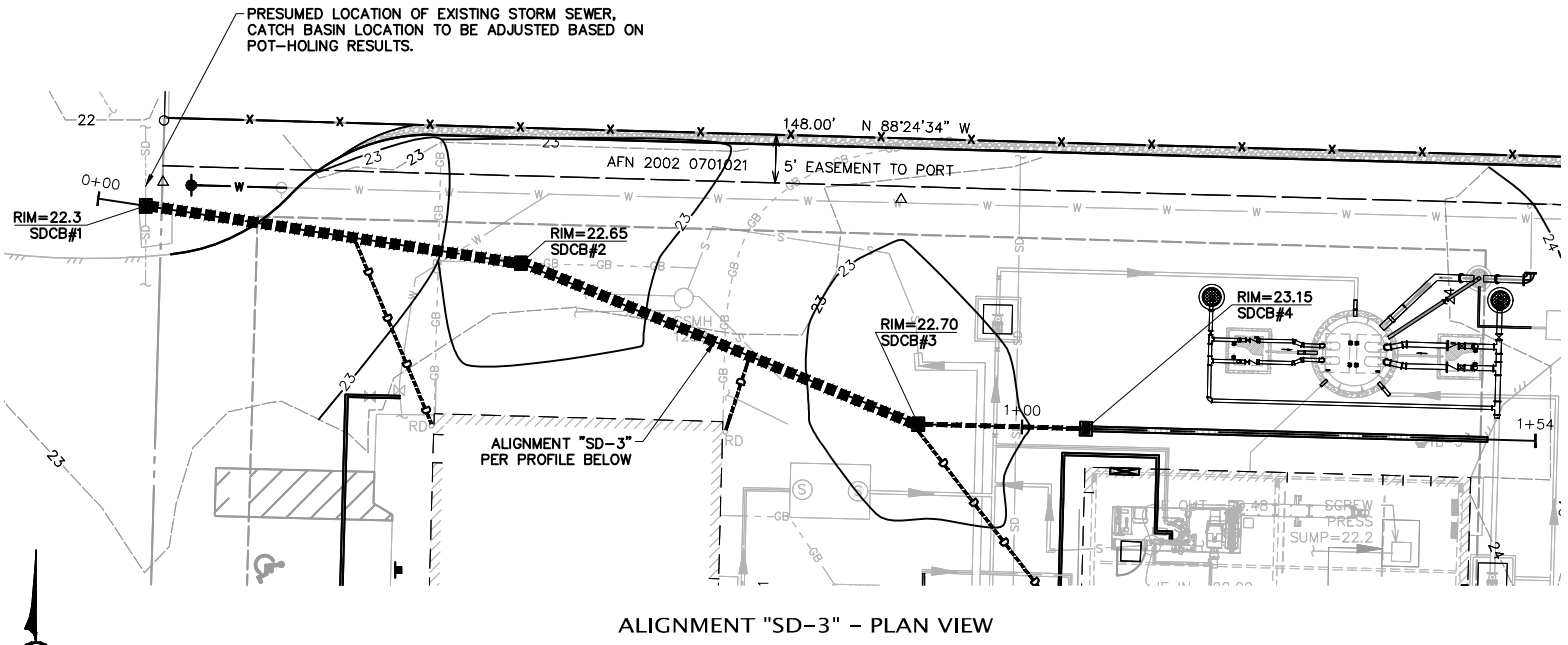
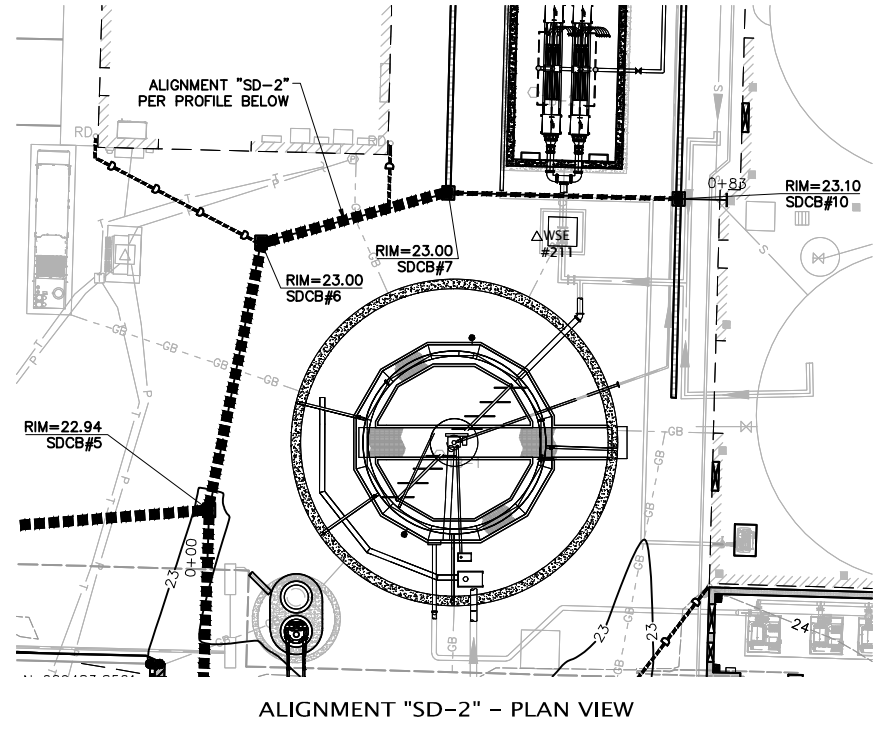
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11-02-2022

SHEET	DATE	SCALE	AS SHOWN	JOB NUMBER
21	NOV 2, 2022	1" = 20'	AS SHOWN	96

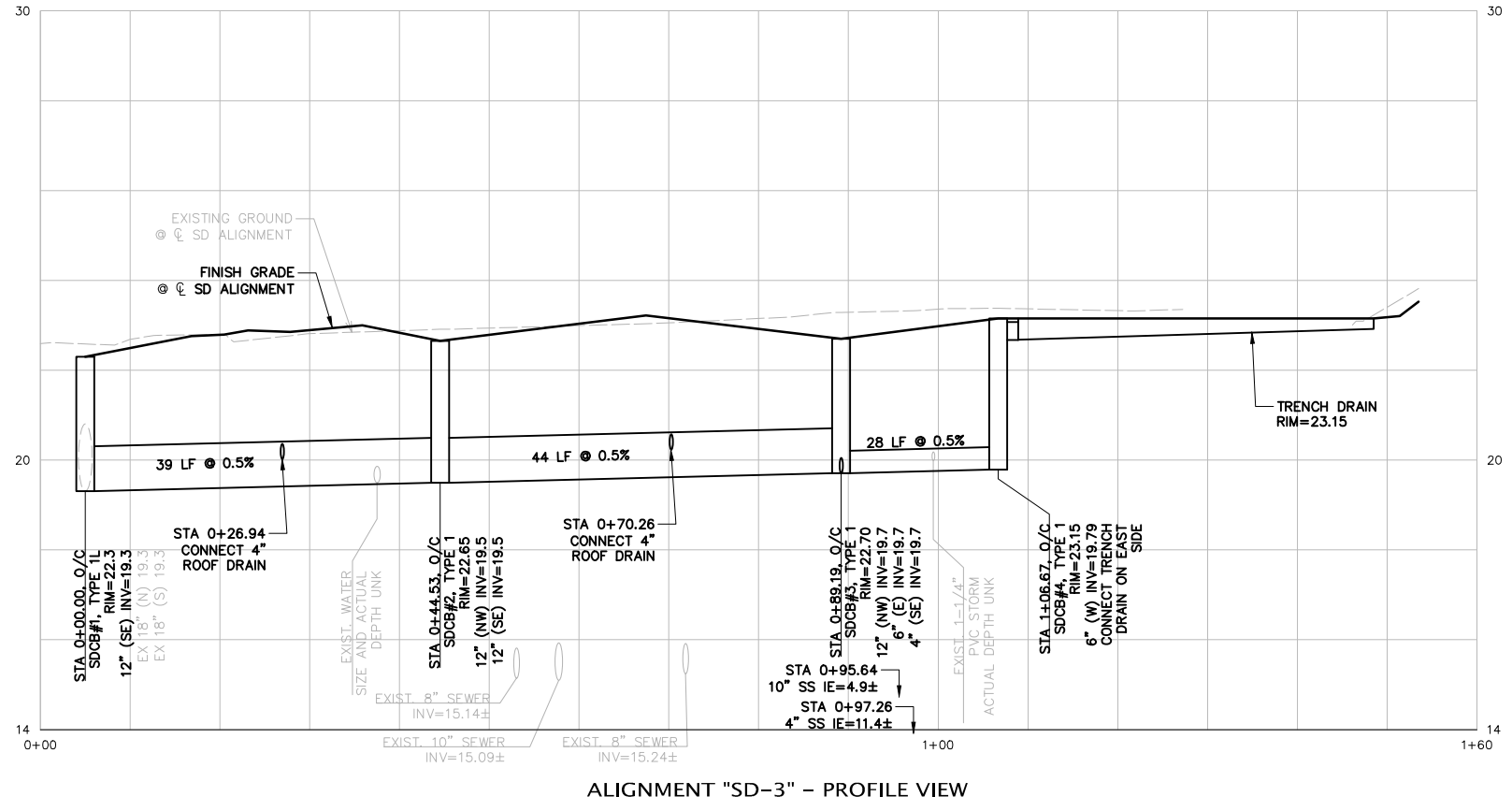
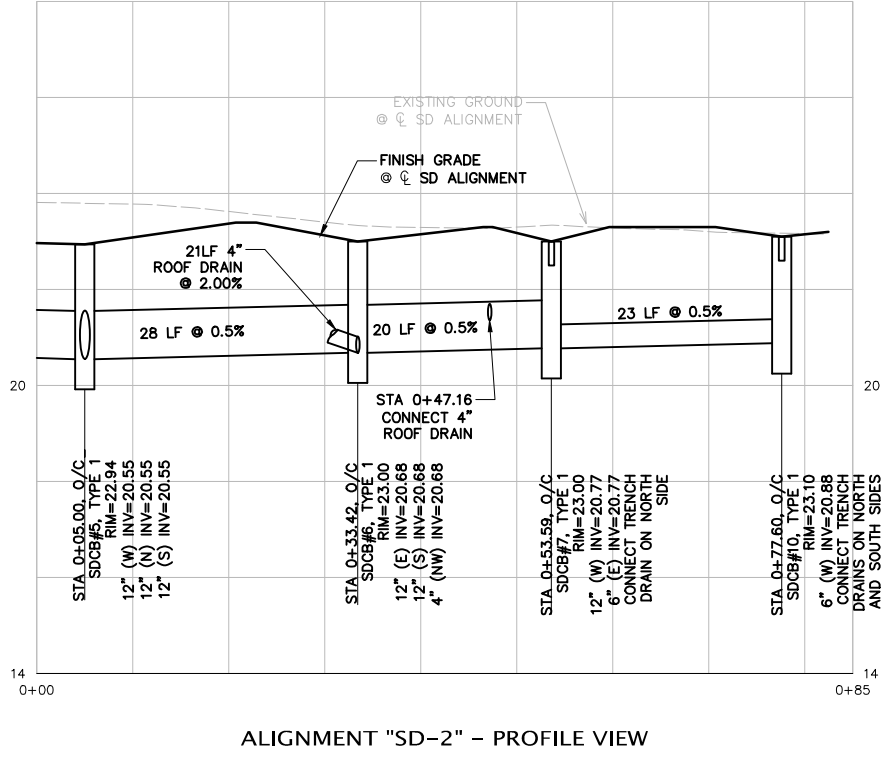
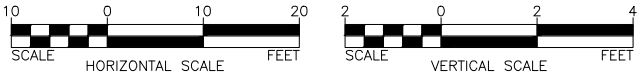
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NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22



BID SET

811 Call 811
two business days before you dig



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DESIGNED BY: JGC

DRAWN BY: JGS

CHECKED BY: AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

STORM DRAINAGE PLAN & PROFILES

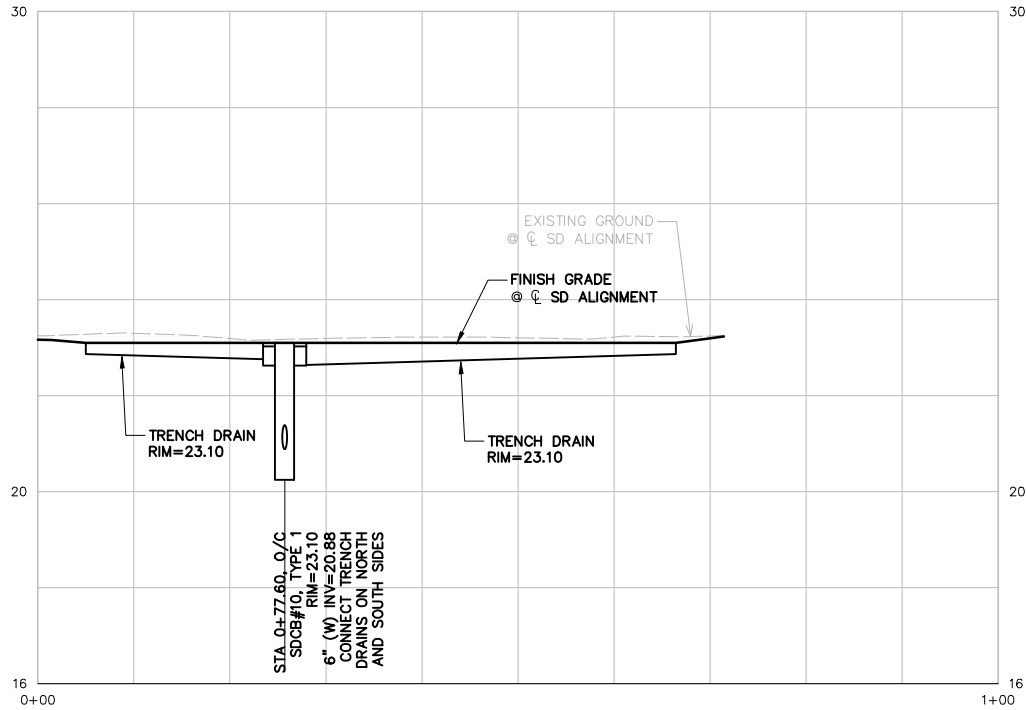
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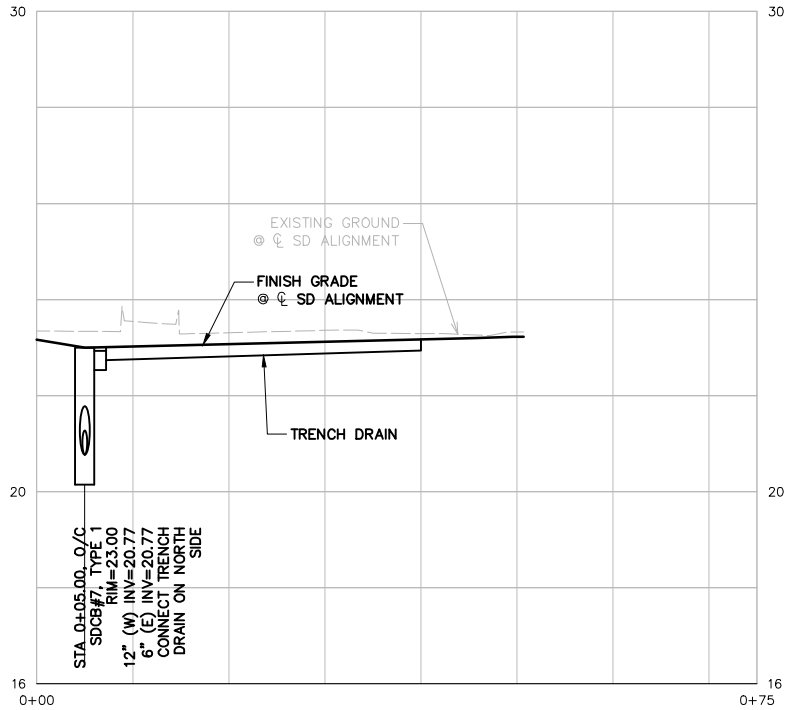
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NOV 2, 2022

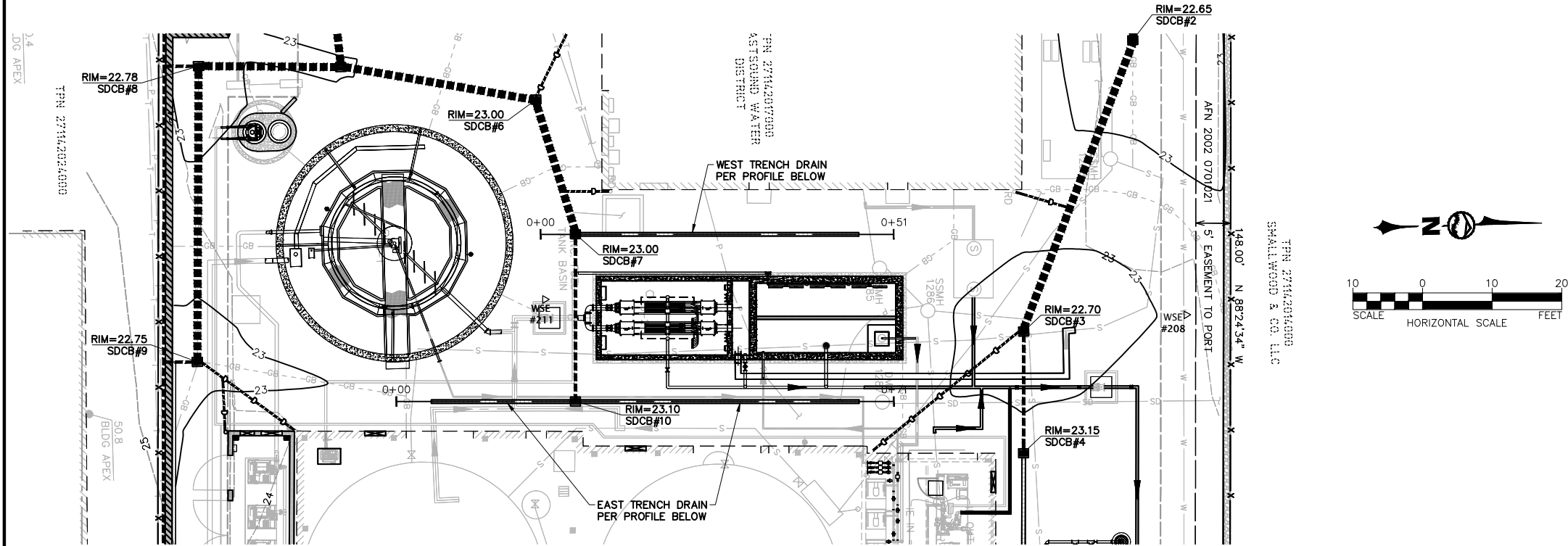
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EAST TRENCH DRAIN – PROFILE VIEW



WEST TRENCH DRAIN – PROFILE VIEW



WEST & EAST TRENCH DRAINS – PLAN VIEW

NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22

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P.E.
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AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

STORM DRAINAGE PLAN & PROFILES

DATE

SCALE

AS SHOWN

JOB NUMBER

23

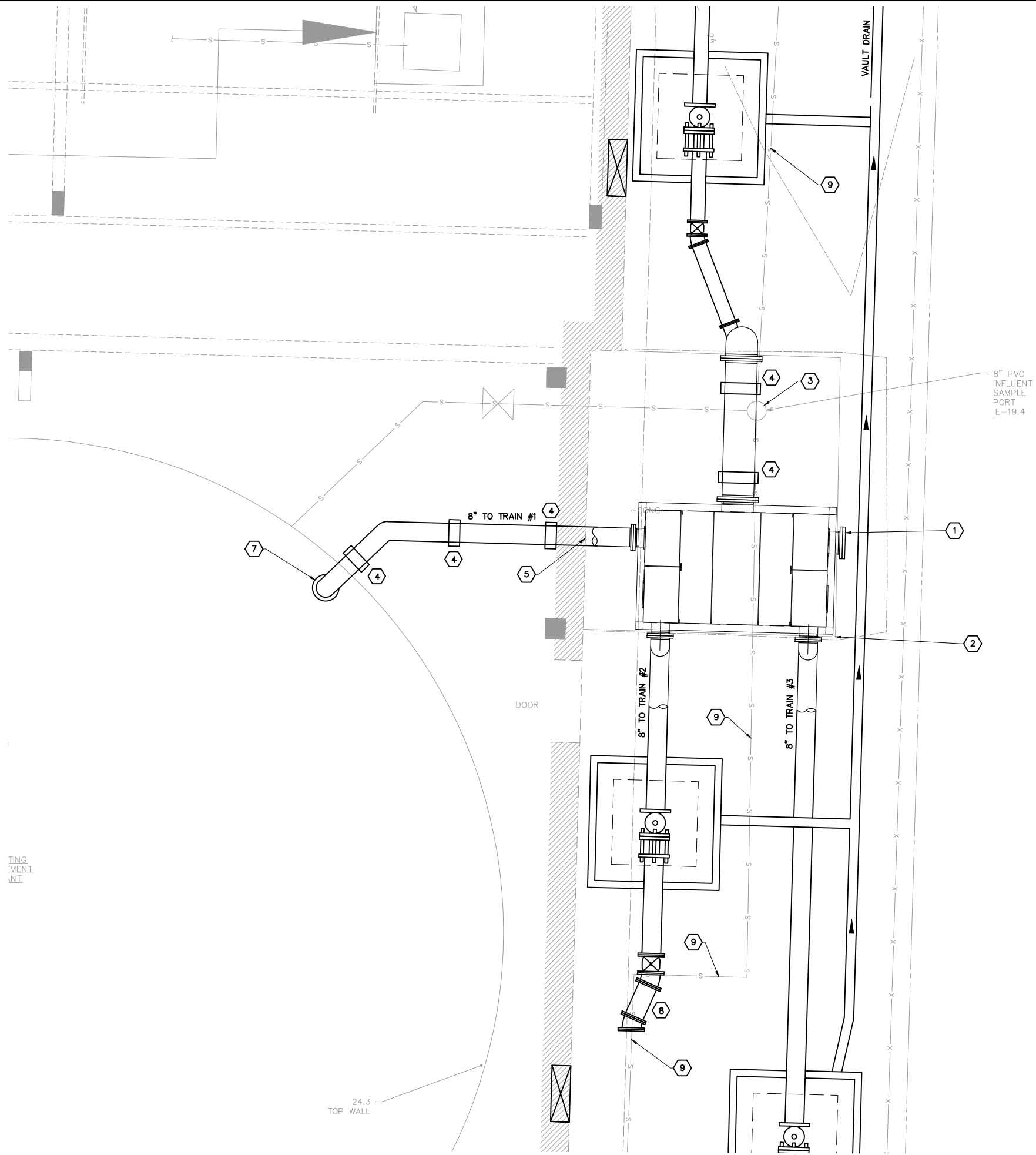
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96

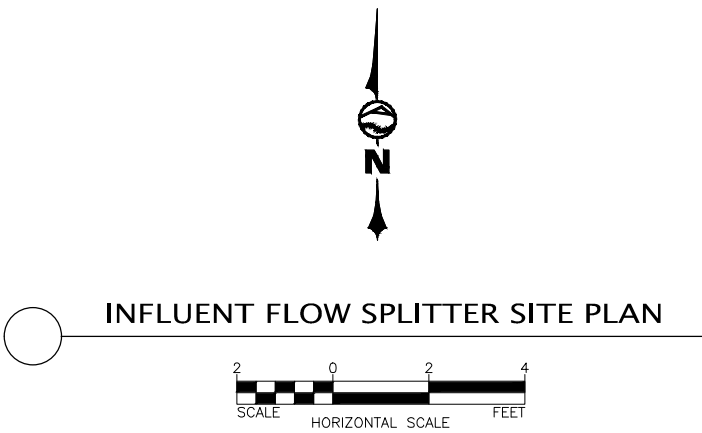
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BID SET

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W:\2020-2020-070 EASTSOUND WWP DESIGN\DWG\2020_2020-070 C3.1- FLOW SPLITTER.DWG - 11/2/2022 3:25 PM - Jeff Smith



CONSTRUCTION SEQUENCE:
INSTALL NEW SPLITTER BOX AND PIPING TO TRAIN #1 AND TRAIN #3 WITHOUT DISTURBING EXISTING PIPING AND PLANT OPERATIONS. WHEN NEW FACILITIES ARE COMPLETE, TESTED, AND READY TO BE PLACED ON-LINE, CONNECT 8" INFLUENT TO NEW INFLUENT PUMP STATION AND DIRECT FLOW TO TRAINS #1 AND #3. THEN DEWATER, ABANDON, AND REMOVE EXISTING INFLUENT PIPING CONFLICTS SO THAT TRAIN #2 PIPING AND FLOW METER VAULT CAN BE INSTALLED AND TRAIN #2 PLACED IN SERVICE. TRAIN #2 DOWNTIME IS TO BE LIMITED TO 5-DAYS MAX (MONDAY-FRIDAY).



NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

- KEYED NOTES**
- 1 = CONNECTION FOR FUTURE TRAIN #4. SEAL WITH 8" BLIND FLANGE
 - 2 = PROPOSED CONCRETE SUPPORT FOR SPLITTER BOX
 - 3 = EXISTING SAMPLING PORT
 - 4 = PIPE SUPPORT PER $\frac{5}{C7.2}$
 - 5 = CAREFULLY CUT HOLE THROUGH BUILDING FOR NEW 8" TRAIN #1 INFLUENT PIPE. SEAL WITH FLASHING
 - 6 = NOT USED
 - 7 = 8" TRAIN #1 INFLUENT, INSTALL ELBOW AND DIRECT FLOW INTO AERATION BASIN ADJACENT TO EXISTING INFLUENT PIPE
 - 8 = CONFIRM ALIGNMENT, SIZE, MATERIAL, AND DEPTH OF EXISTING PIPE TO TRAIN #2 INFLUENT. PROVIDE BENDS & FITTINGS AS NEEDED SO THAT NEW 8" DI PIPING CAN BE CONNECTED TO EXISTING PIPE, AND INFLUENT FLOW IS DIRECTED TO TRAIN #2. ENSURE POSITIVE DRAINAGE THROUGHOUT, FROM SPLITTER BOX TO NEW CONNECTION
 - 9 = POT-HOLE EXISTING 8" PIPING AND CONFIRM ACTUAL ALIGNMENT AND DEPTH PRIOR TO CONSTRUCTION. ADJUST FINAL LAYOUT, AS NEEDED, TO AVOID CONFLICTS



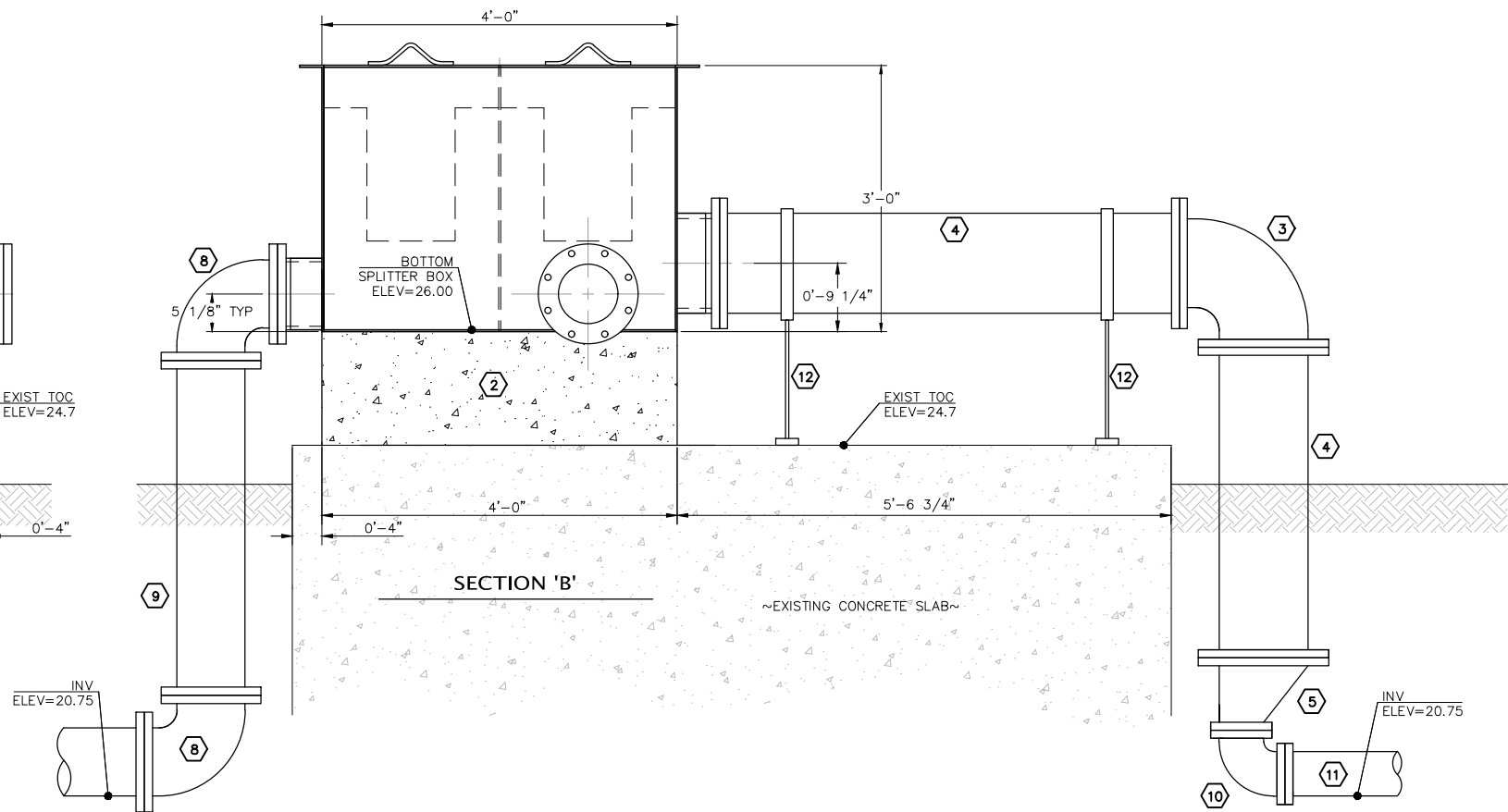
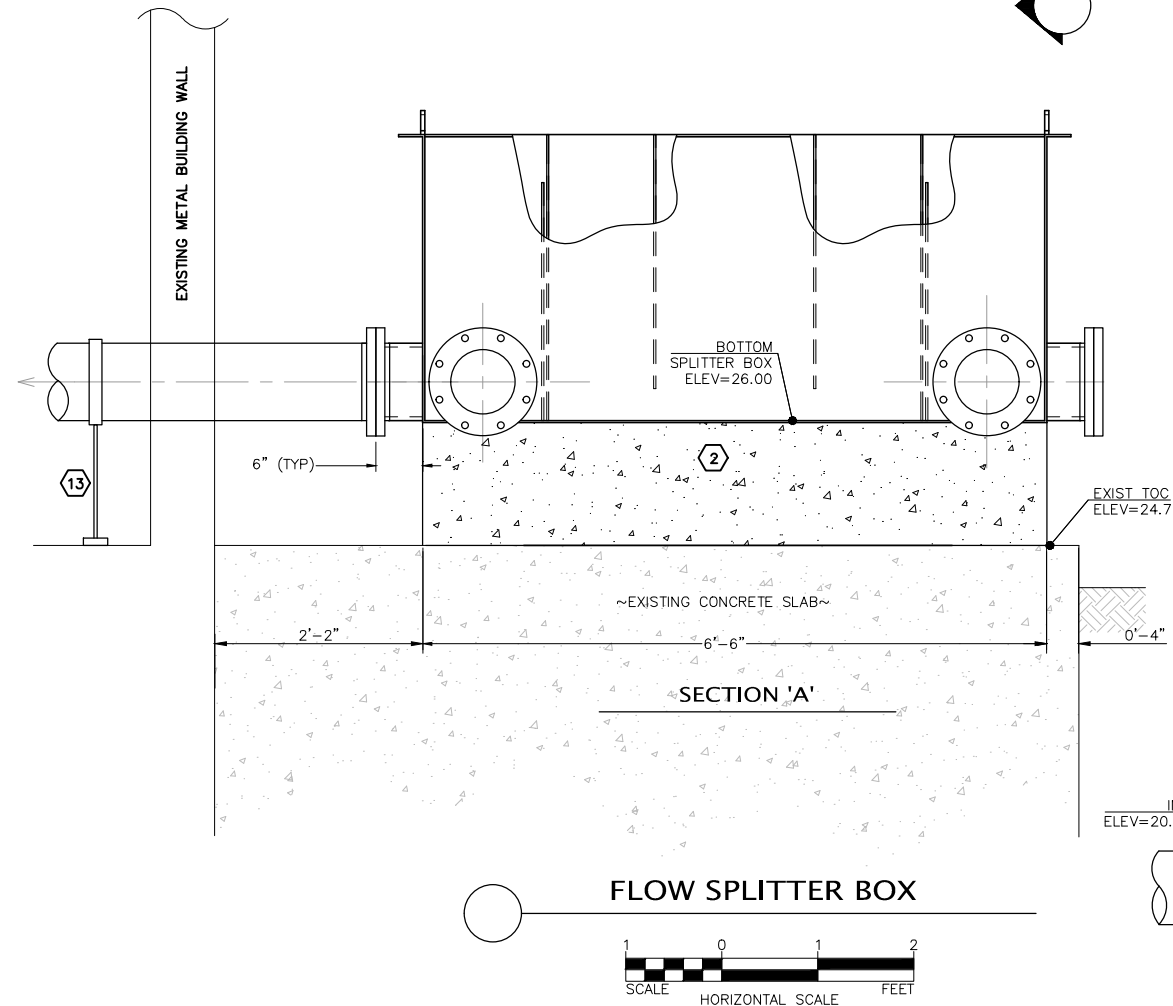
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SHEET	DATE	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JGC	
	SCALE	SAN JUAN COUNTY WASHINGTON			DRAWN BY JGS
	AS SHOWN	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I			CHECKED BY AWL
JOB NUMBER		INFLUENT FLOW SPLITTER SITE PLAN			
26 OF 96					

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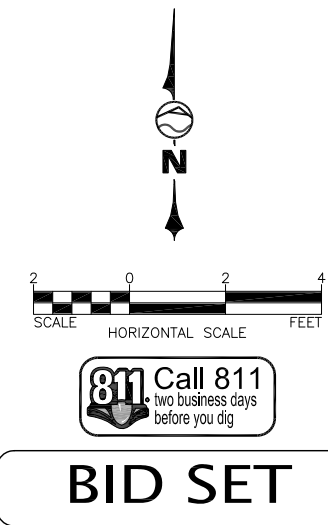
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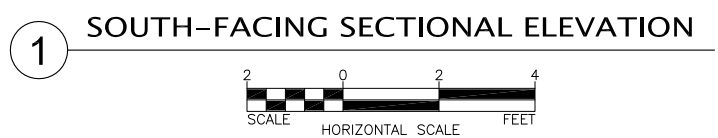
KEYED NOTES

- ① = FUTURE CONNECTION. SEAL WITH BLIND FLANGE
- ② = PROPOSED CONCRETE SUPPORT FOR SPLITTER BOX. APPROXIMATE DIMENSION = 6.5' X 4.0' X 1.3' TALL. CONTRACTOR TO COORDINATE WITH ACTUAL SPLITTER BOX DIMENSIONS, AND ENSURE TOP OF CONCRETE ELEV=26.00. SEE STRUCTURAL DRAWINGS.
- ③ = 12" DI 90° BEND, FL
- ④ = 12" DI FLANGE SPOOL
- ⑤ = 12"x6" DI ECCENTRIC REDUCER, FL
- ⑥ = 6" DI 90° BEND, FL
- ⑦ = FLOW SPLITTER STRUCTURE IS TO BE A PREFABRICATED UNIT CONSTRUCTED OF 316 SS AND SIZED TO MATCH DRAWING UNIT IS TO INCLUDE 4 EQUAL SPLITTER CHAMBERS FOR FLOW TO EACH OF THE 3 TRAINS AND 1 ADDITIONAL TRAIN. UNIT IS TO INCLUDE A REMOVABLE COVER. EACH OF THE 4 FLOW SPLITTER CHAMBERS IS TO INCLUDE AN ADJUSTABLE WEIR. MFR TO BE SMITH & LOVELESS OR APPROVED EQUAL.
- ⑧ = 8" DI 90° BEND, FL
- ⑨ = 8" DI FLANGE SPOOL
- ⑩ = 6" DI 90° BEND, RESTRAINED
- ⑪ = 6" DI PIPE, RESTRAINED
- ⑫ = 12" PIPE SUPPORT PER $\frac{5}{C7.2}$
- ⑬ = 8" PIPE SUPPORT PER $\frac{5}{C7.2}$

[illegible]



NOTES:
1. PARTITIONS TO BE REINFORCED FOR
INDIVIDUAL ZONE DEWATERING. (NOT SHOWN
FOR CLARITY)



811 Call 811
two business days
before you dig

BID SET

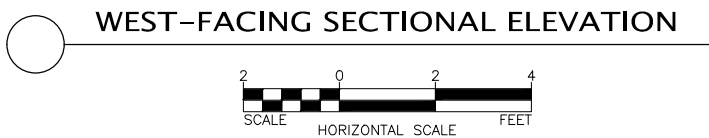


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SHEET	DATE
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	AS SHOWN
	JOB NUMBER
30 OF 96	



BID SET

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- 1 4" D.I. SPOOL, FxMJ
 2 4" D.I. SPOOL, FxF
 3 4" D.I. 90° BEND, FxF
 4 4" D.I. TEE, RESTRAINED
 5 4" CHECK VALVE, FxF
 6 4" GATE VALVE, FxF
 7 4" D.I. 22.5° BEND, RESTRAINED
 8 4" D.I. 45° LATERAL, FLxFL
 9 4"x3" DI REDUCER (IF REQUIRED) FOR PUMP DISCHARGE, FxF
 10 ROTATE TOP ELBOW 11.25° AND PROVIDE OVER-SIZED CORE HOLE, AS NEEDED, TO FIT 90° ELBOW ASSEMBLY IN WET WELL AS SHOWN
 11 4" FLANGED COUPLING ADAPTOR (RESTRAINED)
 12 4" D.I. PIPE, FORCE MAIN, RESTRAINED
 13 PRESSURE GAGE, ISOLATOR, & SHUTOFF VALVE
 14 ADJUSTABLE PIPE SUPPORTS (4 TOTAL)
 15 6"x4" D.I. REDUCER (IF REQ'D.) FOR PUMP DISCHARGE, FxF
 16 4" BURIED GATE VALVE, FLxMJ
 17 PRECAST CONCRETE VAULT 4'-2"L x 4'-2"W x 5'-6"D (MIN. INSIDE DIMENSIONS) UTILITY VAULT CO., MODEL 506-LA, OR EQUAL
 18 3" STAINLESS STEEL VENT PIPE, W/ SCREEN. IE=21.00. SCREEN TO BE 24 MESH STAINLESS STEEL.
 19 TOP SLAB PER

1 S2.4

2 S2.4

 20 36"x36" SINGLE LEAF HATCH, HALLIDAY PRODUCTS, H-20 LOAD RATING, MODEL H1R3636.
 21 STEPS OR LADDER PER WSDOT STD. PLAN B-30.90-01. PROVIDE W/ NON-CORROSIVE MATERIALS.
 22 SUSPENDED CABLE HOOKS, TYP. SEE ELECTRICAL DWGS.
 23 INCOMING CONDUITS
 24 GUIDE RAIL PER PUMP SUPPLIER REQUIREMENTS (TO BE STAINLESS STEEL.)
 25 HAUL CHAIN BY SUPPLIER (TO BE STAINLESS STEEL)
 26 PUMP POWER CORD, LENGTH TO ALLOW PUMP REMOVAL
 27 VERTICAL PIPE BRACE, TYP. SEE DETAIL

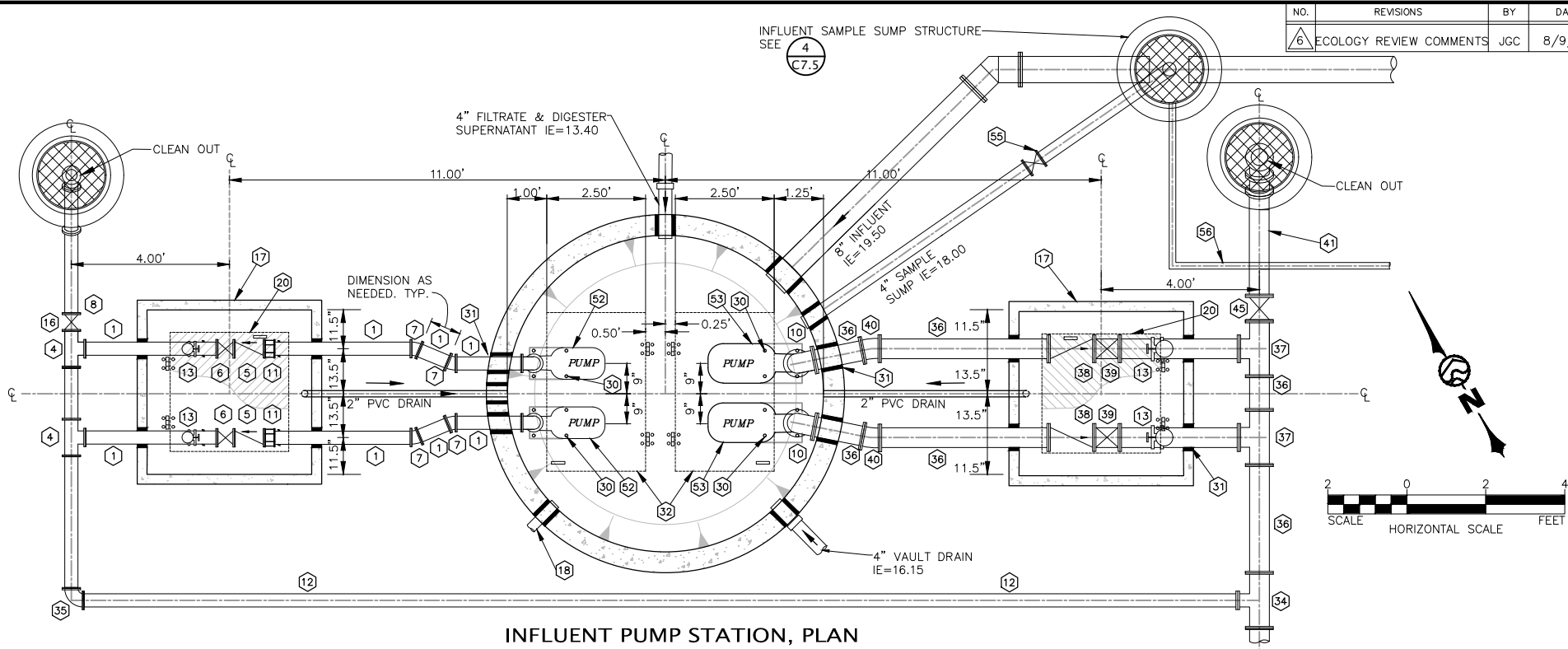
C7.2

 28 PUMPS PER SPECS
 29 MANUFACTURERS DISCONNECT ELBOW
 30 3/4" S.S. ANCHOR BOLT WITH 2" PROJECTION, TYP. (16 TOTAL). LOCATE & SIZE PER MFR. REQUIREMENTS.
 31 INSTALL LINK-SEAL BOOTS, OR EQUAL. TYP. ON ALL PIPE-TO-CONC. CONNECTIONS. INSTALL PER MFR REQUIREMENTS. SEAL ALL VOIDS. (TYP.)
 32 30"x48" SINGLE LEAF HATCH, HALLIDAY PRODUCTS, H-20 LOAD RATING, MODEL H1R3048, LOCATE HATCH OVER PUMPS COORDINATE WITH SUPPLIER
 33 FLAPPER VALVE. CONFIGURE TO CLOSE & SEAL WHEN SUBMERGED
 34 6"x4" D.I. TEE, RESTRAINED
 35 ***VACANT***
 36 6" D.I. SPOOL, RESTRAINED
 37 6" D.I. TEE, FxF
 38 6" CHECK VALVE, FxF
 39 6" GATE VALVE, FxF
 40 6" D.I. 11.25° BEND, RESTRAINED
 41 6" D.I. 45° LATERAL, FLxFL
 42 ***VACANT***
 43 ***VACANT***
 44 6" FLANGED COUPLING ADAPTOR (RESTRAINED)
 45 6" BURIED GATE VALVE, FLxMJ
 46 2"-4" CRUSHED ROCK
 47 24"Ø X 24" LONG CONC. WELL CURB H2 PRECAST, MODEL#WC-24 W/ CONC. ADJ. RING, OR EQUAL.
 48 4" x 3" PVC REDUCER & ADAPTOR AS NEEDED.
 49 ANDREW'S BOSS LOCK 3" FEMALE COUPLE X MALE NPT ALUMINUM.
 50 ANDREW'S BOSS LOCK TYPE DP DUST PLUG, ALUMINUM.
 51 IFCO 813 RING AND COVER, NON-LOCKING
 52 SMALL PUMPS DUPLEX ARRANGEMENT. SEE SPECS
 53 LARGE PUMPS DUPLEX ARRANGEMENT. SEE SPECS
 54 TRANSDUCER. SET 6" ABOVE FLOOR
 55 4" BURIED GATE VALVE
 56 2" SCH 80 CONDUIT FOR INF. SAMPLER INTAKE TUBING

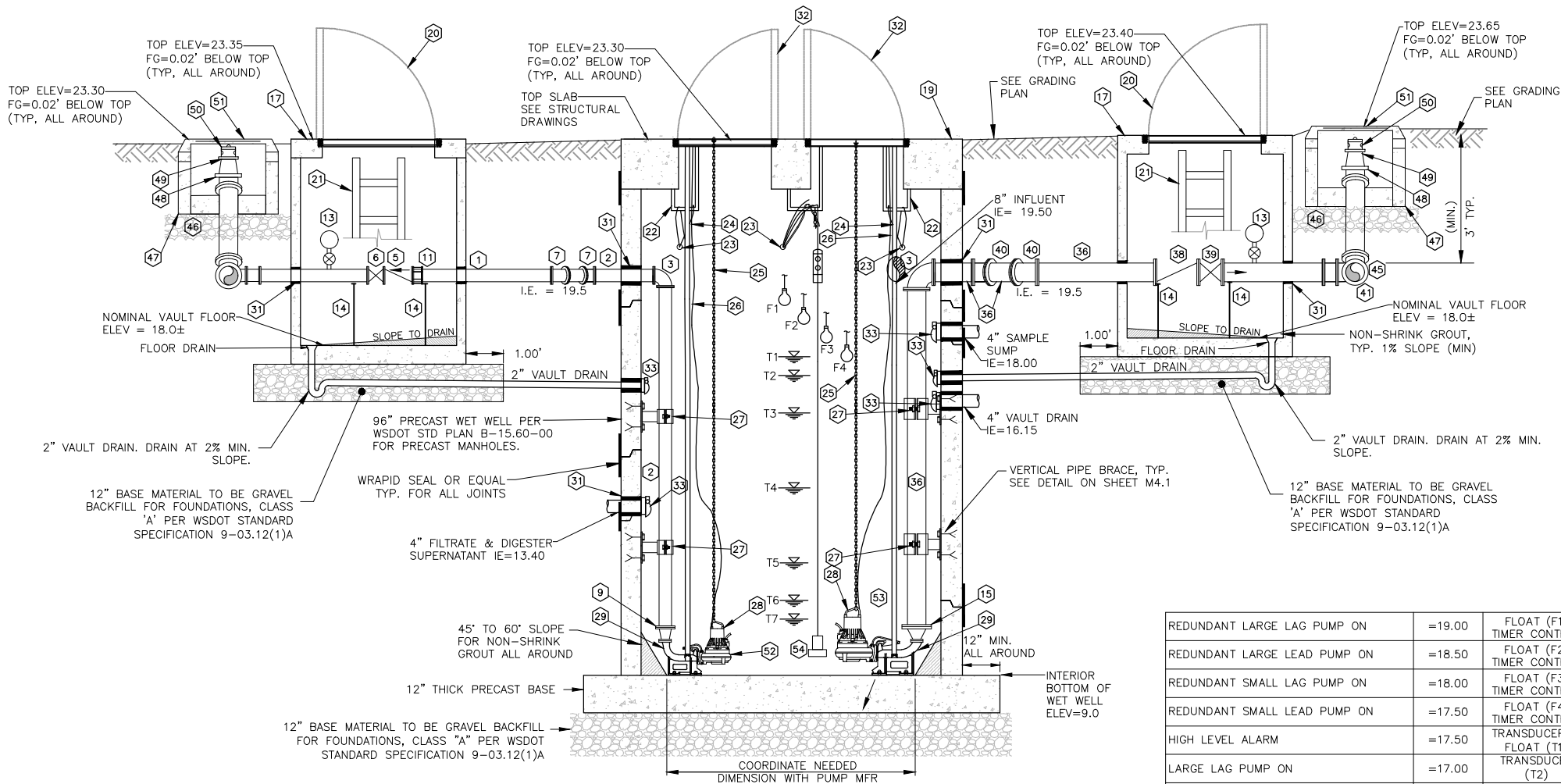


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INFLUENT PUMP STATION, PLAN



INFLUENT PUMP STATION, ELEVATION

REDUNDANT LARGE LAG PUMP ON	=19.00	FLOAT (F1) TIMER CONTROL
REDUNDANT LARGE LEAD PUMP ON	=18.50	FLOAT (F2) TIMER CONTROL
REDUNDANT SMALL LAG PUMP ON	=18.00	FLOAT (F3) TIMER CONTROL
REDUNDANT SMALL LEAD PUMP ON	=17.50	FLOAT (F4) TIMER CONTROL
HIGH LEVEL ALARM	=17.50	TRANSDUCER & FLOAT (T1)
LARGE LAG PUMP ON	=17.00	TRANSDUCER (T2)
LARGE LEAD PUMP ON	=16.00	TRANSDUCER (T3)
SMALL LAG PUMP ON	=14.00	TRANSDUCER (T4)
SMALL LEAD PUMP ON	=12.00	TRANSDUCER (T5)
PUMPS OFF	=11.00	TRANSDUCER (T6)
LOW LEVEL ALARM & REDUNDANT OFF	=10.50	TRANSDUCER (T7)

NO. 6
 REVISIONS ECOLOGY REVIEW COMMENTS
 BY JGC
 DATE 8/9/22

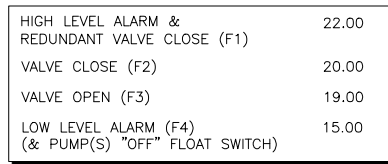
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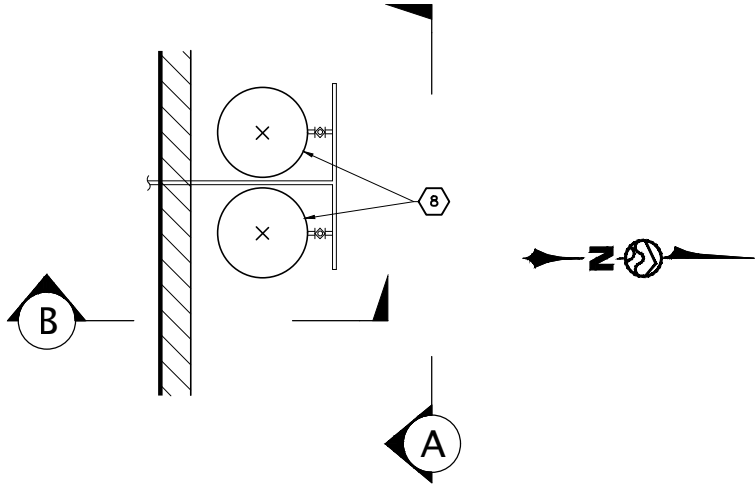
EASTSOUND SEWER AND WATER DISTRICT
 SAN JUAN COUNTY WASHINGTON
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
 INFLUENT PUMP STATION

DATE SCALE AS SHOWN
 JOB NUMBER 96
 SHEET 31 OF

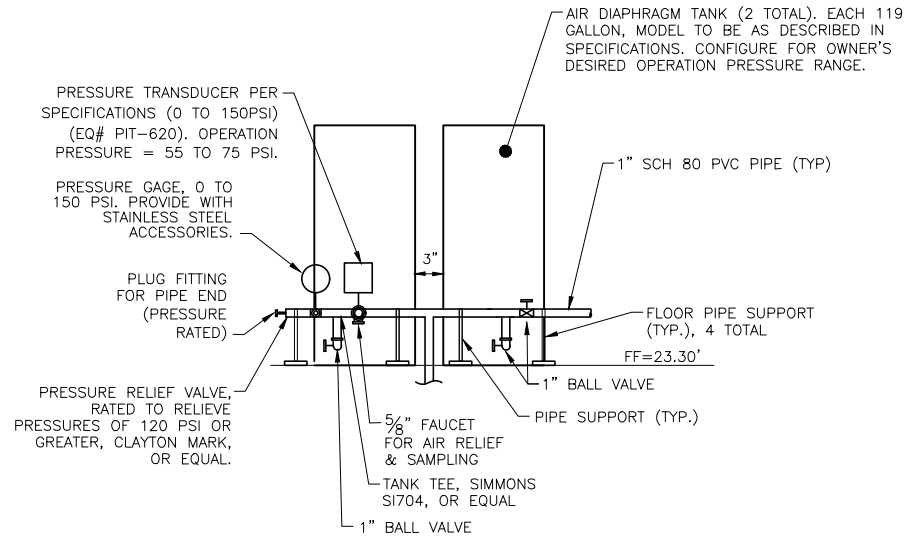


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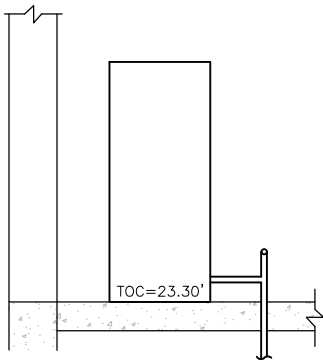


1 2W DIAPHRAGM TANKS – PLAN
 PIPE SUPPORTS NOT SHOWN FOR CLARITY



A 2W DIAPHRAM TANKS – SECTION 'A'
 NOT TO SCALE

- NOTES:
1. FILL ALL PIPING WITH CLEAN WATER (DOMESTIC POTABLE WATER) PRIOR TO STARTING AND TESTING 2W WATER SYSTEM EQUIPMENT.
 2. CONTRACTOR RESPONSIBLE FOR COORDINATING PIPING CONNECTION STYLE (GROOVED, FLANGED, THREADED, ETC.) AND PERFORMING PRESSURE TESTING TO DEMONSTRATE WATER TIGHT CONSTRUCTION.



2W DIAPHRAGM TANK – SECTION 'B'
 VALVE HANDLES AND HOIST CHAINS NOT SHOWN FOR CLARITY.



BID SET

NOV 2, 2022

NO.	REVISIONS	BY	DATE

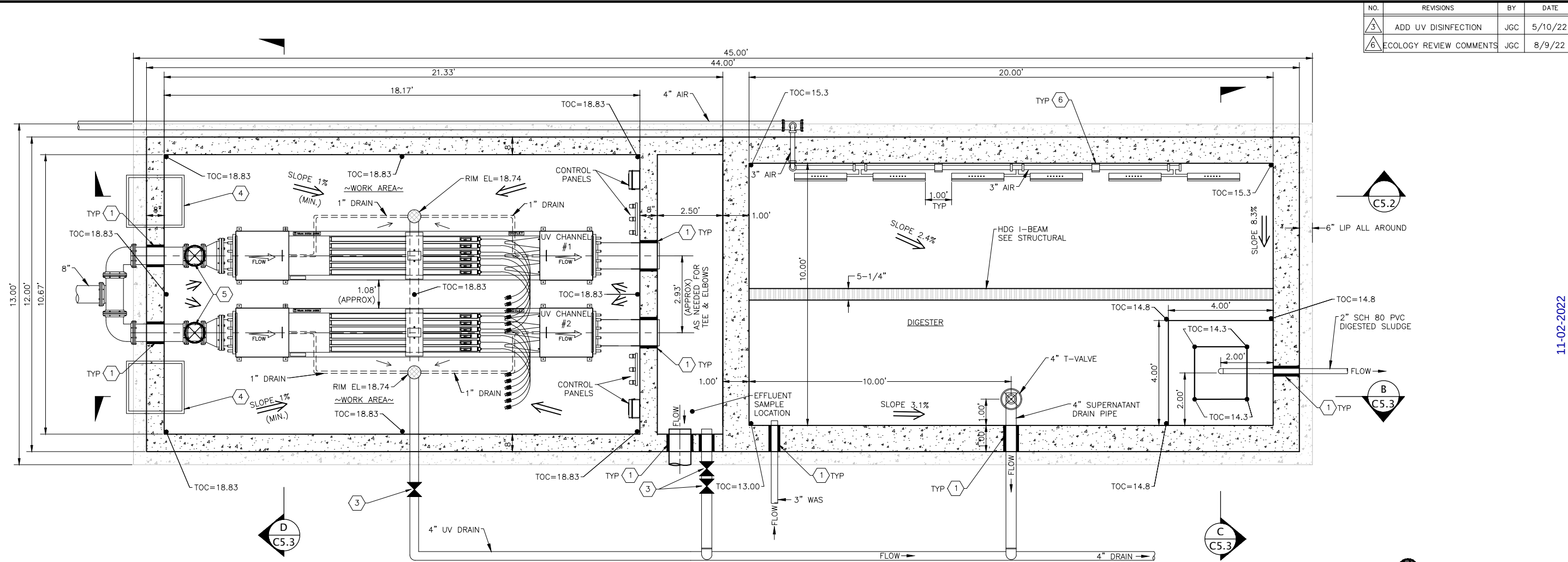
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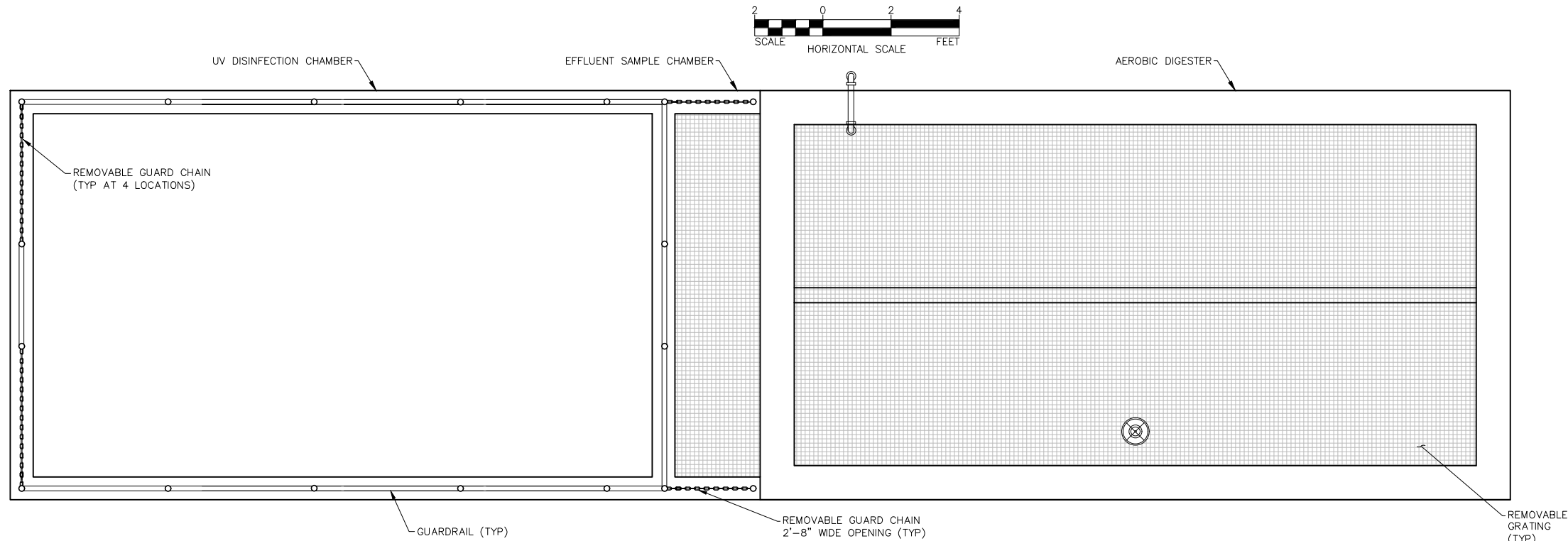
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SHEET	DATE	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY	JCC
	SCALE	SAN JUAN COUNTY	WASHINGTON	DRAWN BY	SNR
	AS SHOWN	WASTEWATER TREATMENT PLANT UPGRADE – PHASE I		CHECKED BY	AWL
	JOB NUMBER	2W EQUIPMENT LAYOUT			
33	OF	96			

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UV DISINFECTION CHAMBER & DIGESTER – PLAN VIEW



UV DISINFECTION CHAMBER & DIGESTER – GRATING & GUARDRAIL PLAN

- KEYED NOTES
- 1 = INSTALL LINK-SEAL BOOTS OR EQUAL. TYP. ON ALL PIPE TO CONCRETE CONNECTIONS. SEAL ALL VOIDS.
 - 2 = BRACKET, SECURELY ANCHORED TO CONC. WALL (TYP)
 - 3 = 4" GATE VALVE (TYP)
 - 4 = ALUMINUM ACCESS LADDER (NOMINAL WIDTH = 2')
 - 5 = 8" GATE VALVE WITH HAND WHEEL
 - 6 = WALL MOUNT PIPE SUPPORT PER 7 C7.2



BID SET

NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

11-02-2022

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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

UV DISINFECTION CHAMBER & DIGESTER PLAN

DATE

SCALE

JOB NUMBER

34

OF

96

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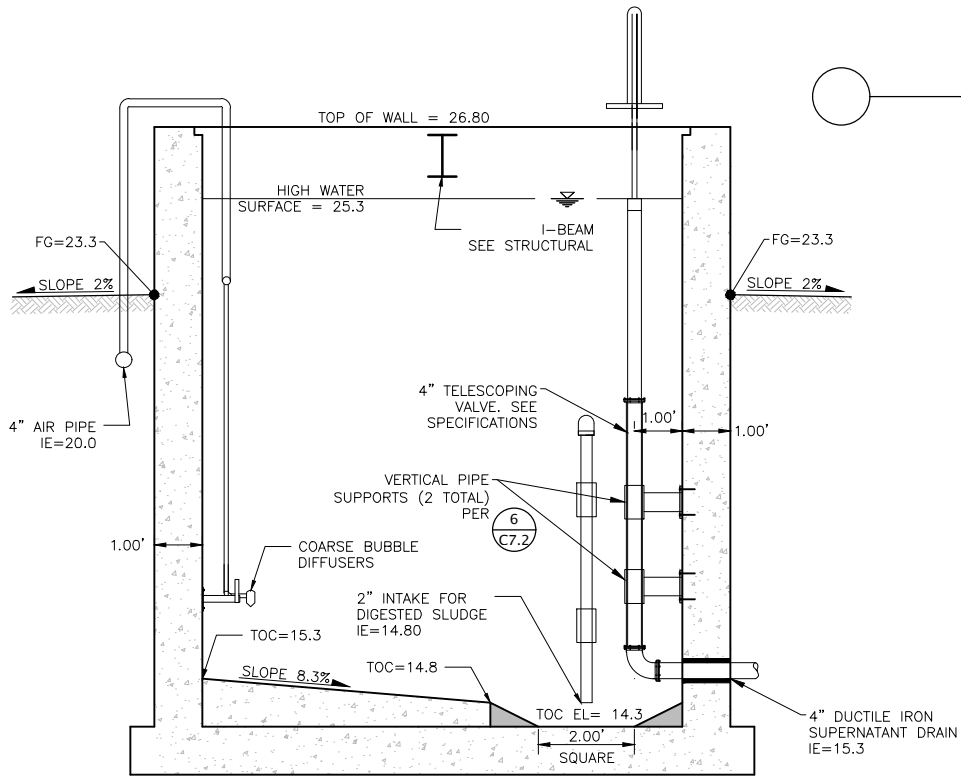
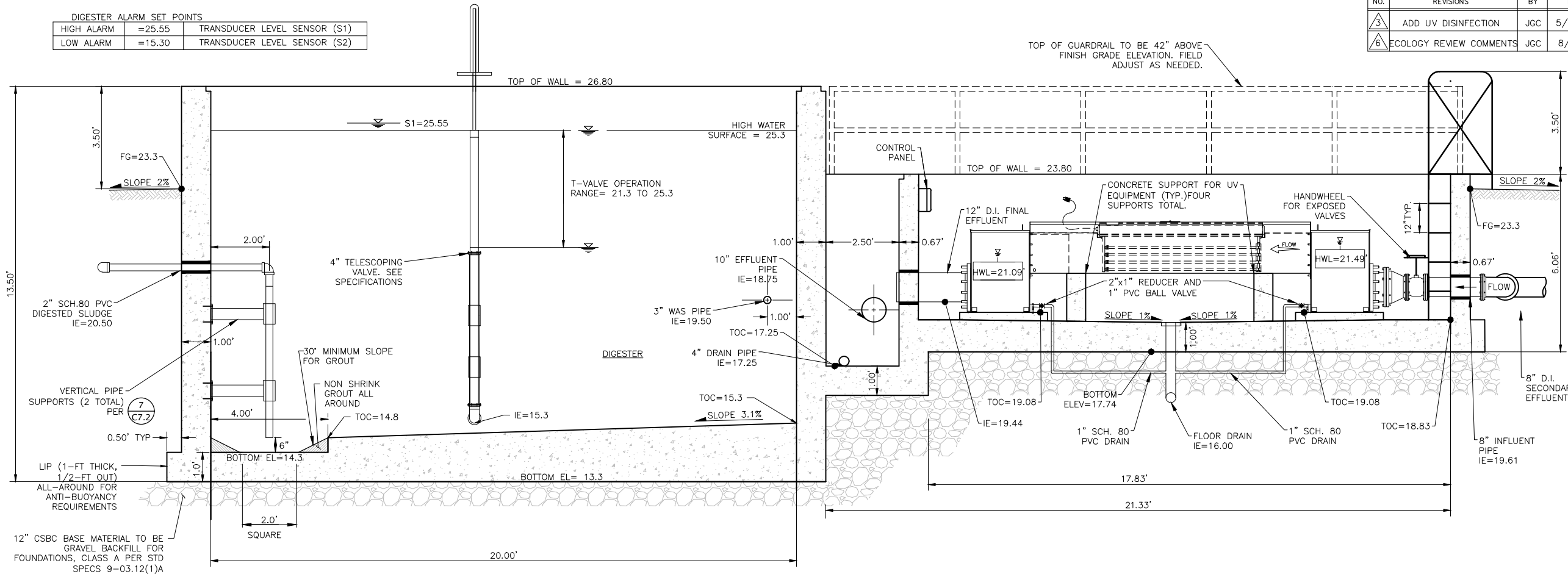
SEAL

11-02-2022

Jeff Smith

DIGESTER ALARM SET POINTS			
HIGH ALARM	=25.55	TRANSDUCER LEVEL SENSOR (S1)	
LOW ALARM	=15.30	TRANSDUCER LEVEL SENSOR (S2)	

NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



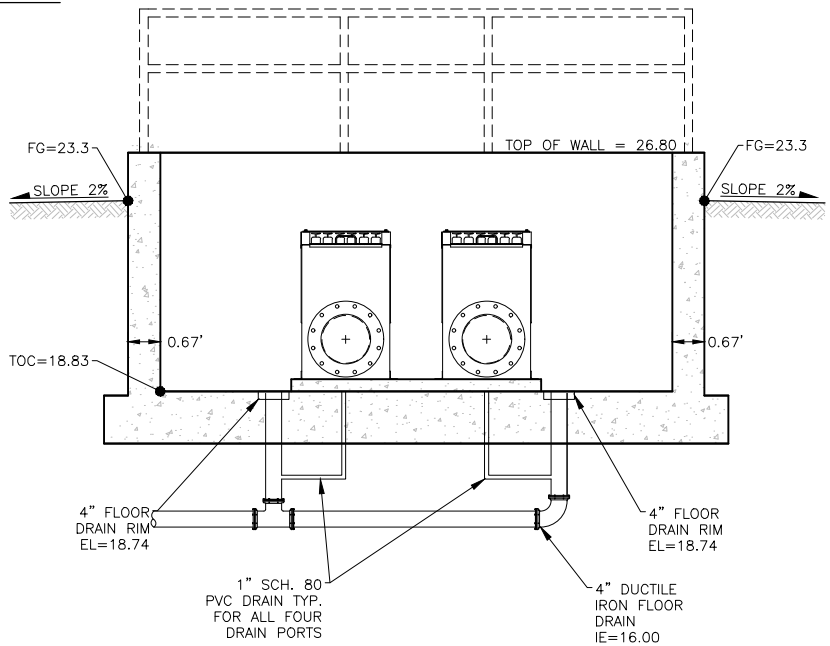
DIGESTER - SECTION VIEW 'C'

UV DISINFECTION & DIGESTER - SECTION VIEW 'B'



811 Call 811
two business days
before you dig

BID SET



UV DISINFECTION CHAMBER - SECTION VIEW 'D'

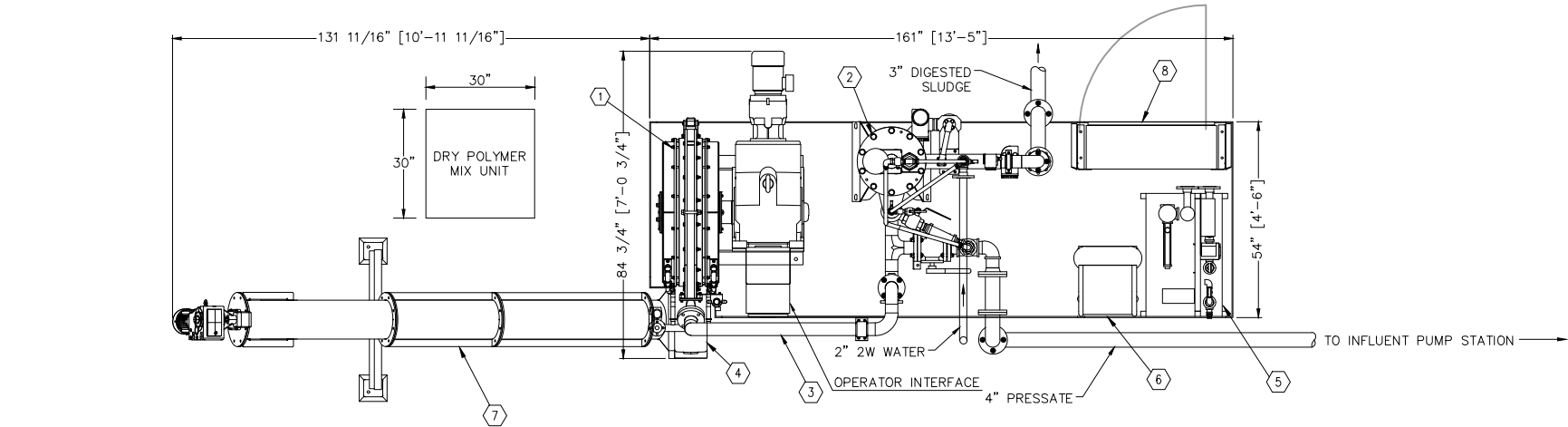


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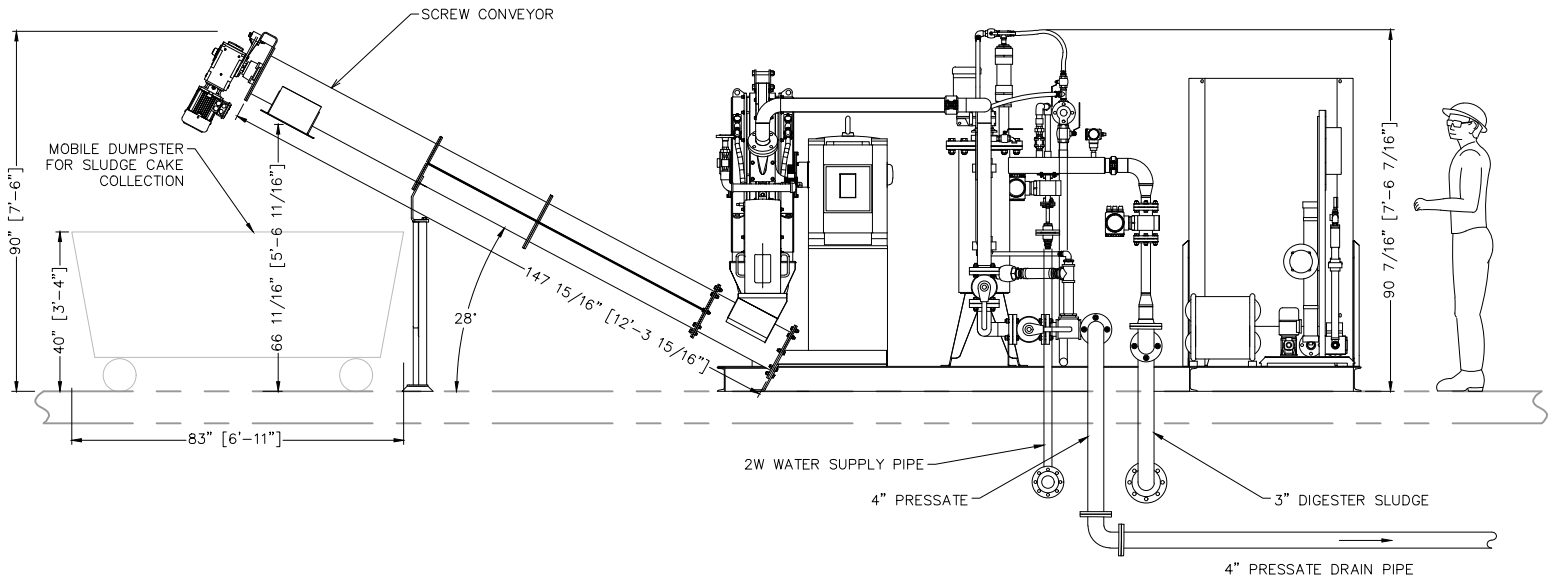
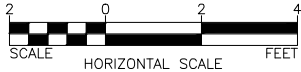
NOV 2, 2022

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

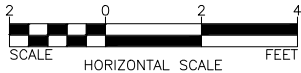
MATERIAL LIST						
ITEM	REF.	QTY	DESIGNATION	MAT.	DIM.	NOTES
1		1	ROTARY PRESS MODEL	STEEL		SECTION 46 76 26
2		1	FLOCCULATOR	STEEL		SECTION 46 76 26
3		1	SLUDGE MANIFOLD AND PIPING	S.S.316		SECTION 46 76 26
4		1	CAKE CHUTE	LLDPE		SECTION 46 76 26
5		1	EMULSION POLYMER MAKE UP SYSTEM			SECTION 46 33 33
6		1	AIR COMPRESSOR			SECTION 46 76 26
7		1	SCREW CONVEYOR EQUIPMENT	S.S.		SECTION 46 12 13
8		1	CONTROL PANEL			SECTION 46 76 26



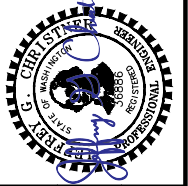
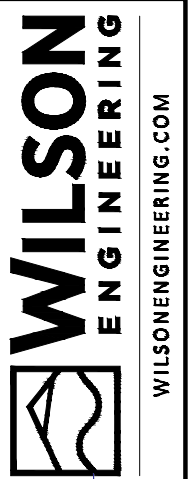
ROTARY FAN PRESS PLAN



ROTARY FAN PRESS ELEVATION



BID SET



DESIGNED BY	JGC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY
 WASHINGTON
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

ROTARY FAN PRESS SYSTEM

SHEET	DATE	SCALE	JOB NUMBER
38	OF	96	



DESIGNED BY	JGC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHTENAW COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

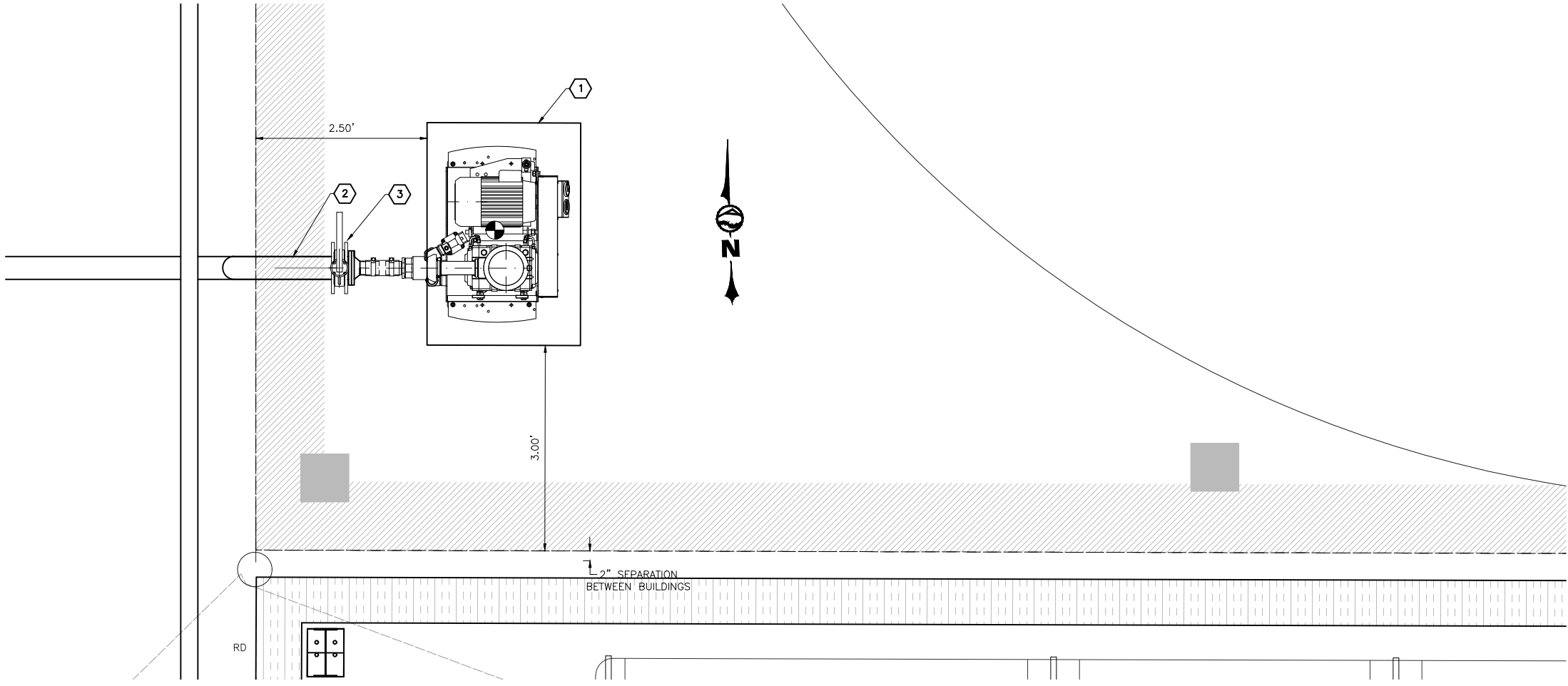
SLUDGE PUMP AND VALVING AREA

SHEET	DATE	
	SCALE	
	AS SHOWN	
	JOB NUMBER	

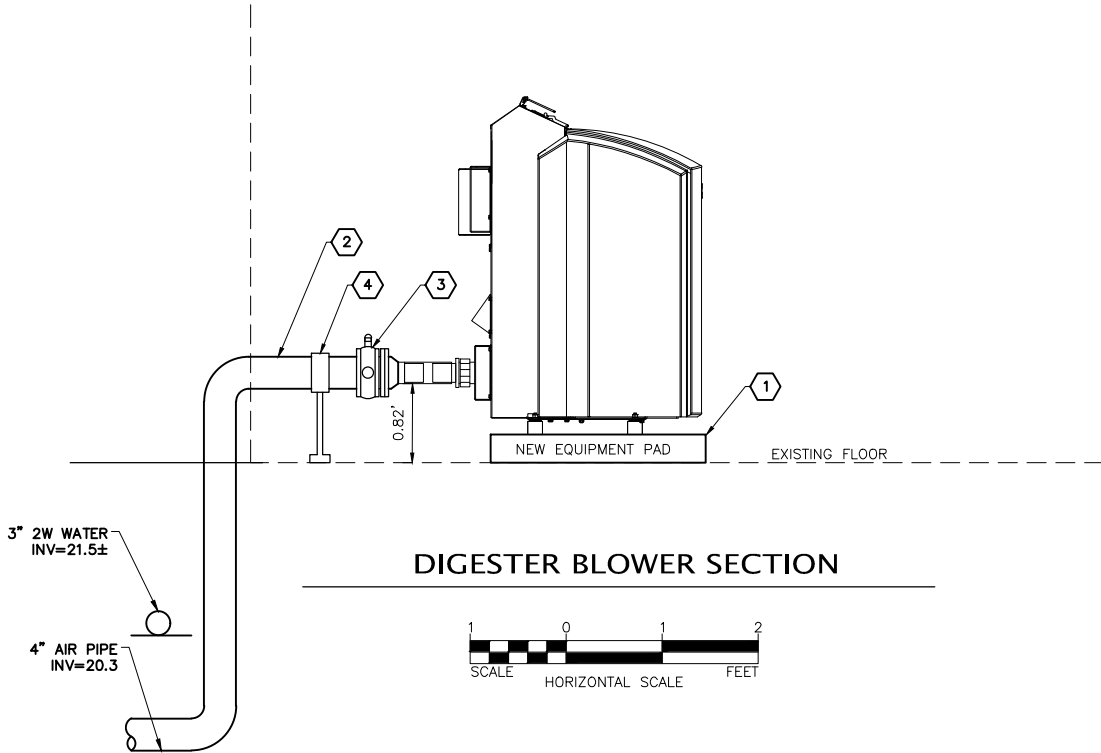
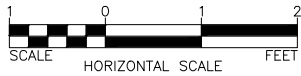
811 Call 811
two business days
before you dig

BID SET

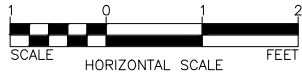
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 W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C6.3 DIGESTER BLOWER.DWG - 11/2/2022 3:26 PM - Jeff Smith



DIGESTER BLOWER



DIGESTER BLOWER SECTION



- KEYED NOTES**
- 1 = 3.25'x2.25' BLOWER EQUIPMENT PAD PER C7.4
 - 2 = 4" AIR PIPING (304 SS)
 - 3 = 4" BUTTERFLY VALVE
 - 4 = PIPE SUPPORT PER C7.2



BID SET

NO.	REVISIONS	BY	DATE

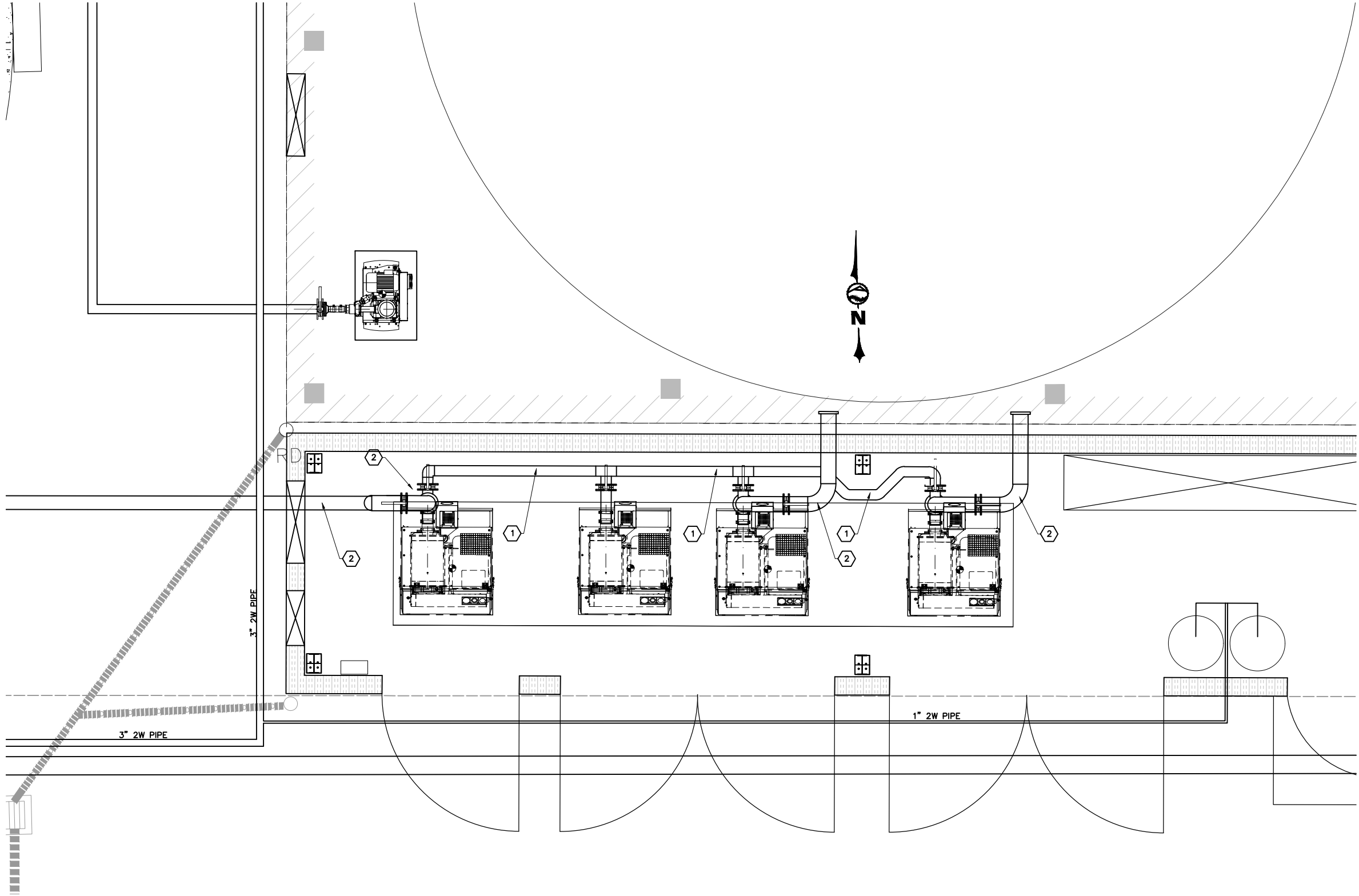
11-02-2022

SHEET	DATE	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY	JCC
	SCALE	SAN JUAN COUNTY	WASHINGTON	DRAWN BY	JGS
AS SHOWN		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		CHECKED BY	AWL
JOB NUMBER		DIGESTER BLOWER			
40 OF 96					

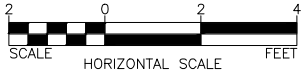
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G. CHRISTOPHER WILSON
PROFESSIONAL ENGINEER
STATE OF WASHINGTON
NO. 34555
MECHANICAL

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 Inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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BIOLOGICAL TREATMENT BLOWERS



- KEYED NOTES**
- 1 = 3" STAINLESS STEEL AIR PIPE. SEE EXPOSED AERATION PIPING, SECTION 33 31 00
 - 2 = 6" STAINLESS STEEL AIR PIPE. SEE EXPOSED & BURIED AERATION PIPING SECTION 33 31 00



BID SET

NO.	REVISIONS	BY	DATE

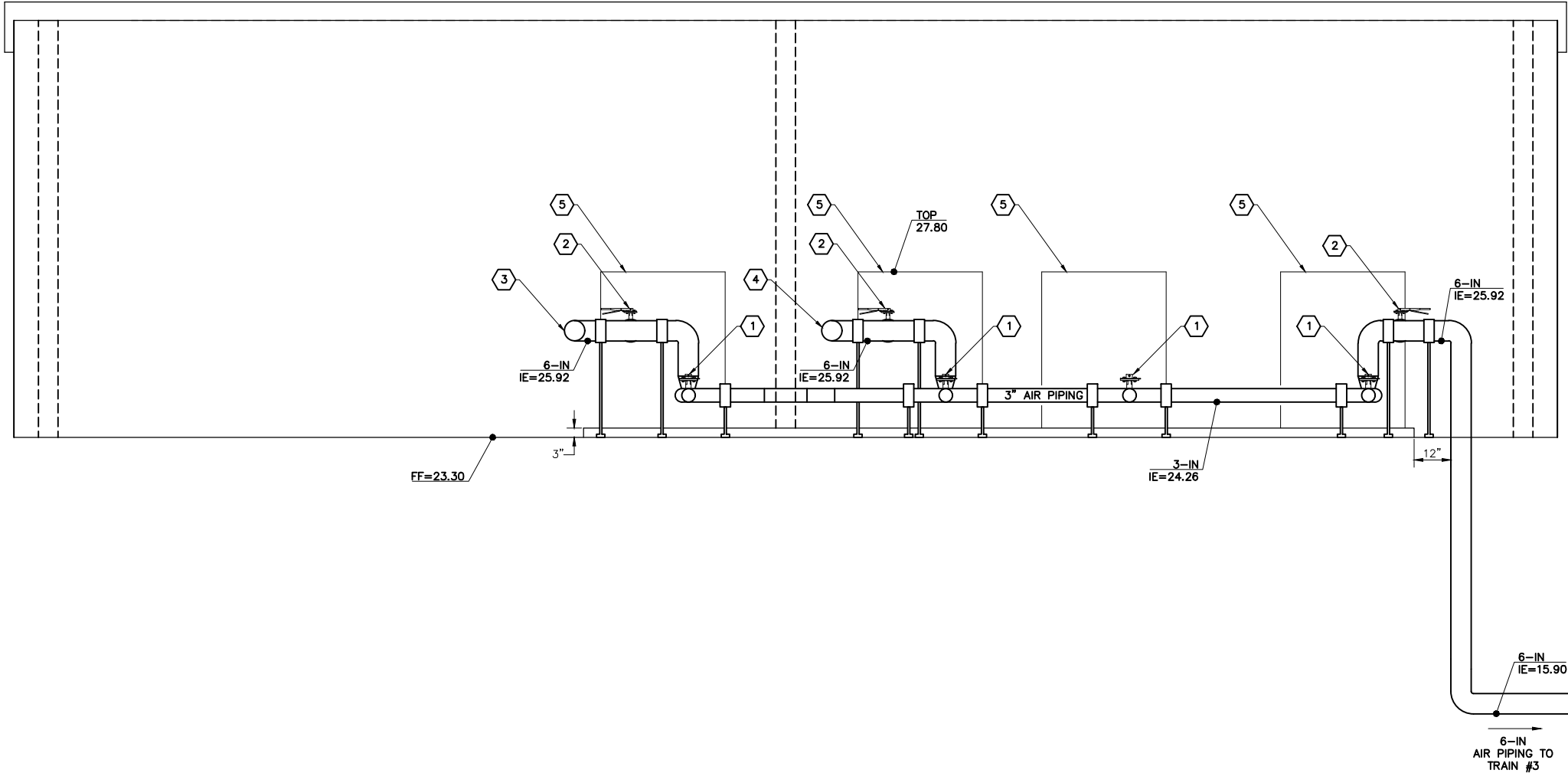
 WILSON ENGINEERING WILSONENGINEERING.COM	 <i>Jeff Smith</i>	DESIGNED BY	JCC
		DRAWN BY	JGS
		CHECKED BY	AWL
EASTSOUND SEWER AND WATER DISTRICT SAN JUAN COUNTY WASTEWATER TREATMENT PLANT UPGRADE - PHASE I BIOLOGICAL TREATMENT BLOWERS		DATE	
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		JOB NUMBER	96
SHEET 41 OF 96			

11-02-2022

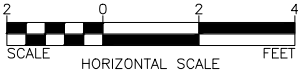
NOV 2, 2022

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NOV 2, 2022



BIOLOGICAL TREATMENT BLOWER ELEVATION



- KEYED NOTES**
- 1 = 3" BUTTERFLY VALVE
 - 2 = 6" BUTTERFLY VALVE
 - 3 = 6" AIR PIPING TO TRAIN #1 (TERMINATE WITH BLIND FLANGE INSIDE EXISTING METAL BUILDING)
 - 4 = 6" AIR PIPING TO TRAIN #2 (TERMINATE WITH BLIND FLANGE INSIDE EXISTING METAL BUILDING)
 - 5 = NEW BLOWER PER SPECS

11-02-2022

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BID SET

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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

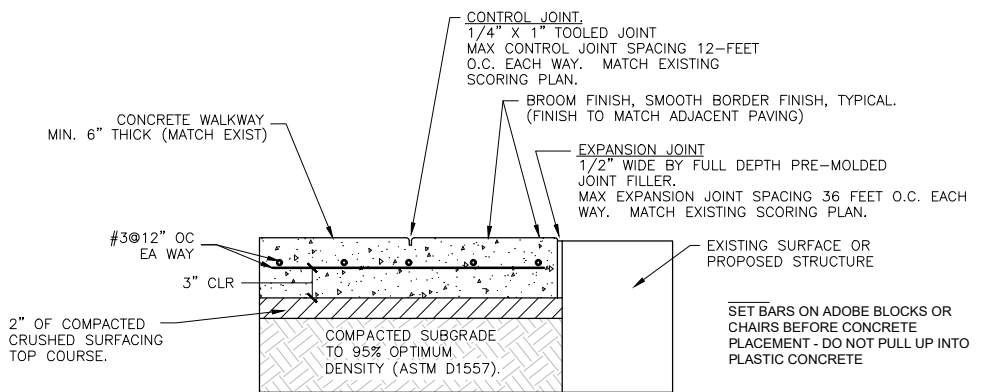
BIOLOGICAL TREATMENT BLOWER ELEVATION

DATE	SCALE	AS SHOWN	JOB NUMBER

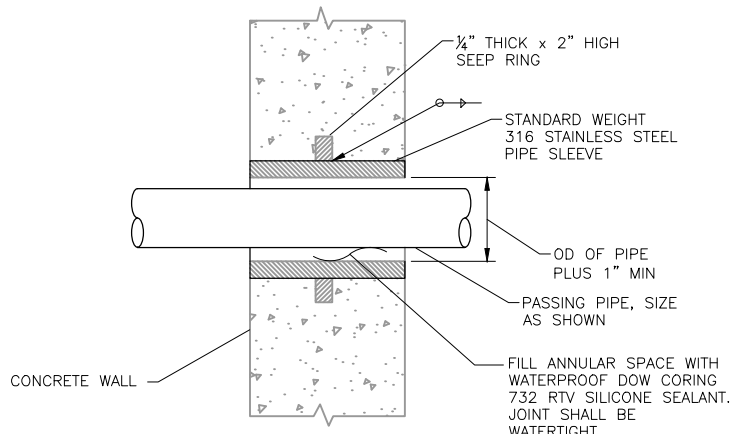
SHEET

42 OF 96

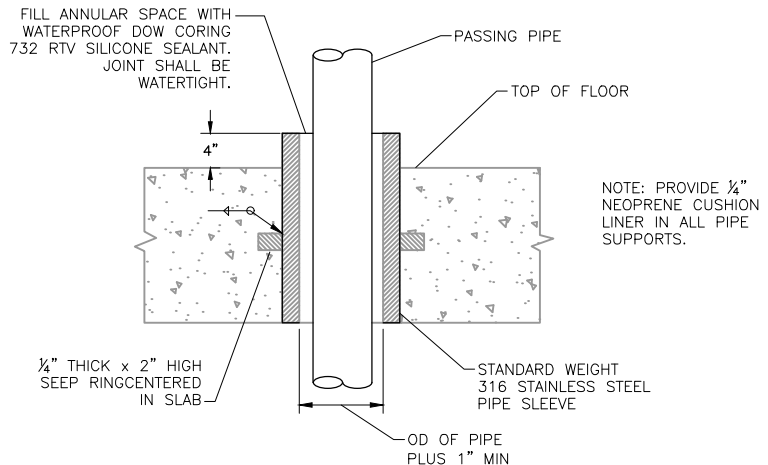
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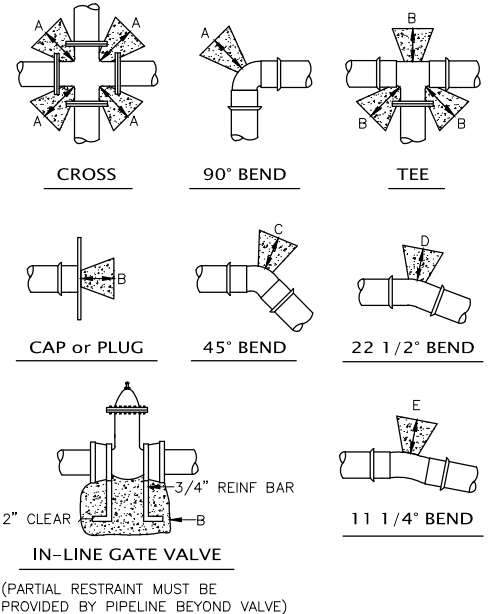
1 CONCRETE WALKWAY DETAIL
 NOT TO SCALE



2 PIPE SLEEVE DETAIL
 NOT TO SCALE



3 FLOOR SLEEVE DETAIL
 NOT TO SCALE



- NOTES:
1. SQUARE FEET OF CONCRETE THRUST BLOCK AREA IS BASED ON 200 P.S.I. INTERNAL PRESSURE, A SOIL SAFE BEARING OF 3000 POUNDS PER SQUARE FOOT AND A FACTOR OF SAFETY OF 1.5.
 2. BEARING AREA MUST BE ADJUSTED FOR INTERNAL PRESSURES AND LOWER SOIL BEARING VALUES.
 3. CONCRETE BLOCKING SHALL BE CAST IN PLACE AND HAVE A MINIMUM OF 1/4 SQUARE FOOT BEARING AGAINST THE FITTING.
 4. BLOCK SHALL BEAR AGAINST FITTINGS ONLY AND SHALL BE CLEAR OF JOINTS TO PERMIT TAKING UP OR DISMANTLING JOINT.
 5. THE CONTRACTOR SHALL INSTALL BLOCKING WHICH IS ADEQUATE TO WITHSTAND FULL TEST PRESSURE AS WELL AS TO CONTINUOUSLY WITHSTAND OPERATING PRESSURE UNDER ALL CONDITIONS OF SERVICE.
 6. STAINLESS STEEL BANDING SHALL BE USED AT 2" PVC VERTICAL BENDS INSTEAD OF 1" RODS. CONTACT ENGINEER FOR SIZING OF THRUST BLOCK AND DETAILS.
 7. ALL BENDS, TEES & CROSSES SHALL INCLUDE RESTRAINED JOINTS (ROMAC GRIPPER) AS WELL AS THRUST BLOCKING.

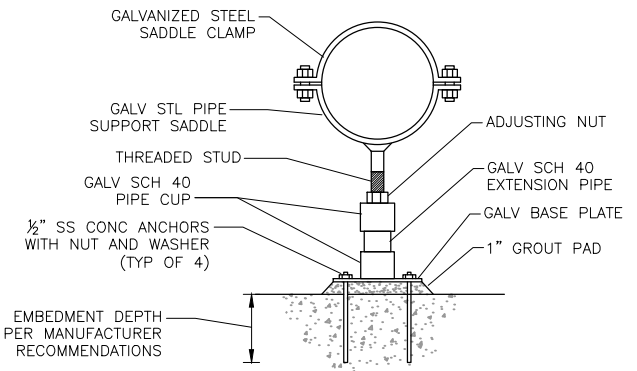
4 FORCEMAIN THRUST BLOCKING SCHEDULE
 NOT TO SCALE

811 Call 811
 two business days
 before you dig

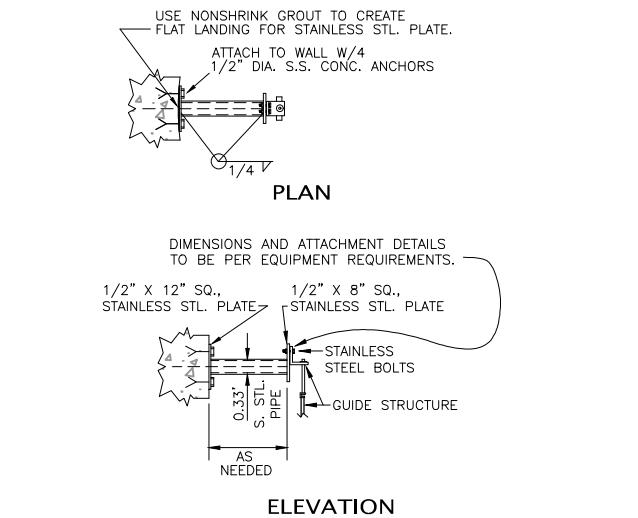
BID SET

- NOTES:
1. TO INSURE PROPER SUPPORT AND STABILITY, AFTER FINAL HEIGHT ADJUSTMENT IS ATTAINED, APPLY TACK WELDS TO BOTH SUPPORT CUPS AND EXTENSION PIPE. USE E70XX ELECTRODE FOR WELDS.
 2. HOT DIP GALVANIZE ALL PARTS AFTER FABRICATIONS.
 3. FIELD PAINT AS SPECIFIED.

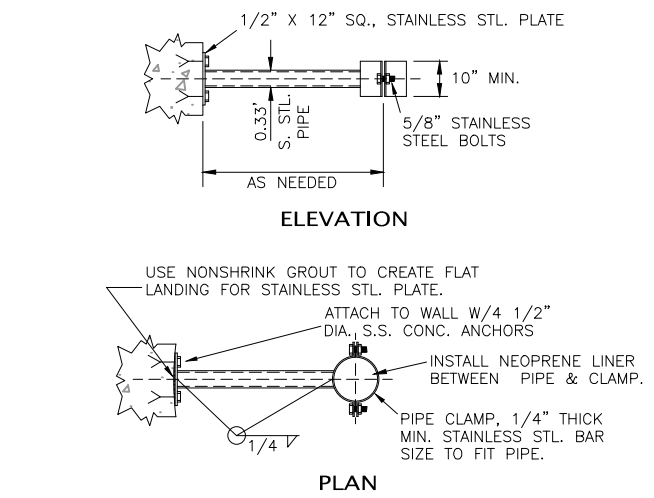
SADDLE SIZE	1/2"x2"
THREADED STUD	1"x6"
CUP ID	2 1/2"
1/4" BASE PLAT	6"x6"
EXTENSION PIPE DIA	2"



5 PIPE SUPPORT DETAIL
 NOT TO SCALE



GUIDE STRUCTURE SUPPORT
 NOT TO SCALE



VERTICAL PIPE BRACE
 NOT TO SCALE

NO. REVISIONS BY DATE

11-02-2022

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EASTSOUND SEWER AND WATER DISTRICT
 WASHINGTON
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

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 DRAWN BY SNR
 CHECKED BY AWL

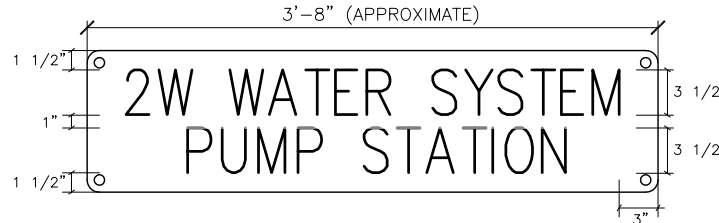
EASTSOUND SEWER AND WATER DISTRICT
 SAN JUAN COUNTY
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DATE SCALE AS SHOWN JOB NUMBER
 44 OF 96

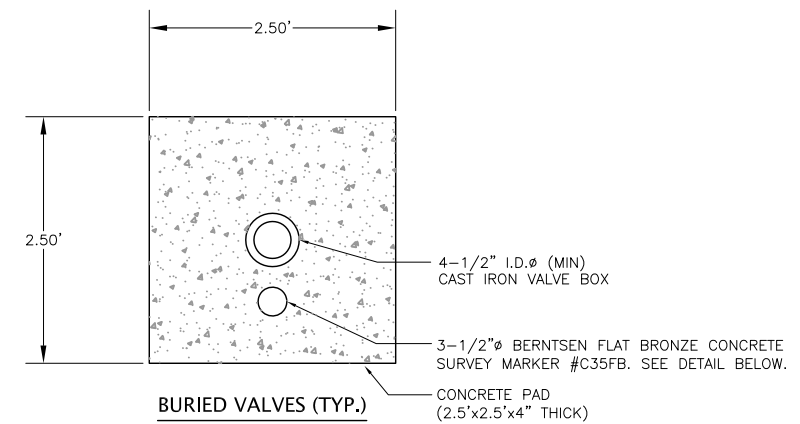
SHEET



1 WWTP TANK SIGN DETAILS
NOT TO SCALE



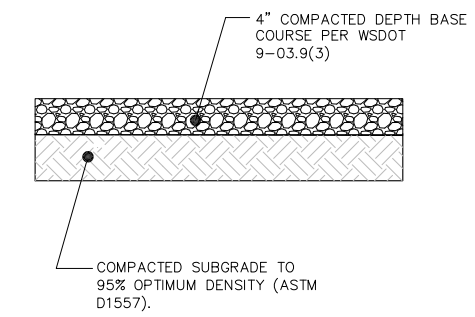
- NOTES FOR SIGNS:**
1. PROVIDE FOUR STAINLESS STEEL CONCRETE ANCHORS PER WALL SIGN.
 2. PROVIDE FOUR 2" S.S. GUARD RAIL FASTENERS PER RAIL SIGN. PROVIDE & ATTACH ALUMINUM BACKING PLATE TO FASTENERS. CENTER BASIN SIGN BETWEEN HORIZONTAL RAILS AND MOUNT TO ALUMINUM BACKING PLATE.
 3. ALL SIGNS ARE TO BE BAKED ENAMEL ON STEEL PLATE WITH BLACK LETTERING ON A WHITE BACKGROUND.
 4. ALL SIGN LOCATIONS ARE TO BE FIELD DETERMINED BY OWNER.



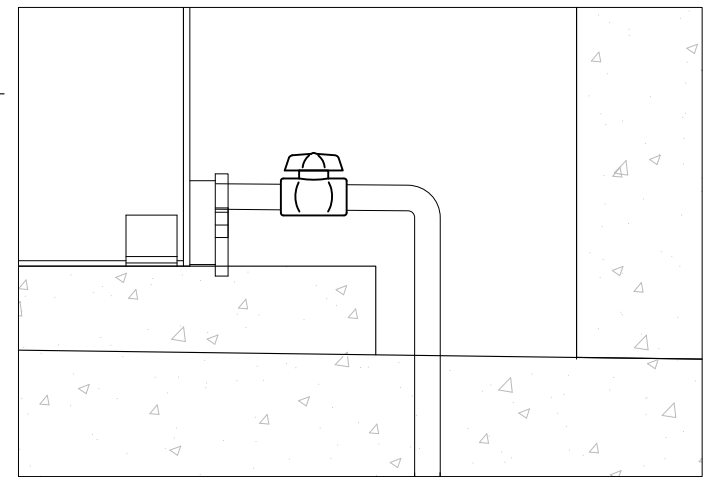
BURIED VALVES (TYP.)

BURIED VALVE IDENTIFICATION LEGEND	
1	6" GATE VALVE FOR INFLUENT FLOW METER ISOLATION
2	8" GATE VALVE FOR TRAIN 2 FLOW METER ISOLATION
3	8" GATE VALVE FOR TRAIN 3 FLOW METER ISOLATION
4	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
5	4" GATE VALVE FOR DRAIN FLOW METER ISOLATION
6	4" GATE VALVE FOR CCB OUTLET BOX DRAIN
7	4" GATE VALVE FOR CCB1 DRAIN
8	4" GATE VALVE FOR CCB 1 DRAIN
9	4" GATE VALVE FOR CCB 1 DRAIN
10	4" GATE VALVE FOR CCB 1 DRAIN
11	4" GATE VALVE FOR CCB 1 DRAIN
12	4" GATE VALVE FOR CCB 2 DRAIN
13	4" GATE VALVE FOR CCB 2 DRAIN
14	4" GATE VALVE FOR CCB 2 DRAIN
15	4" GATE VALVE FOR CCB 2 DRAIN
16	4" GATE VALVE FOR CCB 2 DRAIN
17	8" GATE VALVE FOR EFFLUENT FLOW METER ISOLATION

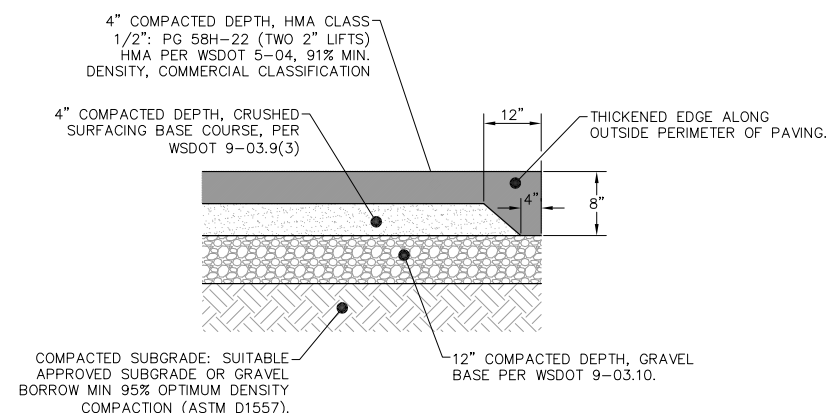
2 BURIED VALVE LABELING SCHEDULE
NOT TO SCALE



5 GRAVEL SURFACING SECTION
NOT TO SCALE

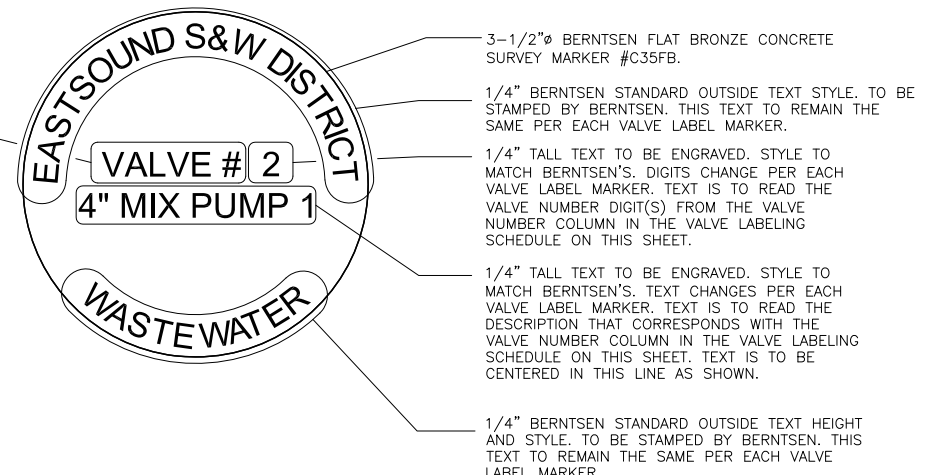


6 1" PVC BALL VALVE
NOT TO SCALE



STANDARD ASPHALT SECTION—FOR AREAS WITH NO CURB

3 ASPHALT CONCRETE PAVEMENT WITH THICKENED EDGE
NOT TO SCALE



4 TYPICAL BURIED VALVE LABEL MARKER
NOT TO SCALE

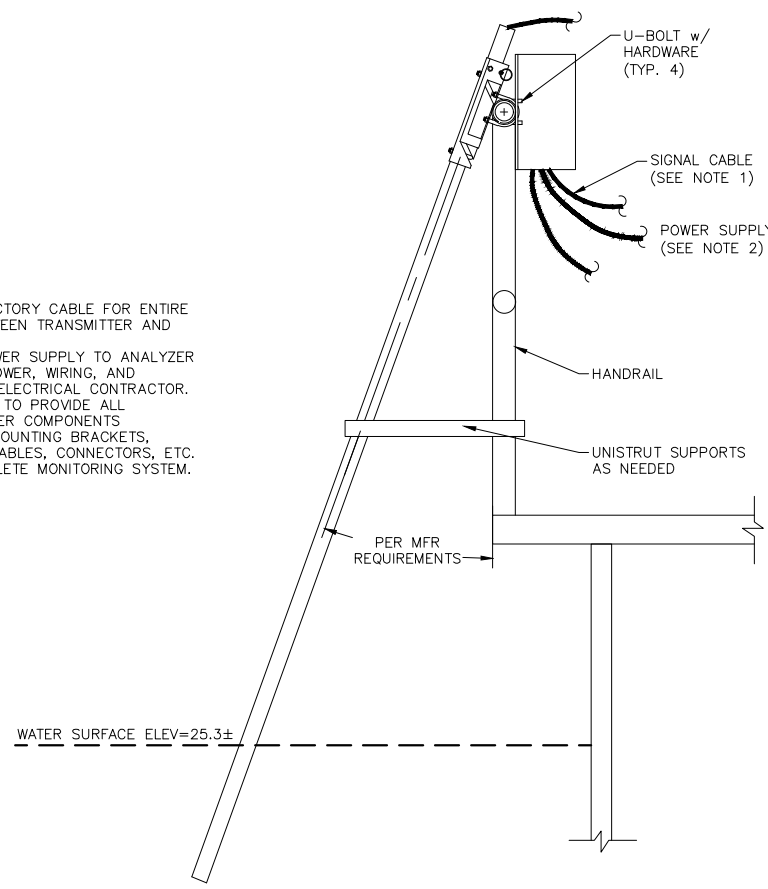


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NOTE:

1. MAINTAIN FACTORY CABLE FOR ENTIRE LENGTH BETWEEN TRANSMITTER AND ELEMENT.
2. 120 VAC POWER SUPPLY TO ANALYZER REQUIRED. POWER, WIRING, AND CONDUIT BY ELECTRICAL CONTRACTOR.
3. CONTRACTOR TO PROVIDE ALL MANUFACTURER COMPONENTS INCLUDING: MOUNTING BRACKETS, ADAPTERS, CABLES, CONNECTORS, ETC. FOR A COMPLETE MONITORING SYSTEM.

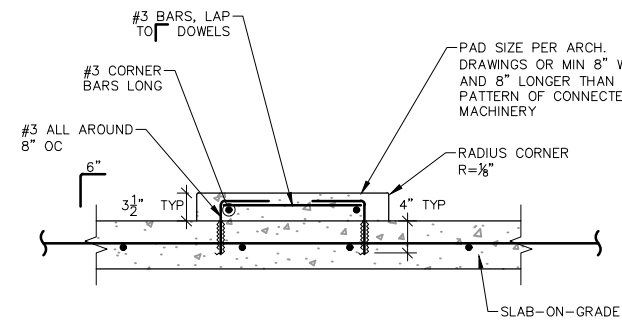


D.O. PROBE ELEVATION (TYP.)

NOT TO SCALE



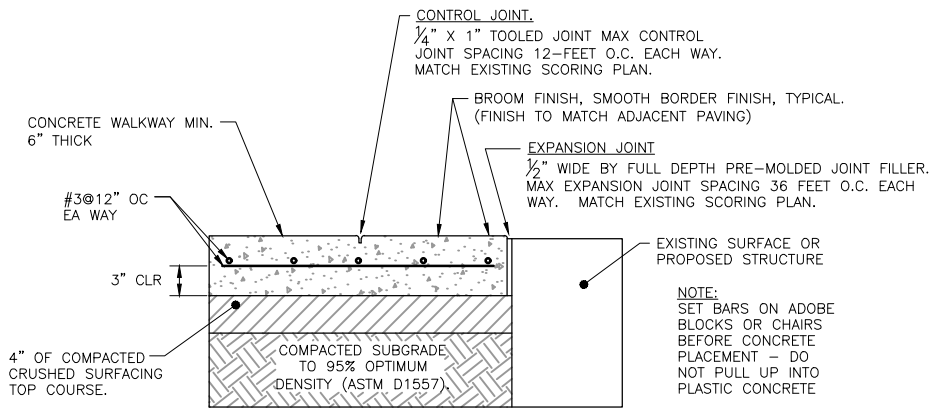
BID SET



NOTE: ADHESIVE FOR DOWELS PER STRUCTURAL NOTES

TYPICAL MECHANICAL EQUIPMENT PAD

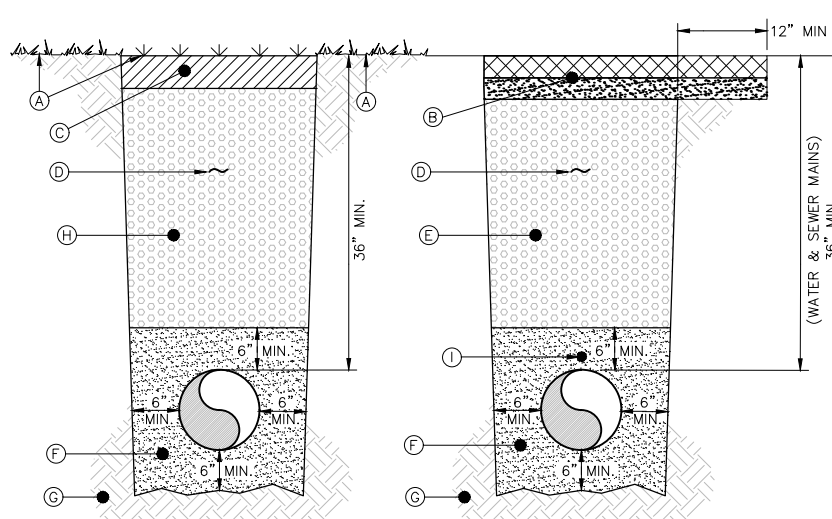
NOT TO SCALE



NOTE: SET BARS ON ADOBE BLOCKS OR CHAIRS BEFORE CONCRETE PLACEMENT - DO NOT PULL UP INTO PLASTIC CONCRETE

CONCRETE PAVEMENT DETAIL

NOT TO SCALE



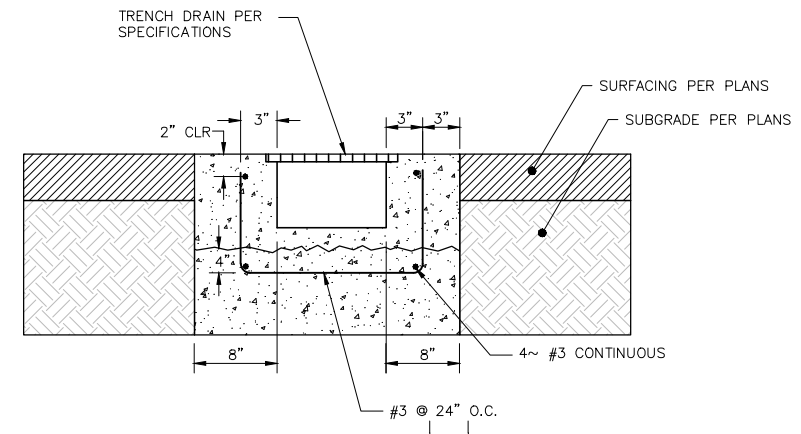
TYPICAL TRENCHING & BACKFILL

NOT TO SCALE

NO.	REVISIONS	BY	DATE

KEYED NOTES

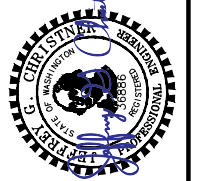
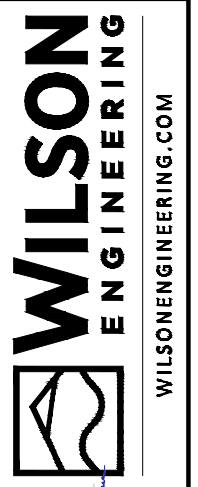
- A. HYDROSEED EXPOSED AREAS.
- B. GRAVEL: NEW CRUSHED SURFACING TOP COURSE (CSTC) LAYER PER WSDOT 9-03.9(3). PAVED: HMA CL 1/2" APPLIED IN 2 LIFTS MIN. TO THICKNESS MATCHING EX. SECTION (4" MIN.) PER WSDOT 5-04.2, WITH 2" CSTC LAYER BELOW.
- C. NEW TOPSOIL LAYER MIN 4".
- D. 2" METALLIC DETECTOR TAPE 8" TO 12" BELOW FINISH GRADE.
- E. BANK RUN GRAVEL BACKFILL PER WSDOT 9-03.19 COMPACTED TO 95% MAX. DENSITY INSIDE RIGHT-OF-WAY.
- F. PIPE ZONE GRAVEL BEDDING PER WSDOT 9-03.12(3) COMPACTED TO 95% MAX. DENSITY.
- G. UNDISTURBED NATIVE MATERIAL.
- H. SELECT NATIVE BACKFILL COMPACTED TO 95% MAX. DENSITY.
- I. #10 AWG INSULATED TRACER WIRE ON NONMETALLIC PIPE (IN ADDITION TO 2" DETECTOR TAPE)



TRENCH DRAIN DETAIL

NOT TO SCALE

11-02-2022



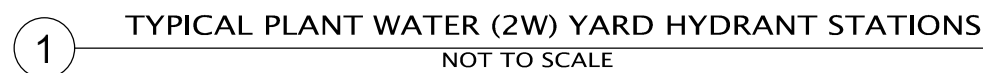
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CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

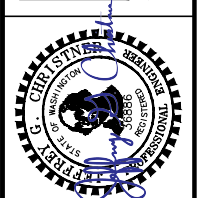
CIVIL DETAILS

DATE	SCALE	JOB NUMBER
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JGC	SNR	AWL

EASTSOUND SEWER AND WATER DISTRICT

WA
COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

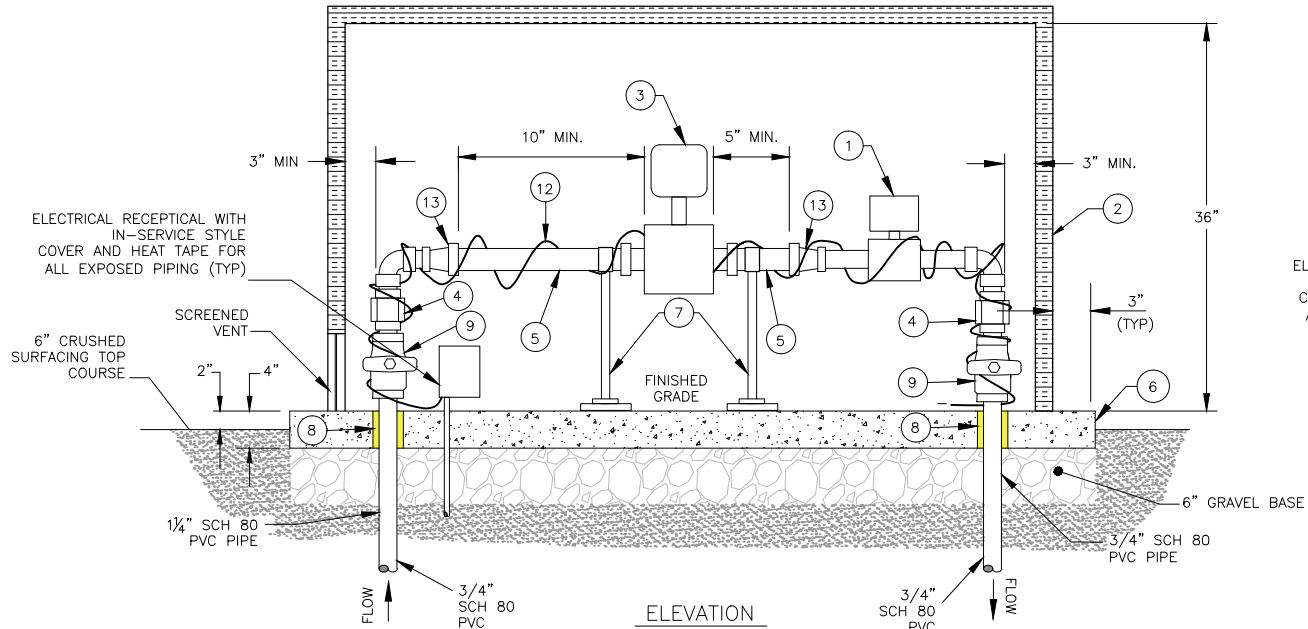
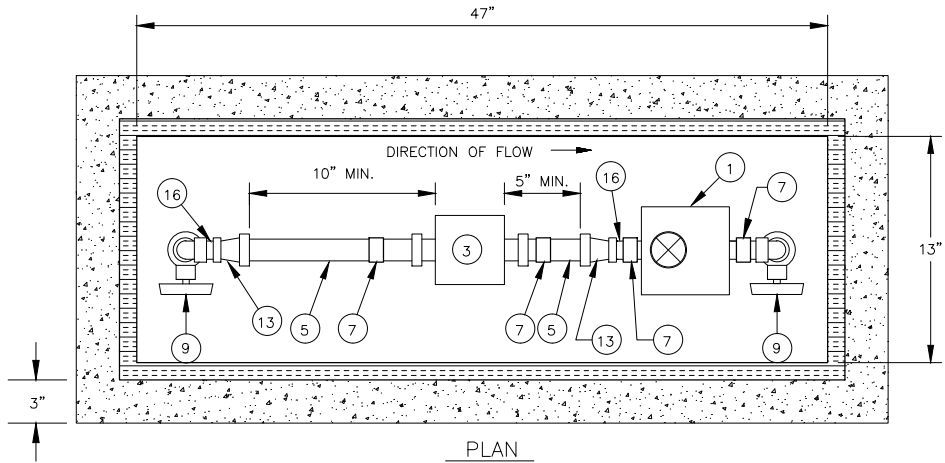
CIVIL DETAILS

SHEET	DATE
	SCALE
	AS SHOWN
47 OF 96	JOB NUMBER

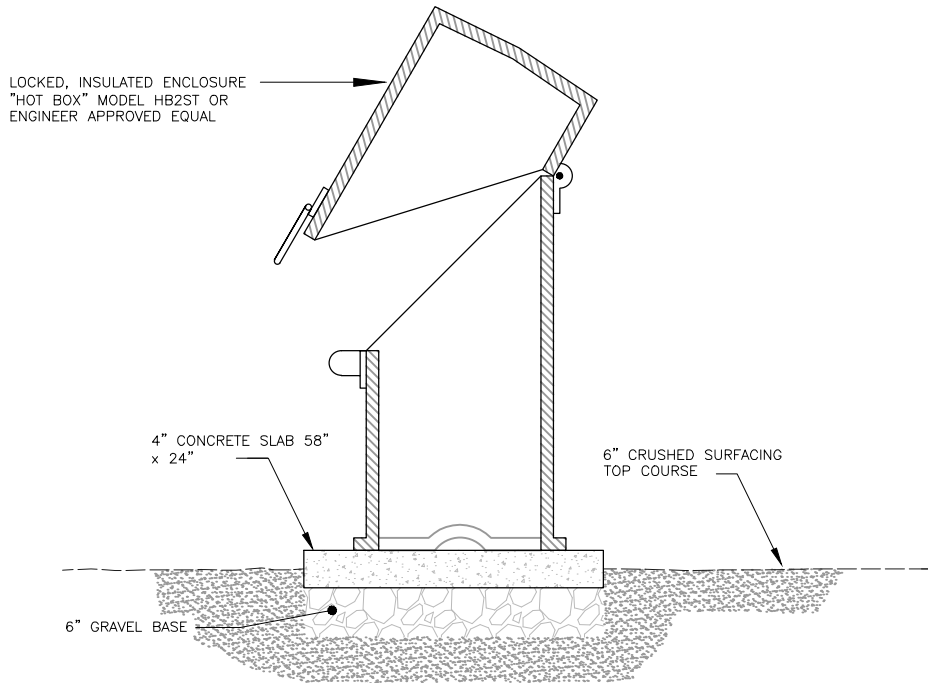
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MATERIAL LIST

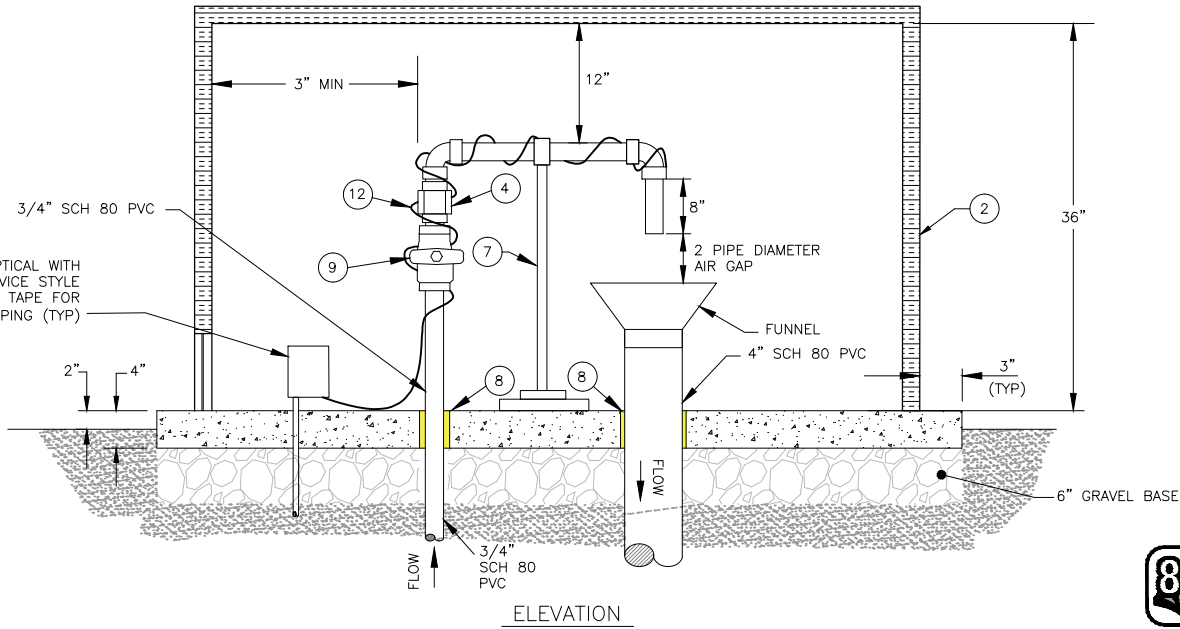
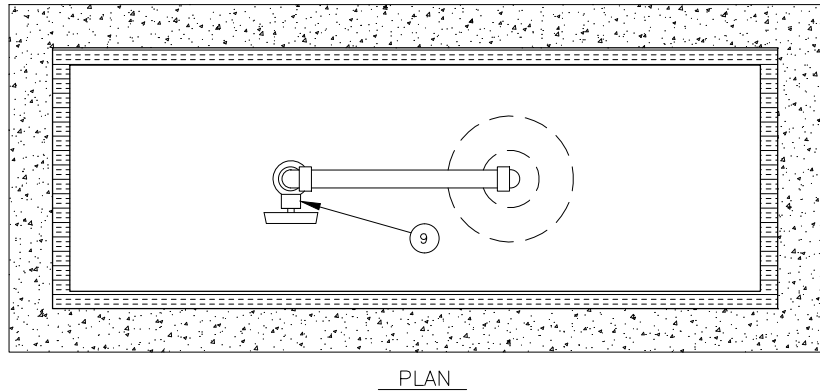
- SOLENOID VALVE, ASCO REDHAT 2 WAY SERIES 8221. SLOW CLOSING SOLENOID VALVE TO PROTECT AGAINST WATER HAMMER. NORMALLY CLOSED.
MAX WATER DIFFERENTIAL PRESSURE = 150 PSI.
SIZE = 3/4 INCH.
POWER= 120V AC, 6.1 WATTS
ORIFICE SIZE = 3/4 INCH.
BODY = BRASS, CATALOG #8221G005.
STAINLESS STEEL CORE TUBE, PLUGNUT, SPRINGS, AND PISTON.
ENCLOSURE = WATERTIGHT NEMA 4X. CORROSION RESISTANT. IP66. OUTDOOR RATING
- INSULATED AND HEATED ENCLOSURE. "HOT-BOX" OR DISTRICT APPROVED EQUIVALENT.
- 1" FLOW METER PER ELECTRICAL DESIGN REQUIREMENTS.
- 3/4" UNION. DEMONSTRATE REMOVAL AND RE-ATTACHMENT.
- PIPING TO BE SCH 80 PVC PIPE, 1 INCH DIAMETER.
- 4" THICK, 3,000 PSI, CONCRETE SLAB. 58"Lx24"Wx4" MIN. THICKNESS.
- SUPPORTS PER PIPE SUPPORT DETAIL C7.2
- FILL ANNULAR SPACE WITH DOW CORNING 732 MULTI-PURPOSE SEALANT. JOINT SHALL BE WATERTIGHT.
- 3/4" PVC MANUAL BALL VALVE
- CHECK VALVE, 1.25 INCH, SPEARS MODEL NO. S1780 WITH EPDM SEALS, 150 PSI PRESSURE RATING, STAINLESS STEEL SPRING ASSISTED OPERATION. SELF-CLOSING STYLE DESIGNED FOR HORIZONTAL INSTALLATION/OPERATION.
- FLOW CONTROL ASSEMBLY. ORENCO 1.25 INCH MODEL FCA. INCLUDE SPARE ORIFICE PLATES WITH THE FOLLOWING ORIFICE SIZES (IN INCHES): 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90 0.95, 1.00 AND 1.05.
- 90 W, 120 V, SINGLE PHASE HEAT TRACE TAPE.
- 1" x 3/4" EXPANSION FITTING



HOT BOX #1 - FLOW METER & ISOLATION VALVE
NOT TO SCALE



HOT BOX #2 - AIR GAP PIPING
NOT TO SCALE



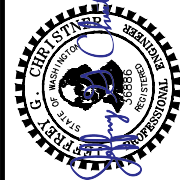
HOT BOX #2 - AIR GAP PIPING
NOT TO SCALE

NOTES:

- ALL WORK PERFORMED ID TO MEET MANUFACTURER'S RECOMMENDATIONS, AND MOST RECENT LOCAL AND STATE PLUMBING CODES.
- ALL CORROSIVE SURFACES SHALL BE CLEANED, PRIMED AND PAINTED WITH A MINIMUM OF (2) COATS OF EPOXY PAINT, AS APPROVED BY THE OWNER.
- ALL PIPING INSIDE HOT BOX #1 AND #2 TO BE SCH. 80 PVC PIPE

NO.	REVISIONS	BY	DATE

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CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

CIVIL DETAILS

DATE	SCALE	JOB NUMBER
	AS SHOWN	
SHEET	48	OF 96



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